The Möbius function at $N - p$

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Abstract

For even N tending to infinity, we deduce cancellation of $\mu(N-p)$ over primes and equal densities of its two nonzero signs from a zero-free input and specified arguments of Lü, Wang and Wu. We extend the determinant reduction to both positive-argument families $\mu(\varepsilon p + h)$, uniformly for $\varepsilon = \pm 1$ and $0 < |h| \le x(\log x)^C$, with error $O_C(\pi(x)/\sqrt{\log\log x})$ under the inputs H1 and H2′ stated below. We give the corresponding progression result for $q \le (\log x)^{C_q}$ and a relative rough-value result for $z \le \exp((\log x)^a)$, every fixed $0 < a < 1/2$, with even h. All have Liouville counterparts. The small-shift and reflected small-radical cases retain their weaker analytic inputs. We also give radical criteria using a uniform real-zero gap or classical exceptional-conductor avoidance.

Changes in version 2. This version adds signed affine shifts, progressions, an explicit functional error rate, rough shifted values, and two radical criteria. The proofs include a positive active-prime remainder bound, uniform packing of disjoint auxiliary weights, and a direct sifted Vaughan identity. The original Goldbach, small-radical, small-shift and Liouville statements are retained with their stated hypotheses.

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1 Introduction

All logarithms are natural. The Möbius function is denoted by μ, and $\lambda(n) = (-1)^{\Omega(n)}$ is the Liouville function on positive integers. Here Ω counts prime factors with multiplicity. Write $\mathrm{rad}(n) = \prod_{p\mid n} p$ and let $\pi(x)$ count the primes at most x. Put

$$A_{\mathcal{N}} = \prod_{p\nmid\mathcal{N}} \left(1 - \frac{1}{p(p-1)}\right), \qquad A_h = \prod_{p\nmid h} \left(1 - \frac{1}{p(p-1)}\right). \tag{1.1}$$

These products are at least Artin's constant $0.3739\ldots$.

We specify the inputs to the statements. The assertion H2′ is that the arguments of LWW Sections 4.2–4.7, including the proofs of Propositions 4.3–4.5 and Proposition 4.2, and Appendix A, (A.2)–(A.4), are correct as written [12, pp. 17–39, 48–50]. The assertion H1 is

$$L(s,\chi) \ne 0 \quad (\Re s > 7/8) \quad \text{for every Dirichlet character } \chi, \tag{1.2}$$

with the principal pole at $s = 1$ allowed. This is the Dirichlet conclusion of [18, Section 1, Theorem 1.1, p. 4]. For $L = \log\mathcal{N}$ and fixed A_0, write H1$^\star(C_{\mathrm{zf},1}, C_{\mathrm{zf},2}; A_1)$ at $\mathcal{N}$ for the following assertion: for every $k \le L^{A_0}$ and every primitive character χ^* of conductor $q^* \mid k\mathcal{N}$ with $q^* > L^{A_1}$,

$$L(\sigma+it,\chi^*) \ne 0 \quad \left(\sigma \ge 1 - C_{\mathrm{zf},1}\frac{\log L}{L}, \quad |t| \le L^{C_{\mathrm{zf},2}}\right). \tag{1.3}$$

For every fixed choice of these constants, (1.2) implies (1.3) at all sufficiently large $\mathcal{N}$, since $1 - C_{\mathrm{zf},1} \log L/L > 7/8$.

Theorem G (Goldbach shift)**.** *Assume* H2′ *and* H1⋆ *at every sufficiently large even* $\mathcal{N}$*, with the fixed constants chosen in Theorem 6.13 and Section 10. Then, as* $\mathcal{N} \to \infty$ *through even integers,*

$$\sum_{p<\mathcal{N}} \mu(\mathcal{N}-p) = o(\pi(\mathcal{N})), \qquad \#\{p<\mathcal{N} : \mu(\mathcal{N}-p)=\kappa\} = \left(\frac{A_{\mathcal{N}}}{2} + o(1)\right)\pi(\mathcal{N}) \quad (\kappa=\pm 1). \tag{1.4}$$

In particular these conclusions are deduced from H2′ *and* (1.2)*. The errors are uniform in even* $\mathcal{N}$ *and ineffective.*

Theorem S (Small radical)**.** *Assume* H2′*. For each fixed* $C>0$*, the conclusions* (1.4) *hold as* $\mathcal{N}\to\infty$ *through even integers satisfying* $\mathrm{rad}(\mathcal{N}) \le (\log \mathcal{N})^C$*, using Iwaniec's region* (6.70) *and the Burgess bound* (6.38)*. The errors are uniform in this class for fixed* C *and ineffective.*

Theorem H (Small shifts)**.** *Assume the arguments of LWW as written:* H2′ *and, in addition, Sections 2.3, 3, 5 and 6 [12, pp. 6–14, 39–47]. For each fixed* $C>0$*, uniformly in integers* $0<|h|\le(\log x)^C$*,*

$$\sum_{\substack{p\le x\\ p+h\ge 1}} \mu(p+h) = o(\pi(x)), \qquad \#\{p\le x : p+h\ge 1,\ \mu(p+h)=\kappa\} = \left(\frac{A_h}{2}+o(1)\right)\pi(x) \quad (\kappa=\pm1). \tag{1.5}$$

The errors are ineffective.

Theorem 1.1 (Liouville versions)**.** *Under the respective hypotheses of Theorems G, S and H, their conclusions hold with* λ *in place of* μ *and each sign density replaced by* $1/2+o(1)$*. The domain* $p+h\ge 1$ *is retained in the small-shift statement.*

Affine notation and classical inputs. All statements below evaluate the arithmetic functions only on positive integers. For $\varepsilon\in\{-1,1\}$ put

$$\mathcal{P}_{\varepsilon,h}(x) = \{p\le x : \varepsilon p + h \ge 1\}, \qquad \Pi_{\varepsilon,h}(x) = |\mathcal{P}_{\varepsilon,h}(x)|, \qquad E(x) = (\log\log x)^{-1/2}. \tag{1.6}$$

Here and below p, r in products over primes are positive primes, $P^-(1)=\infty$, and the theorem range parameters C, C_q, a are fixed before $x\to\infty$. Band locations vary only within the uniform families of Section 8.2. In addition to the hypotheses specified in each theorem, we use the prime number theorem, Siegel–Walfisz, Bombieri–Vinogradov for primes and for Möbius on reduced classes with endpoint maxima, Montgomery–Vaughan mean values, fixed exponent-pair bounds for logarithmic phases, and the dyadic Möbius estimate of [14, Lemma 3.9(ii)]. Their uses, interval extensions and Liouville consequences are given in the proofs.

Theorem A (Uniform affine shifts)**.** *Assume* H1 *and* H2′*. For each fixed* $C\ge 0$*, uniformly for* $\varepsilon=\pm1$ *and integers* $0<|h|\le x(\log x)^C$*, and for* $f\in\{\mu,\lambda\}$*,*

$$\sum_{p\in\mathcal{P}_{\varepsilon,h}(x)} f(\varepsilon p+h) = O_C(\pi(x)E(x)). \tag{1.7}$$

For either $\kappa=\pm1$ *the Möbius sign main term is* $A_h\Pi_{\varepsilon,h}(x)/2$*, and the Liouville sign main term is* $\Pi_{\varepsilon,h}(x)/2$*, with the same error as in* (1.7)*.*

For the progression statement let $\Pi_{q,a_0;\varepsilon,h}(x)$ count the primes in (1.6) with $p \equiv a_0 \pmod q$. Define

$$\mathfrak{A}(q,a_0;\varepsilon,h) = \prod_{r\nmid qh}\left(1-\frac{1}{r(r-1)}\right)\prod_{\substack{r\mid q\\ r\mid \varepsilon a_0+h}}\left(1-\frac{1}{r}\right) \cdot \prod_{\substack{r^2\mid q\\ r^2\mid \varepsilon a_0+h}} 0. \tag{1.8}$$

The last product sets the value to zero when a square is forced; an empty product is one.

Theorem B (Progressions). *Under the hypotheses of Theorem A, for each fixed $C_q \ge 0$, uniformly for $q \le (\log x)^{C_q}$ and $(a_0,q)=1$,*

$$\sum_{\substack{p\in\mathcal{P}_{\varepsilon,h}(x)\\ p\equiv a_0\ (q)}} f(\varepsilon p+h) = O_{C,C_q}\left(\frac{\pi(x)E(x)}{\varphi(q)}\right). \tag{1.9}$$

The sign main terms are $\mathfrak{A}(q,a_0;\varepsilon,h)\Pi_{q,a_0;\varepsilon,h}(x)/2$ for Möbius and $\Pi_{q,a_0;\varepsilon,h}(x)/2$ for Liouville, with the error in (1.9).

Theorem C (Explicit rate). *In Theorems A, B and D the displayed function $E(x)=(\log\log x)^{-1/2}$ is valid with fixed K, fixed positive gaps between bands in each weight, and fixed analytic precision. The same rate holds for the weaker-input A/D branches specified below. In particular it gives a rate for G and S on their respective classes and for the shift range of H. The constants and thresholds may be ineffective.*

For nonzero even h and $z \ge 2$ define

$$\begin{aligned} V_h(z) &= \prod_{\substack{r\le z\\ r\nmid h}}\left(1-\frac{1}{r-1}\right),\\ U_h(z) &= V_h(z)\prod_{\substack{r>z\\ r\nmid h}}\left(1-\frac{1}{r(r-1)}\right). \end{aligned} \tag{1.10}$$

Theorem D (Rough affine values). *Assume* H1 *and* H2′. *For fixed $C \ge 0$ and $0<a<1/2$, uniformly for nonzero even h in the A range, $\varepsilon=\pm1$ and $2\le z\le \exp((\log x)^a)$,*

$$\sum_{p\in\mathcal{P}_{\varepsilon,h}(x)} f(\varepsilon p+h)\mathbf{1}_{P^-(\varepsilon p+h)>z} = O_{a,C}(V_h(z)\pi(x)E(x)), \qquad f=\mu,\lambda. \tag{1.11}$$

For each $\kappa=\pm1$, the corresponding sign main terms are $U_h(z)\Pi_{\varepsilon,h}(x)/2$ and $V_h(z)\Pi_{\varepsilon,h}(x)/2$, respectively, with the error in (1.11).

The A and D errors are on the displayed ambient scales. They are not relative errors in an arbitrarily short eligible interval. For example $\varepsilon=1$, $h=-x$ and integral x give an empty set in (1.6). At fixed z the tail product in $U_h(z)$ is retained. Uniformly in h, $0 \le 1-U_h(z)/V_h(z) \ll 1/z$.

Weaker-input branches. For each fixed $C_0 > 0$, A and D on $0 < |h| \le (\log x)^{C_0}$ require H2′ and the listed classical inputs alone. For $\varepsilon = -1$, $x = h = \mathcal{N}$ even, A and D instead require H2′ and the sufficiently strong fixed instance of H1⋆ selected by the reflected energy proof, together with Burgess. On $\mathrm{rad}(\mathcal{N}) \le (\log \mathcal{N})^{C_0}$, Iwaniec and Burgess supply this branch without H1. The D branch always retains even h and fixed $a < 1/2$. These assertions concern A and D; no progression refinement of these weaker-input branches is asserted.

Theorem F (Radical criteria)**.** *Fix $A_0, A_1, C_1, C_2 > 0$, interpreting C_1, C_2 as the constants in (1.3), and write $L = \log \mathcal{N}$.*

1. *Suppose* LS(c)*: for some fixed $c > 0, q_0$, every primitive real character of conductor $q \ge q_0$ and every real zero $\beta \in (0,1)$ satisfy $1-\beta \ge c/\log q$. For each prescribed nonnegative function $\eta(L) \to 0$, Iwaniec's region and* LS(c) *imply the selected* H1⋆ *uniformly on*

$$\log \mathrm{rad}(\mathcal{N}) \le \eta(L) L/\log L. \tag{1.12}$$

   The threshold may depend on η. In particular one may take $\mathrm{rad}(\mathcal{N}) \le \exp(L/(\log L)^2)$.

2. *Let $c_{\mathrm{LP}} > 0$ be a Landau–Page constant, choose*

$$0 < \kappa \le \frac{\min(c_{\mathrm{LP}}, 1/(4\cdot 10^4))}{32 C_1}, \quad \ell = \log X, \quad R_X = \exp(\kappa \ell/\log \ell), \quad Q_X = 8(2\ell)^{A_0} R_X. \tag{1.13}$$

   Let q_X be the conductor of the possible Landau–Page character at Q_X. The selected H1⋆ *holds for all sufficiently large even $X \le \mathcal{N} \le 2X$ with $\mathrm{rad}(\mathcal{N}) \le R_X$ if the exception is absent, if $q_X \le \ell^{A_1}$, or if*

$$q_X/\gcd(q_X, \mathcal{N}) > (\log \mathcal{N})^{A_0}. \tag{1.14}$$

   This analytic criterion uses Iwaniec and Landau–Page.

After the precision required by the reflected energy proof is fixed, these criteria give reflected A and D under H2′*, Burgess and the listed classical inputs, with* LS(c) *needed only in the first criterion. Choose κ after that precision is fixed. The analytic criteria themselves do not use* H2′*.*

Under H1 and H2′, A contains G and the shift range of H; C quantifies both. The separate G, S, H and Liouville statements above keep their original hypotheses. A correlation at two distinct nonzero shifted arguments remains an open problem for the present method.

Related averaged and fixed-shift statements. Lichtman [10, Theorem 1.1] states cancellation averaged over positive shifts $h \le H < X$ when $\log H/\log\log X \to \infty$. Lichtman and Teräväinen [11, Corollary 1.4] state an almost-all-shifts result for $(\log X)^{1+\epsilon} \le H \le X$. Carella [2, Theorem 1.1] states, for a small fixed nonzero shift a, a bound $O_{c,a}(x(\log x)^{-c})$ for every fixed $c > 1$; its Theorem 1.2 gives a two-shift statement under its Hypothesis 3.1. Tao and Teräväinen [19, Theorem 1.6] state hybrid correlations at distinct fixed natural-number shifts, with at most two von Mangoldt factors and a fixed number of Liouville factors, on scales governed by the conductor and quality of a hypothetical Siegel zero. The real-zero bounds stated by Banks [1] and Kerr [9] motivate the separately named assumption LS(c); here that assumption is used in exactly its displayed form. For integer summands, Mangerel [13] states $|\sum_{1\le n<N} \lambda(n)\lambda(N-n)| < N-1$ for all sufficiently large N. The rough-summand statement in [5] gives prescribed Liouville signs for two rough integer summands. The prime-factor law in [17] states convergence of normalized logarithms of the prime factors of $p-1$ to the Poisson–Dirichlet law with parameter one. For ordinary short intervals, Matomäki, Pandey, Pliego, Teräväinen and Wang [15] state Möbius cancellation at exponent $19/35+\epsilon$ and Gowers uniformity of the von Mangoldt function minus its Cramér model at exponent $3/5+\epsilon$.

Background. In their preprint submitted on 14 October 2026, Lü, Wang and Wu [12] state in Theorem 1 that $\sum_{p\le x}\mu(p+2)=o(\pi(x))$, and in Theorem 2 give the sign densities $A_2/2$ among all primes. Their abstract describes the conjecture as "dating back at least to Hildebrand's work in 1989"; Section 1 attributes its appearance in print to Hildebrand's 1989 paper [12, p. 3]. Their Section 4.4 describes the construction of its auxiliary operators as similar to Sections 4.1–4.3 of OpenAI's *The Poisson–Dirichlet law for prime predecessors*, dated 24 September 2026 [17]; their Appendix A says that its root-counting proof follows the argument of that paper's Lemma 3.3, with an explicit error for arbitrary squarefree S [12, pp. 26, 47]. The author's related papers, *Shifted primes with a prescribed parity of the number of prime factors* [6] and *Sums of two rough numbers with prescribed Liouville signs* [5], are dated 9 October 2026 in their repository entries.

Proof inputs and changes to the argument. The scope of H2′ is exactly the one stated above. The sifted Vaughan identity, the small-conductor interval estimate, Type I, finite moment transfer and positive population counts are proved here from their named classical inputs. Their versions in LWW Propositions 2.1–2.3, 3.1 and 6.1 provide reference points [12, pp. 4–14, 46–47]. The signed operator, memory and collision arguments use H2′ after the changes listed in Appendix A. The original statement H retains its additional assumptions on LWW Sections 2.3, 3, 5 and 6; Section 11 also gives the stronger small-shift branch using the reconstructed estimates.

The reflected energy theorem uses Perron, Montgomery–Vaughan, Pólya–Vinogradov, Burgess at $r=2$, and the small-conductor Möbius input. Iwaniec's exact 3/4-power region is stated in Input 6.15. The general affine major arcs instead use a hybrid character/time mean value and the Mellin scale $\max(1,X/|h|)$. Finite checks of selected algebraic models and identities are described in Appendix B.

Parameters and outline. All precision exponents are fixed before the large-parameter limit. The coefficient-loss constants and the selected zero-free constants are chosen separately. The quantitative removal of weights fixes K and each within-weight band gap, and packs $b_x\asymp\log\log x$ disjoint copies with a common threshold. Sections 2–5 give the shared determinant reduction, signed root count and Type II transport. Section 6 gives the reflected and general analytic interfaces and the radical criteria. Sections 7– 10 provide Type I, relative moments, densities and assembly. The small-shift and Liouville sections record their specific changes. Section 13 supplies the additional character and relative-measure arguments for B.

2 Setting and determinant reduction

The reflected subsections retain the version 1 notation: $\mathcal{N}$ is the global bound and x is a prime-dyad scale. Subsections headed "Changes for general h" use global bound x and dyad scale y. In each case L is the logarithm of the global bound, and the dyadic logarithm differs by $O_B(\log L)$. The normalized bases and later estimates are shared through the explicit specialization $\Delta=H=\mathcal{N}$.

Definition 2.1 (Parameters and coefficients)**.** For the Goldbach shift, $\mathcal{N}$ is even and $L=\log\mathcal{N}$. Fix $B>2$, let $p\in(x,2x]$, and put

$$\begin{aligned}&\frac{\mathcal{N}}{2L^B}\le x\le\frac{\mathcal{N}}{2},\quad Z=\exp(L^{3/4}),\quad U=V=x^{1/5},\\&X=MN\asymp x,\quad M,N\ge x^{1/5}/2,\quad MN\le 2x\le\mathcal{N}.\end{aligned}\tag{2.1}$$

The letter N always denotes the length of the Vaughan variable n. Write $R_Z(n)$ for the indicator that every prime divisor of n exceeds Z. Fix $0<\rho<1$ and

$$\frac{1}{10}<a_1<\cdots<a_K<\theta<\frac{1}{2},\quad \mathcal{P}_i=\{p:\exp(L^{a_i})<p\le\exp(2L^{a_i}),\ p\nmid\mathcal{N}\},\quad V_i=\sum_{p\in\mathcal{P}_i}\frac{1}{p}. \tag{2.2}$$

Put $\mathcal{A}=\{p_1\cdots p_K:p_i\in\mathcal{P}_i\}$, $\omega_i(n)=\#\{p\in\mathcal{P}_i:p\mid n\}$ and $\omega(n)=\sum_i\omega_i(n)$. The weight of LWW (2.13), p. 7, is

$$W(n)=\prod_{i=1}^{K}\frac{\rho^{\omega_i(n)-1}\omega_i(n)}{V_i}. \tag{2.3}$$

With j_*,r_* fixed in Section 5.3, let $J=\lfloor L^{j_*}\rfloor$, $R=\lceil L^{r_*}/2\rceil$, and let d_0,D have the meanings of LWW Section 4.2, pp. 17–21. The real smooth dyadic cutoff η is supported on $[1,4]$ (LWW Section 4.1, p. 15), so its product scale Y has $a,b\in[Y,4Y]\subset[Y,16Y]$. Define

$$\Omega=[d_0YM,16d_0YM]\times[N,2N]. \tag{2.4}$$

For $|v|\le L^{2B}$, the coefficients are

$$\begin{aligned}\alpha_m&=\mu(m)R_Z(m)\mathbf{1}_{(m,\mathcal{N})=1}m^{iv},\\ \beta_n&=c_V(n)R_Z(n)\mathbf{1}_{(n,\mathcal{N})=1}n^{iv},\qquad c_V(n)=\sum_{\substack{\ell\mid n\\ \ell>V}}\Lambda(\ell).\end{aligned} \tag{2.5}$$

Thus $|\alpha_m|\le1$, $|\beta_n|\le L$, and the untwisted coefficient $c_V(n)R_Z(n)\mathbf{1}_{(n,\mathcal{N})=1}$ is nonnegative. For a character ξ of modulus dividing $k\mathcal{N}$, put

$$\begin{aligned}B_\xi(t)&=\frac{1}{Y\prod_i V_i}\sum_{b\in\mathcal{A}}\eta(b/Y)e(-\lambda_0 b/Y)\xi(b)b^{it},\\ M_\xi(t)&=\frac{1}{M}\sum_{M<m\le2M}\alpha_m\xi(m)m^{it},\qquad N_\xi(t)=\frac{1}{N}\sum_{N<n\le2N}\beta_n\xi(n)n^{it}.\end{aligned} \tag{2.6}$$

Here $e(t)=\exp(2\pi it)$, $k\le L^{A_0}$, $|\lambda_0|\le2L^{A_0}$, $T_0=\lceil L^{B_1}\rceil$, and $|B_\xi(t)|\le C_B=16\|\eta\|_\infty/\prod_i V_i$. We write the arc displacement as λ_0 to distinguish it from the Liouville function. The major-arc target, at a fixed requested precision A, is

$$\frac{1}{\varphi(\mathcal{N})}\sum_{\chi\bmod\mathcal{N}}\int_{|t|\le T_0}|B_{\chi\psi_2}(t)M_{\chi\psi}(t)N_{\chi\psi}(t)|^2\,dt\ll_A\frac{L^{-A}}{\mathcal{N}}. \tag{α^*}$$

The fixed twists ψ,ψ_2 are those constructed in Section 6.1. The trivial bound is $L^{O(1)}/\mathcal{N}$; $1/\mathcal{N}$ is the normalization.

Remark 2.2 (Ranges). Primes $p\le\mathcal{N}/L^B$ contribute $O(\mathcal{N}/L^B)$ to the bounded target sum, and $O(\mathcal{N}L^{1-B})$ with $\log p$. The rest splits into $O(\log L)$ dyadic intervals in (2.1). Before separating the product cutoff, $\mathcal{N}-mn\in[\mathcal{N}-2x,\mathcal{N}-x)\cap(0,\mathcal{N})$. For fixed $a\asymp Y$ the positive cofactor belongs to

$$\left[\frac{\mathcal{N}-2x}{a},\frac{\mathcal{N}-x}{a}\right], \tag{2.8}$$

an interval of length $x/a\asymp X/Y$. As a varies, its centre varies, and the common positive support has $O(\mathcal{N}/Y)$ integers. Terms with $0<\mathcal{N}-mn<Y$ vanish, since W requires a factor from every band. No dyadic condition on $\mathcal{N}-p$ is needed.

Proposition 2.3 (Sifted Vaughan reduction)**.** *Let $y \in [x/(\log x)^B, x]$, $U = V = y^{1/5}$, and $Z = \exp((\log x)^\gamma)$ for fixed $\gamma < 1$. For any bounded function F on the positive integers set*

$$T_1(F) = \sum_{d \le U} \mu(d) R_Z(d) \sum_{y < dk \le 2y} R_Z(k)(\log k) F(dk), \tag{2.9}$$

$$T_2(F) = \sum_{\substack{d \le U \\ \ell \le V}} \mu(d)\Lambda(\ell) R_Z(d\ell) \sum_{y < d\ell k \le 2y} R_Z(k) F(d\ell k), \tag{2.10}$$

$$T_3(F) = \sum_{\substack{m > U, n > V \\ y < mn \le 2y}} \mu(m) R_Z(m) c_V(n) R_Z(n) F(mn), \quad c_V(n) = \sum_{\substack{\ell \mid n \\ \ell > V}} \Lambda(\ell). \tag{2.11}$$

Then $0 \le c_V(n) \le \log n$ and

$$\sum_{y < p \le 2y} (\log p) F(p) = T_1(F) - T_2(F) + T_3(F) + O\big(\|F\|_\infty [Z + \sqrt{y}(\log(2y))^2]\big). \tag{2.12}$$

The same holds for a partial dyad by including its endpoint in F.

Proof. For Dirichlet convolution $*$, $\log = 1 * \Lambda$ and $\mu * 1 = \delta_1$. Splitting at U, V gives the exact identity

$$\Lambda = \Lambda_{\le V} + \mu_{\le U} * \log - \mu_{\le U} * \Lambda_{\le V} * 1 + \mu_{>U} * \Lambda_{>V} * 1. \tag{2.13}$$

The three convolution terms combine to $\mu * 1 * \Lambda_{>V} = \Lambda_{>V}$. On the dyad the first term vanishes. Multiplication by $R_Z(n)F(n)$ and complete multiplicativity of R_Z yield $T_1 - T_2 + T_3$. Deleting the sifting indicator from the von Mangoldt sum costs $O(\|F\|_\infty Z)$: in $(y, 2y]$ a fixed prime has at most one power, and $\sum_{p \le Z} \log p \ll Z$. Deleting all higher prime powers, including bases above Z, costs at most $O(\|F\|_\infty \sqrt{y}(\log(2y))^2)$.

In T_2 the outer product $d\ell \le y^{2/5}$ has coefficient bounded by $\log(d\ell)$. Both variables in T_3 exceed $y^{1/5}$. Dyadic localization and smoothing therefore produce

$$T = \sum_{m,n} \alpha_m \beta_n F(mn), \qquad M, N \ge y^{1/5}/2, \quad MN \asymp y, \quad |\alpha_m| \le 1, \quad |\beta_n| \ll \log x. \tag{2.14}$$

For a partial dyad $(y, t]$, write its bounded factor as $F_0(n)\mathbf{1}_{n \le t}$ and expose the full product cutoff $\mathbf{1}_{y < mn \le t}$ before the rectangular reduction; for a full dyad take $t = 2y$. Smooth this product cutoff as a function of mn/y at relative width $\delta = (\log x)^{-B_s}$. The positivity endpoint of the shifted target remains in F_0. On each rectangle choose the shorter variable as the counting variable. An endpoint strip contains $O(\sum_{m \asymp M}(\delta y/m + 1)) = O(\delta y + \sqrt{y})$ pairs, with fixed logarithmic coefficient losses. Mellin inversion of the smooth cutoff has a fixed power of δ^{-1} as its L^1 norm; truncate at $|v| \le \delta^{-2}$, raising the fixed integration-by-parts order for any desired logarithmic precision. The coefficients acquire only m^{iv}, n^{iv}. The shifted target, its positivity endpoint and its roughness multiplier are kept intact and are never differentiated. The extra factor $\log k$ in T_1 is treated by real-weight partial summation after its endpoint-uniform estimate. □

Remark 2.4 (Changes in version 2: prime decomposition). The displayed identity replaces the version 1 invocation of LWW Proposition 2.1, which lies outside H2′. The two prime-power corrections and the boundary-strip count apply to both orientations, character twists and the rough target. For the reflected specialization take $x = \mathcal{N}$ and use y for the dyad variable denoted by x in (2.1). The fixed Mellin cutoff exponent in (2.5) can be increased to the required $2B_s$ before choosing the energy height; this only changes the fixed constants denoted there by B.

Lemma 2.5 (Puncturing and unit support)**.** *The punctured bands satisfy $V_i = \log 2 + o(1)$, uniformly in $\mathcal{N}$. The restrictions $(m,\mathcal{N}) = (n,\mathcal{N}) = 1$ in (2.5) cost $O_A(XL^{-A})$ for every fixed A. The remaining variables are units modulo $k\mathcal{N}$ whenever $k \le L^{A_0}$.*

Proof. The removed reciprocal mass is at most

$$\sum_{\substack{p\mid\mathcal{N}\\ p>\exp(L^{0.1})}} \frac{1}{p} \le \omega_{\mathrm{all}}(\mathcal{N})e^{-L^{0.1}} \le Le^{-L^{0.1}}, \tag{2.15}$$

where ω_{all} counts all distinct prime divisors. Thus LWW (2.14)–(2.15), p. 7, apply with the same limits. A common prime of a sifted variable and $\mathcal{N}$ exceeds Z. The number of discarded pairs is bounded, up to an absolute constant, by

$$\sum_{\substack{p\mid\mathcal{N}\\ p>Z}} \frac{X}{p} \le \frac{X\omega_{\mathrm{all}}(\mathcal{N})}{Z} \le XLe^{-L^{3/4}}. \tag{2.16}$$

The bound $|\alpha_m\beta_n| \ll L$ absorbs this into the claimed error. The restrictions preserve the coefficient and sifting hypotheses of Proposition 4.2. Every prime factor of k is below Z, so the sifted variables are also units modulo k. Every list prime and every $a, b \in \mathcal{A}$ is a unit modulo $\mathcal{N}$ by puncturing, and modulo k because the bands lie above L^{A_0}. □

Proposition 2.6 (The cofactor square)**.** *Specialize the bounded function in Proposition 2.3 to*

$$F(n) = \begin{cases} \mu(\mathcal{N}-n)W(\mathcal{N}-n), & 1 \le n < \mathcal{N},\\ 0, & \text{otherwise.}\end{cases}$$

After a smooth dyadic partition $\eta(a/Y)$, the resulting (2.14) has the form

$$T = \frac{(-1)^K}{\prod_i V_i} \sum_{\ell_0\ge 1} \mu(\ell_0)\rho^{\omega(\ell_0)} \sum_{\substack{p_i\in\mathcal{P}_i,m,n\\ mn+a\ell_0=\mathcal{N}}} \alpha_m\beta_n + O(XLe^{-L^{0.1}}). \tag{2.17}$$

Its dyadic pieces satisfy

$$|T_Y|^2 \ll \frac{\mathcal{N}}{Y}Q_Y^+ \le \frac{\mathcal{N}}{Y}Q_Y^{\mathrm{all}}, \qquad Q_Y^{\mathrm{all}} = \sum_{\substack{\ell_0\in\mathbb{Z}\\ \ell_0\ne 0}} \left| \frac{1}{\prod_i V_i} \sum_{\substack{a\in\mathcal{A},m,n\\ \mathcal{N}-mn=a\ell_0}} \eta(a/Y)\alpha_m\beta_n \right|^2 \ge 0. \tag{2.18}$$

Here Q_Y^+ is the same square restricted to $\ell_0 \ge 1$. All cofactors in Q_Y^{all} satisfy $|\ell_0| \ll \mathcal{N}/Y$.

Proof. If $\mathcal{N} - mn = a\ell_0$ and $(a,\ell_0) = 1$, then $\mu(a\ell_0) = (-1)^K\mu(\ell_0)$ and $\omega(a\ell_0) = K + \omega(\ell_0)$. Also $(a,m) = 1$: a common list prime would divide $\mathcal{N}$. When $(a,\ell_0) > 1$, some list prime has square dividing $\mathcal{N} - mn$. For fixed m this is one residue class of n modulo that square, and the count in LWW (4.3), p. 15, gives $O(XLe^{-L^{0.1}})$. The coprimality restriction can be removed in the expansion at the same cost; its new terms have that square-divisibility property. Cauchy–Schwarz uses $|\mu(\ell_0)\rho^{\omega(\ell_0)}| \le 1$ and the common range $O(\mathcal{N}/Y)$ from Remark 2.2. Each square is nonnegative, proving $Q_Y^+ \le Q_Y^{\mathrm{all}}$. On a supported rectangle $mn \ll MN \le \mathcal{N}$, whence $|\mathcal{N} - mn| \ll \mathcal{N}$. The cofactor 0 is impossible because $mn = \mathcal{N}$ contradicts $(mn,\mathcal{N}) = 1$ for $\mathcal{N} > 1$. Only this upper-bound quadratic form introduces negative cofactors; F and the expansion of its target always use positive arguments. □

Lemma 2.7 (Disjoint lists and determinant equation)**.** *Terms with $(a,b)>1$ may be removed from Q_Y^{all} at a cost $O_A(\mathcal{N}YL^{-A})$. For $(a,b)=1$ the exact determinant equation is*

$$b(\mathcal{N}-mn)=a(\mathcal{N}-rs) \quad\Longleftrightarrow\quad t:=bmn-ars=\mathcal{N}(b-a). \tag{2.19}$$

Equality on the left is equality of the two integer cofactors, with no positivity condition. In particular $\mathcal{N}\mid t$ and $|t|\ll\mathcal{N}Y$, while $t'=t/\mathcal{N}=b-a$ has size $O(Y)$.

Proof. The elementary divisor-square estimate proved in (5.14), specialized to $\Delta=\mathcal{N}$, has the form

$$\sum_{\ell_0\in I_\ell}\tau(\mathcal{N}-\ell\ell_0)^2\ll(X/Y)L^3,\qquad |I_\ell|\ll X/Y \tag{2.20}$$

where I_ℓ is the translated cofactor interval for fixed $\ell\asymp Y$, with $\mathcal{N}-\ell\ell_0$ in the positive dyadic product range. The argument depends on $d\mid\mathcal{N}-\ell\ell_0\Rightarrow(d,\ell)\mid\mathcal{N}$; since $(\ell,\mathcal{N})=1$, it gives $(d,\ell)=1$, exactly the needed hypothesis. For the all-integer square use the full interval $|\ell_0|\ll\mathcal{N}/Y$ and $|\mathcal{N}-\ell\ell_0|$ in its absolute counts. Its length costs at most $\mathcal{N}/X\ll L^B$ relative to the displayed local estimate. The absolute estimates in Section 5.1 and the common-list saving then give the stated $\mathcal{N}YL^{-A}$ after increasing the fixed precision. Reducing (2.19) modulo a and b, and using $(a,b)=1$, proves divisibility by a,b of the corresponding numerators and equality of their integer quotients. □

2.1 Changes for general h: the common square

For the general statements x denotes the global prime bound, $L=\log x$, and $y\ge x/L^B$ is a prime-dyad scale. In (2.14) put $X=MN\asymp y$ and

$$\Delta=-\varepsilon h,\qquad H=|h|,\qquad W_0=X+H,\qquad H/X\ll L^{B+C},\qquad H/(d_0YX)=o(1). \tag{2.21}$$

Use the bands and weight (2.3), punctured at $p\mid h$. The common retuned parameters are specified in Section 5.3; their band ceiling is $\theta<1/2$. The puncturing and unit-support proof above applies with $\omega_{\mathrm{all}}(H)=O_C(L)$ and $Z=\exp(L^{3/4})$. In particular the coefficient sequences remain sifted and bounded.

Proposition 2.8 (Affine cofactor square)**.** *In (2.14) take $F(n)=f(\varepsilon n+h)W(\varepsilon n+h)$ when $\varepsilon n+h\ge1$ and $F(n)=0$ otherwise, for $f=\mu$ or $f=\lambda$. On unit Vaughan support set $w=\varepsilon mn+h=\varepsilon(mn-\Delta)$ and require $w\ge1$. The selected-prime expansion gives a sum over $w=a\ell$, $\ell\ge1$, after the square-overlap error $O_{K,\rho}(XL\exp(-cL^{0.1}))$. After a smooth dyadic partition in a,*

$$|T_Y|^2\ll(W_0/Y)\,Q_Y, \tag{2.22}$$

where the nonnegative enlarged square is

$$Q_Y=\sum_{j\in\mathbb{Z}\setminus\{0\}}\left|\frac{1}{\prod_i V_i}\sum_{mn-\Delta=aj}\eta(a/Y)\alpha_m\beta_n\right|^2. \tag{2.23}$$

For coprime a,b, equality of the integer cofactors is equivalent to

$$bmn-ars=\Delta(b-a). \tag{2.24}$$

For D, define $P_h(z)=\prod_{r\le z,\ r\nmid h}r$ and $\mathcal{R}_{z,h}(n)=1_{(n,P_h(z))=1}$. If every selected list prime exceeds z, then

$$\mathcal{R}_{z,h}(a\ell) = \mathcal{R}_{z,h}(\ell). \tag{2.25}$$

Multiplying the target by $\mathcal{R}_{z,h}$ leaves exactly the same upper-bound square (2.23).

Proof. Puncturing at $r \mid h$ changes a band's reciprocal mass by at most $O(L\exp(-L^{0.1}))$. Sifted variables sharing a prime with h share one exceeding Z; counting their multiples costs $O(XL^2/Z)$, at every fixed logarithmic precision. Each list prime is coprime to m. Hence its square dividing w selects one residue class of n, and $N \gg r^2$ allows the bound $O(N/r^2)$. The bounded-weight square-overlap calculation in the reflected proof applies, including its separate Liouville expansion; it uses no sign of the affine coefficient. Once overlaps are removed, the common factor is $f(\ell)\rho^{\omega(\ell)}$ of modulus at most one. The positive cofactors lie in $[1, O(W_0/Y)]$. Cauchy gives the square over cofactors of the permitted sign, and adding the other sign adds nonnegative terms, producing (2.23). A zero cofactor would mean $mn = \Delta$; for $H > 1$ this contradicts unit support, and for $H = 1$ the dyadic product range excludes it. The equation (2.24), reduced modulo a, b, forces both divisibilities because $(a, b) = 1$; it then gives equality of integer quotients. Finally $(a, P_h(z)) = 1$ proves (2.25), so the extra roughness indicator stays in the common bounded cofactor factor. □

For actual primes $p \nmid h$, the shifted value is coprime to h. Thus $\mathcal{R}_{z,h}(\varepsilon p + h)$ equals the full roughness indicator. The omitted primes dividing h contribute $O(L)$ without logarithmic weight and $O(L^2)$ with it.

3 Blocks and relative coordinates

Endpoint positions, with the list data of LWW attached, are

$$P = (Dbm, s)^T, \qquad Q = (Dar, n)^T, \qquad \det(P, Q) = D(bmn - ars) = Dt. \tag{3.1}$$

All endpoint coordinates, and D, a, b, are units modulo $\mathcal{N}$.

3.1 The major-arc congruence

Let $\mathfrak{M}_1$ be LWW's union of arcs with denominators $k \le L^{A_0}$ and width $2L^{A_0}/Y$ [12, Section 4.1, p. 16]. Define

$$\begin{aligned} \mathfrak{M}_{\mathcal{N}} &= \{\theta \in \mathbb{T} : \mathcal{N}\theta \in \mathfrak{M}_1\}, \\ H_{\mathcal{N}}(t; a, b) &= \psi_1(t/(\mathcal{N}Y)) \int_{\mathfrak{M}_{\mathcal{N}}} e(\theta(t - \mathcal{N}(b - a)))\, d\theta, \\ Q_Y^{\mathrm{maj},\mathcal{N}} &= \prod_i V_i^{-2} \sum_{a,b,m,n,r,s} \eta(a/Y)\eta(b/Y)\alpha_m\overline{\alpha_r}\beta_n\overline{\beta_s} H_{\mathcal{N}}(bmn - ars; a, b). \end{aligned} \tag{3.2}$$

The cutoff ψ_1 is the fixed smooth cutoff of LWW, equal to 1 on $[-4, 4]$.

Lemma 3.1 (Covering map)**.** *For every integer t,*

$$\int_{\mathfrak{M}_{\mathcal{N}}} e(\theta(t - \mathcal{N}(b - a)))\, d\theta = \mathbf{1}_{\mathcal{N}\mid t} \int_{\mathfrak{M}_1} e(\phi(t/\mathcal{N} - b + a))\, d\phi. \tag{3.3}$$

In particular $|\mathfrak{M}_{\mathcal{N}}| = |\mathfrak{M}_1| \ll L^{3A_0}/Y$.

Proof. For a fundamental domain of $\mathfrak{M}_1$, the change of variables gives

$$\int_{\mathfrak{M}_{\mathcal{N}}} F(\theta)\,d\theta = \frac{1}{\mathcal{N}} \sum_{j \bmod \mathcal{N}} \int_{\mathfrak{M}_1} F((\phi+j)/\mathcal{N})\,d\phi.$$

For the integrand in (3.3),

$$F((\phi+j)/\mathcal{N}) = e(\phi t/\mathcal{N})e(jt/\mathcal{N})e(-\phi(b-a))e(-j(b-a)).$$

The last factor is 1 and the j-sum is $\mathcal{N}\mathbf{1}_{\mathcal{N}|t}$. This also makes the identity independent of the chosen representatives. □

The raw-minus-major kernel is

$$\begin{aligned} K_{\mathcal{N}}(P,Q) = \frac{d_0}{D}\eta(a/Y)\eta(b/Y)\psi_1(t/(\mathcal{N}Y)) \\ \cdot \left\{ \mathbf{1}_{t=\mathcal{N}(b-a)} - \int_{\mathfrak{M}_{\mathcal{N}}} e(\theta(t-\mathcal{N}(b-a)))\,d\theta \right\} \rho^{(\omega(P_1)+\omega(Q_1)-2K(J+1))/2}. \end{aligned} \tag{3.4}$$

Both terms vanish unless $\mathcal{N} \mid t$. On that congruence the brace is

$$\mathbf{1}_{t'=b-a} - \int_{\mathfrak{M}_1} e(\phi(t'-b+a))\,d\phi, \qquad t' = t/\mathcal{N}, \tag{3.5}$$

with cutoff $\psi_1(t'/Y)$, the transformed kernel of LWW Section 4.2, p. 18.

3.2 Lines and their sizes

Definition 3.2 (Unit endpoints and labelled blocks)**.** Let $\mathcal{C}_{\mathcal{N}}$ consist of $P \in \Omega$ primitive in $\mathbb{Z}^2$ with $(P_1P_2, \mathcal{N}) = 1$. For a unit $\varrho \bmod \mathcal{N}$, put

$$\Lambda_\varrho = \{(u,v) \in \mathbb{Z}^2 : v \equiv \varrho u \pmod{\mathcal{N}}\}. \tag{3.6}$$

This has basis $(1,\varrho), (0,\mathcal{N})$ and index $\mathcal{N}$. The slope class of a unit-coordinate endpoint is $\varrho(P) = P_2\overline{P_1} \bmod \mathcal{N}$.

Lemma 3.3 (Block diagonality)**.** *For unit-coordinate endpoints,*

$$\mathcal{N} \mid \det(P,Q) \iff \varrho(P) = \varrho(Q), \qquad \mathcal{C}_{\mathcal{N}} = \coprod_{\varrho \in (\mathbb{Z}/\mathcal{N}\mathbb{Z})^\times} (\mathcal{C}_{\mathcal{N}} \cap \Lambda_\varrho). \tag{3.7}$$

Thus the endpoint kernel is block diagonal. Each full lattice is regarded as a labelled copy when the state space is enlarged.

Proof. Invert P_1Q_1 in the determinant congruence. Since $(D,\mathcal{N}) = 1$, $\mathcal{N} \mid t$ is equivalent to $\mathcal{N} \mid \det(P,Q)$. The unit restriction makes the slope class unique. □

Lemma 3.4 (Block size)**.** *The number of vectors in $\Omega \cap \Lambda_\varrho$ primitive in Λ_ϱ is*

$$O\left(1 + \frac{d_0 YMN}{\mathcal{N}}\right). \tag{3.8}$$

In the application $d_0YMN/\mathcal{N} \to \infty$ and $MN \le \mathcal{N}$, so this is $O(d_0YMN/\mathcal{N}) \le O(d_0Y)$, uniformly in ϱ.

Proof. If the block is occupied, choose a lattice-primitive P_0 in it. For any Q in the block,

$$\det(P_0, Q) \in \mathcal{N}\mathbb{Z}, \qquad |\det(P_0, Q)| \le 2 \cdot 16 d_0 Y M \cdot 2N = 64 d_0 Y M N \le 64 d_0 Y \mathcal{N}.$$

There are $O(1 + d_0 Y M N/\mathcal{N})$ possible determinants. For each, two solutions differ by an element of $\mathbb{Q}P_0 \cap \Lambda_\varrho = \mathbb{Z}P_0$; their first coordinates permit at most 16 solutions. Here $\mathbb{Q}$ denotes the rational numbers. In the application $\mathcal{N}/(d_0 Y M N) \ll L^B/(d_0 Y) \to 0$. □

This partition supplies pieces of the size of LWW's slope intervals [12, Section 4.6, p. 36]. Physical slope locality is replaced by relative-coordinate bounds.

3.3 A normalized basis

Let $P = (u, v)^T \in \mathcal{C}_\mathcal{N}$, and choose $ud - vc = 1$, $0 \le c < u$. Define

$$\begin{aligned} &c' \equiv \mathcal{N} c \pmod{u}, \quad 0 \le c' < u, \qquad c'v \equiv -\mathcal{N} \pmod{u}, \quad \beta_P = c'/u, \\ &d'' = (\mathcal{N} + c'v)/u, \qquad P^* = (c', d'')^T, \qquad g_\mathcal{N}(P) = \begin{pmatrix} u & c' \\ v & d'' \end{pmatrix}, \quad \det g_\mathcal{N}(P) = \mathcal{N}. \end{aligned} \tag{3.9}$$

Equivalently

$$P^* = \mathcal{N}(c, d)^T - \lfloor \mathcal{N} c/u \rfloor P, \qquad g_\mathcal{N}(P) = g E_\mathcal{N} U_0, \quad g \in \mathrm{SL}_2(\mathbb{Z}), \quad E_\mathcal{N} = \mathrm{diag}(1, \mathcal{N}), \tag{3.10}$$

where U_0 is the corresponding unimodular shear.

Lemma 3.5 (Relative coordinates)**.** *Writing $\varrho = \varrho(P)$ and $Q = g_\mathcal{N}(P)z$, one has:*

1. $\Lambda_\varrho = \mathbb{Z}P \oplus \mathbb{Z}P^*$.
2. *With $\tau_\mathcal{N}(z) = z_1 + \beta_P z_2$,*

$$\det(P, Q) = \mathcal{N} z_2, \quad Q_1 = u\tau_\mathcal{N}(z), \quad Q_2 = v\tau_\mathcal{N}(z) + \frac{\mathcal{N} z_2}{u}. \tag{3.11}$$

3. *Q is primitive in Λ_ϱ precisely when $(z_1, z_2) = 1$. It is primitive in $\mathbb{Z}^2$ precisely when also $(Q_1, \mathcal{N}) = 1$.*
4. *For primitive z and every prime $p \nmid \mathcal{N}$,*

$$p \mid Q_1 \quad \Longleftrightarrow \quad [z]_p = \ker(u, c') \in \mathbb{P}^1(\mathbb{F}_p). \tag{3.12}$$

5. *For $Q \in \Omega$,*

$$|z_2| \le \frac{64 d_0 Y M N}{\mathcal{N}} \le 64 d_0 Y, \qquad |z_1| \le 16 + |z_2| \le 2^7 d_0 Y. \tag{3.13}$$

Conversely every integer z gives $Q \in \Lambda_\varrho$.

Proof. The two columns belong to Λ_ϱ and have its determinant. Formula (3.11) follows from $d'' = \mathcal{N}/u + \beta_P v$. The lattice bijection proves lattice-primitivity. At $p \mid \mathcal{N}$ the matrix has rank one, with unit-coordinate column P, and $Q \equiv 0 \pmod{p}$ precisely when $p \mid Q_1$. At $p \nmid \mathcal{N}$ the matrix is invertible, so $Q \equiv 0$ precisely when $z \equiv 0$. Also $(u, c') = 1$, since $c'v \equiv -\mathcal{N} \pmod{u}$ and $(u, \mathcal{N}) = 1$. This proves the kernel assertion for primitive z. Finally use the determinant bound in Lemma 3.4, $Q_1/u \le 16$ and $0 \le \beta_P < 1$. □

The continuous extension and its comparison with LWW are

$$g_{\mathcal{N}}(u,v,\beta)=\begin{pmatrix} u & \beta u \\ v & \mathcal{N}/u+\beta v\end{pmatrix},\quad \tau_{\mathcal{N}}(z)=z_1+\beta z_2,\quad g'(u,v,r)=\begin{pmatrix} u & -2ur \\ (u-v)/2 & r(v-u)+1/u\end{pmatrix}. \tag{3.14}$$

Thus $-2r$ is replaced by β and $1/u$ by $\mathcal{N}/u$; both entries are retained exactly. The ratio in LWW (4.26), p. 21, becomes $\beta_P=\{\mathcal{N}r_P\}$. Multiplication by $\mathcal{N}$ preserves Lebesgue measure on $\mathbb{T}$.

3.4 Enlarged states and intrinsic ratios

Definition 3.6 (States and good-state ratios)**.** The ambient position space is the disjoint union of labelled sets

$$\mathcal{S}'_\varrho=\{(\varrho,Q): Q\in\Omega\cap\Lambda_\varrho,\ Q \text{ primitive in } \Lambda_\varrho\}, \tag{3.15}$$

with LWW's lists attached. For such a state choose $W\in\Lambda_\varrho$ with $\det(Q,W)=\mathcal{N}$ and put

$$\beta_Q=\frac{W_1}{Q_1}\pmod 1. \tag{3.16}$$

The good-state tests of LWW (4.14)–(4.16), pp. 18–19, use this ratio at every state.

Lemma 3.7 (Intrinsic ratio and endpoint compression)**.** *Definition 3.6 is independent of the complement and of the chosen block basis. In relative coordinates,*

$$\beta_Q\equiv\frac{\tau_{\mathcal{N}}(w)}{\tau_{\mathcal{N}}(z)}\pmod 1\quad\big(Q=g_{\mathcal{N}}(P)z, W=g_{\mathcal{N}}(P)w, \det(z,w)=1\big). \tag{3.17}$$

For a unit endpoint it equals $\{\mathcal{N}r_Q\}$. LWW's test that z_1 is odd is replaced by endpoint support $(Q_1Q_2,\mathcal{N})=1$, which is not a parity statement about the basis used here. The restricted-root trace in LWW Section 4.6 is valid in this space.

Proof. A lattice-primitive Q has a determinant-$\mathcal{N}$ complement; any two such complements differ by an integer multiple of Q. Two oriented bases differ by $\mathrm{SL}_2(\mathbb{Z})$, proving (3.17) and basis independence. At a unit endpoint the normalized complement in (3.9) gives $\{\mathcal{N}r_Q\}$. A nonunit vector can belong to several full lattices, giving a different labelled state in each. Such a vector is not primitive in $\mathbb{Z}^2$, by Lemma 3.5; the endpoint vector vanishes there because its coefficients and lists are supported on units. For example, $\mathcal{N}=2$, $P=(3,1)$, $P^*=(1,1)$ and $z=(0,1)$ give a unit endpoint with z_1 even. Thus no assertion z_1 odd is made after this shear.

What the moment argument uses is a completion at a root, the primitive-coordinate equivalence

$$D\mid\det(z,z')\iff [z]_p=[z']_p\quad(p\mid D),$$

for squarefree D, and the ratio at the current state. All hold here. The diagonal operators S,G preserve the endpoint pairing under enlargement. For positive B_0, the matrix

$$\big(\langle u_P,B_0^R u_Q\rangle\big)_{P,Q\in\mathcal{C}_{\mathcal{N}}\cap\Lambda_\varrho}\succeq 0,\qquad \lambda_{\max}\le d_\varrho:=\sum_{P\in\mathcal{C}_{\mathcal{N}}\cap\Lambda_\varrho}\langle u_P,B_0^R u_P\rangle \tag{3.18}$$

is a principal submatrix of a Gram matrix. Spectral Hölder in LWW (4.65), p. 36, is applied in the ambient space before this restriction. The sum $\sum_\varrho d_\varrho$ is exactly the root average over $\Omega\cap\mathcal{C}_{\mathcal{N}}$ counted below. No estimate for the full trace on added nonunit states is required. The second good-state test in LWW (4.47)–(4.48), p. 30, is applied at the source position Q, using β_Q. For the exceptional-measure bound, multiplication by $\mathcal{N}$ preserves Haar measure; for the subsequent cube count we count β directly. □

3.5 Changes for general h: signed bases and slope cells

The unit blocks now have modulus $H = |h|$, including the single block $\mathbb{Z}^2$ when $H = 1$. The determinant orientation is $\Delta = -\varepsilon h$.

Lemma 3.8 (Signed normalized basis)**.** *Let $P = (u, v)^T$ have $u, v > 1$, $(u, v) = (uv, H) = 1$. Define*

$$0 \le c < u, \quad cv \equiv -\Delta \pmod{u}, \quad d = (\Delta + cv)/u, \quad \beta = c/u, \quad g_\Delta(P) = \begin{pmatrix} u & c \\ v & d \end{pmatrix}. \tag{3.19}$$

It is an oriented basis of $\Lambda_\varrho = \{(s, t) : t \equiv \varrho s \pmod{H}\}$, $\varrho = v\bar{u} \pmod{H}$, with determinant Δ. For $Q = g_\Delta(P)z$ and $\tau_\beta(z) = z_1 + \beta z_2$,

$$Q_1 = u\tau_\beta(z), \qquad Q_2 = v\tau_\beta(z) + \Delta z_2/u, \qquad \det(P, Q) = \Delta z_2. \tag{3.20}$$

Lattice-primitivity is $(z_1, z_2) = 1$; plane primitivity also requires $(Q_1, H) = 1$. For primitive z and $r \nmid H$,

$$r \mid Q_1 \quad \Longleftrightarrow \quad [z]_r = \ker(u, c) \pmod{r}. \tag{3.21}$$

Proof. The congruence defining c gives $ud - vc = \Delta$ and $(u, c) = 1$. Both columns lie in the index-H lattice, and their index is H. This proves the basis assertion for either sign. Matrix multiplication gives (3.20). At a prime outside H the matrix is invertible; at a prime dividing H it has rank one with a first column whose coordinates are units. The primitivity and kernel proofs in Lemma 3.5 therefore apply with these signed entries. □

For odd u, v, reduction modulo two gives

$$Q_1, Q_2 \text{ odd} \quad \Longleftrightarrow \quad \begin{cases} z_1 \text{ odd}, \ z_2 \text{ even}, & H \text{ odd}, \\ z_1 + cz_2 \text{ odd}, & H \text{ even}. \end{cases} \tag{3.22}$$

The even-H case uses the labelled enlargement of Section 3.4 with unit endpoint support; it does not impose a parity test on z_1. For odd H retain the odd-odd physical projection, equivalently the first line of (3.22). Nonunit intermediate positions remain labelled lattice-primitive states. At every source state use a determinant-Δ lattice complement to define its intrinsic ratio. For $\det(z, w) = 1$ this ratio and its derivative are

$$\beta_Q = \frac{\tau_\beta(w)}{\tau_\beta(z)} \pmod{1}, \qquad \partial_\beta \frac{\tau_\beta(w)}{\tau_\beta(z)} = \frac{1}{\tau_\beta(z)^2}. \tag{3.23}$$

Changing orientation changes the ratio's sign and preserves both good-state tests' distances to integers. In particular both tests are imposed at the current source, including added nonunit states.

Lemma 3.9 (Uniform local cells)**.** *Set $D_0 = d_0 Y$ and retain $\Omega = [D_0 M, 16 D_0 M] \times [N, 2N]$. In each labelled block partition physical slopes into half-open intervals of width $H/(D_0 M^2)$. Each cell contains $O(D_0)$ lattice-primitive positions, and an edge with determinant $O(HD_0)$ interacts with only $O(1)$ neighboring cells. A path of at most $2R$ such edges from a root has relative coordinates*

$$|(z_j)_2| \ll RD_0, \qquad |(z_j)_1| \ll 1 + RD_0. \tag{3.24}$$

All constants are independent of H.

Proof. Fix a lattice-primitive P in one cell. For another Q in it, $|\det(P,Q)| \le 256HD_0$ and the determinant is in $H\mathbb{Z}$. For each of its $O(D_0)$ possible values, differences of solutions belong to $\mathbb{Q}P \cap \Lambda_\varrho = \mathbb{Z}P$; the fixed ratio of first coordinates leaves $O(1)$ choices. This proves occupancy. An edge changes slope by $O(H/(D_0M^2))$. Sum the slope changes along the path and multiply by the two first coordinates to get $|\det(P,P_j)| \ll RHD_0$. Divide by Δ in (3.20), and use $\tau_\beta(z_j) \in [1/16,16]$ and $0 \le \beta < 1$, to prove (3.24). □

This cell partition extends the version 1 whole-block partition to every H. Its occupancy improves the earlier unnormalized small-shift bound $O(|h|d_0Y)$ to $O(d_0Y)$. The finite checks in Appendix B include these identities and the signed swapped complement below.

4 Root counting

Remark 4.1 (Changes in version 2: signed root count). The root count below replaces the version 1 positive-determinant Lemma R. It counts the normalized signed basis directly, includes all three swapped-complement branches, and permits every fixed subpower modulus exponent below one.

Lemma 4.2 (Lemma R, signed form)**.** *Put $U_0 = d_0YM$, $V_0 = N$ and assume $U_0, V_0 \ge x^\delta$, $0 < H < U_0V_0$, $H \le xL^C$. Let S be odd, squarefree, coprime to H, with $S \le \exp(C_0L^\sigma)$ for any fixed $\sigma < 1$. For a matrix $g_0 = \left(\begin{smallmatrix} u_0 & c_0 \\ v_0 & d_0' \end{smallmatrix}\right)$ modulo S of determinant Δ, and intervals $I_1 \subset [1,16]$, $I_2 \subset [1,2]$, $I_3 \subset [0,1]$, the formula is*

$$\begin{aligned} &\#\left\{(u,v): \begin{array}{l} (u,v)=1,\ (uv,2H)=1,\ u/U_0 \in I_1,\ v/V_0 \in I_2, \\ \beta = c/u \in I_3,\ g_\Delta(u,v) \equiv g_0 \pmod S \end{array}\right\} \\ &\quad = \frac{c_{2H}U_0V_0|I_1||I_2||I_3|}{|\mathrm{SL}_2(\mathbb{Z}/S\mathbb{Z})|} + O(U_0V_0x^{-c\delta}), \end{aligned} \tag{4.1}$$

where

$$c_{2H} = \zeta(2)^{-1} \prod_{r \mid 2H} \frac{r-1}{r+1}. \tag{4.2}$$

The error is uniform in the signed Δ, intervals, and residue class, with a threshold allowed to depend on the fixed σ, C_0. Prescribed first-row kernel lines at $r \mid S$ instead have main mass $c_{2H}\prod_{r\mid S}(r+1)^{-1}$ and a power-saving error with a possibly smaller fixed $c > 0$.

Proof. The shorter-coordinate case. Fix $u \equiv u_0 \pmod S$, $(u, 2H) = 1$, and replace the Appendix A inverse graph by

$$\mathcal{R}_u = \{(c,v) \pmod{uS} : c \equiv c_0\ (S),\ v \equiv v_0\ (S),\ cv \equiv ud_0' - \Delta\ (uS)\}. \tag{4.3}$$

The last right-hand side is a unit at every prime dividing u, including the primes shared by u, S, because $u_0d_0' - c_0v_0 \equiv \Delta$ and $(S,H) = 1$. Exactly as in the CRT calculation of LWW (A.3), with $S_1 = (u,S)$, $S_2 = S/S_1$, and $\gamma S_2 \equiv 1 \pmod{uS_1}$, its exponential sum is bounded by

$$\frac{1}{S_1} \sum_{j \bmod S_1} |K_{uS_1}(\gamma r + ju, \gamma s(ud_0' - \Delta))|. \tag{4.4}$$

Here $K_m(r,s) = \sum_{y \bmod m,(y,m)=1} e((ry + s\bar{y})/m)$ is the Kloosterman sum. The common-divisor factor is at most $S_1(r,s,u)$, independently of the size or sign of Δ. The stated Appendix A estimates (A.2)–(A.4) therefore give the same discrepancy $O(u^{0.99}S^2)$ for every rectangle. Its total mass is

$$|\mathcal{R}_u| = u \prod_{p\mid u,\ p\nmid S} (1-1/p). \tag{4.5}$$

Sieve $(v, 2H) = 1$ by $e \mid \mathrm{rad}(2H)$. Each such e is coprime to uS; replacing v by ev' replaces the inverse-graph constant by a unit multiple and leaves the rectangle error unchanged. Period splitting gives a main term

$$\frac{V_0|I_2||I_3|}{eS^2} \prod_{p\mid u, p\nmid S} (1-1/p) \tag{4.6}$$

and error $O((V_0/(uSe)+1)u^{0.99}S^2)$. Summing the signed divisor sieve multiplies the main term by $\prod_{p\mid 2H}(1-1/p)$ and costs at most $2^{\omega_{\mathrm{all}}(2H)} = x^{o(1)}$ in the error.

For the u average expand its Euler weight as $\sum_{\ell\mid u,(\ell,S)=1} \mu(\ell)/\ell$ and its unit condition by $d \mid \mathrm{rad}(2H)$. Counting the fixed residue modulo S gives the factor $1/\operatorname{lcm}(\ell, d)$. Sum first over d:

$$\sum_{d\mid \mathrm{rad}(2H)} \frac{\mu(d)}{\operatorname{lcm}(\ell,d)} = \begin{cases} 0, & (\ell, 2H) > 1, \\ \ell^{-1} \prod_{p\mid 2H}(1-1/p), & (\ell, 2H) = 1. \end{cases} \tag{4.7}$$

This proves the convergent tail bound $O(1/U_0)$ after the additional $\mu(\ell)/\ell$; it avoids an unnecessary divisor-count factor in the main term. The local Euler factors and $|\mathrm{SL}_2(\mathbb{Z}/S\mathbb{Z})| = S^3 \prod_{p\mid S}(1-p^{-2})$ give precisely c_{2H} above. The summed rectangle error is

$$O\big(x^{o(1)} S^2 (V_0 U_0^{0.99} + U_0^{1.99})\big) = O(U_0 V_0 x^{-c_0\delta}), \tag{4.8}$$

for a fixed $c_0 > 0$, since $U_0 \le V_0$ and $S = x^{o(1)}$. No comparison $H \asymp x$ was used; the polynomial upper bound for H suffices for the divisor cost.

Swapping the coordinates. If $V_0 < U_0$, write $\epsilon_0 = \Delta/(uv)$; it lies strictly between -1 and 1. Since $d/v = \beta + \epsilon_0$, let $k = \lfloor \beta + \epsilon_0 \rfloor \in \{-1, 0, 1\}$. The normalized complement of $(v, u)^T$ with determinant Δ is

$$\widetilde{c} = (1+k)v - d, \qquad \widetilde{d} = (1+k)u - c, \qquad \widetilde{\beta} = 1 + k - \beta - \epsilon_0. \tag{4.9}$$

Here $0 < \widetilde{c} < v$: an endpoint value would imply $v \mid \Delta$, excluded by $v > 1$ and $(v, H) = 1$. Direct multiplication gives $v\widetilde{d} - u\widetilde{c} = \Delta$. For each of the three k branches use the corresponding fixed matrix residue and the interval

$$((1+k) - I_3 - \Delta/(uv)) \cap [0, 1). \tag{4.10}$$

Their total length is $|I_3|$, since they describe one reflected interval modulo 1. Partition $I_1 \times I_2$ into relative cells of side $x^{-c_0\delta/4}$. On a cell $\Delta/(uv)$ varies by $O(x^{-c_0\delta/4})$, because its absolute value is less than one. Enlarge or contract the *unclipped* interval by that amount before intersecting with $[0, 1)$; this includes branches that are empty at a corner but become nonempty inside the cell. The shorter-coordinate discrepancy is uniform in interval length. Summing it over $O(x^{c_0\delta/2})$ cells leaves a power saving; the boundary main mass is $O(U_0 V_0 x^{-c_0\delta/4})$. This proves (4.1), with smaller c. The negative branch $k = -1$ is essential and is not present in the positive-Delta proof.

At a prime $p \mid S$, the determinant-Delta matrix coset has $p(p^2-1)$ elements. A prescribed first-row kernel admits $p-1$ nonzero first rows and p second rows each, giving probability $1/(p+1)$. CRT multiplies the probabilities. Summing per-matrix errors costs at most $S^3 = x^{o(1)}$, absorbed by decreasing c. This proves the line version. The factor at 2 is already in c_{2H}, so is not inserted again.

□

For the reflected specialization $\Delta = H = \mathcal{N}$ even, write

$$c_{\mathcal{N}} = c_{2H} = \frac{1}{\zeta(2)} \prod_{r|\mathcal{N}} \frac{r-1}{r+1} \le 1. \tag{4.11}$$

Thus (4.1) also supplies the matrix and line counts used in the reflected transport below. Only the upper bound on this density is used.

5 Transport of the Type II argument

Lemma 5.1 (Reduced determinant and boundary geometry)**.** *For $Q = g_{\mathcal{N}}(P)z$, $Q' = g_{\mathcal{N}}(P)z'$ in one labelled block,*

$$\det(g_{\mathcal{N}}(P)z, g_{\mathcal{N}}(P)z') = \mathcal{N}\det(z,z'), \qquad t' = \frac{\det(Q,Q')}{\mathcal{N}D} = \frac{\det(z,z')}{D}. \tag{5.1}$$

The kernel in relative coordinates is (3.5) with cutoff $\psi_1(\det(z,z')/(DY))$. On allowed positions and edges,

$$\frac{1}{16} \le \tau_{\mathcal{N}}(z) \le 16, \qquad |\det(z,z')| \ll d_0 Y, \qquad |z| \ll d_0 YMN/\mathcal{N} \le d_0 Y. \tag{5.2}$$

For $s = u/(d_0 YM)$, $t_0 = v/N$, the position boundaries are

$$s\tau_{\mathcal{N}}(z) \in \{1, 16\}, \qquad E_2 = t_0\tau_{\mathcal{N}}(z) + \frac{z_2\mathcal{N}}{d_0 YMNs} \in \{1,2\}. \tag{5.3}$$

The second term is bounded by 64 in absolute value on allowed positions; its s derivative is $O(1)$, and $\partial_{t_0} E_2 = \tau_{\mathcal{N}}(z) \ge 1/16$.

Proof. The determinant identities follow from $\det g_{\mathcal{N}} = \mathcal{N}$ and (3.1). The remaining bounds follow from Lemma 3.5 and the cutoff in (3.4). In the last boundary formula $s \in [1,16]$, so the extra term and its s derivative are $O(1)$. The full boundary has $\partial_\beta E_2 = t_0 z_2$, with $|\partial_\beta E_2| \ll \exp(O_K(L^{j_*+\theta}))$. Transversality, rather than smallness of this term, is what the cube count uses. □

The full dictionary is Table 3 in Appendix A. It includes the all-integer square, the coefficient conjugation and the unit-coordinate support. The position cutoffs in LWW Sections 4.2–4.7 are sharp; η is a cutoff in the list product. There is no smooth cutoff in the second position coordinate.

Proposition 5.2 (Root averaging)**.** *Using the arguments in* H2′*, the block version of LWW Proposition 4.4 [12, pp. 22–24] is*

$$\begin{aligned} &\frac{1}{d_0 YMN} \sum_{P \in \Omega \cap \mathcal{C}_{\mathcal{N}}} \langle u_P, (\mathcal{A}_{\mathcal{N}} \mathcal{A}_{\mathcal{N}}^*)^R u_P \rangle \\ &\quad = c_{\mathcal{N}} \int_1^{16} \int_1^2 \int_0^1 M_R^{(\mathcal{N})}(d_0 YMs, Nt_0, \beta)\, d\beta\, dt_0\, ds + O_A(L^{-2AR}). \end{aligned} \tag{5.4}$$

Here $M_R^{(\mathcal{N})}$ is LWW's M_R with g' replaced by $g_{\mathcal{N}}$ and LWW's coordinate-parity test replaced by the labelled state space of Section 3.4.

Proof. A closed path from a root $P \in \mathcal{C}_{\mathcal{N}}$ has $P_j = g_{\mathcal{N}}(P) z_j$, with z_j primitive, $z_0 = z_{2R} = (1,0)^T$ and $|z_j| \le 2^7 d_0 Y$. List divisibility becomes $[z_j]_p = \ell_p(P) = \ker(u, c') \bmod p$. Its good-state ratio is (3.17) at the current state. Apply the path expansion and cube argument of LWW Proposition 4.4 with Lemma 4.2 in place of its root count. The measure (4.29), p. 23, becomes $c_{\mathcal{N}}\, d(u/(d_0 Y M))\, d(v/N)\, d\beta$, and each prescribed line has mass $1/(p+1)$.

The absolute coefficient mass remains $\exp(O_K(L^{r_*+j_*+\theta}))$, since $|z_j| \ll R d_0 Y$. The model Bonferroni remainder in this retuned application uses Lemmas 5.6–5.7; arithmetic discrepancy still uses the crude coefficient count. The retained Bonferroni truncation has $S \le \exp(O_K(L^{1-7j_*}))$. By (5.3) and

$$\partial_\beta \frac{\tau_{\mathcal{N}}(w)}{\tau_{\mathcal{N}}(z)} = \frac{\det(z,w)}{\tau_{\mathcal{N}}(z)^2} = \frac{1}{\tau_{\mathcal{N}}(z)^2} \quad (\det(z,w) = 1), \tag{5.5}$$

the boundary surfaces have a transverse partial derivative of modulus at least $1/256$ and gradients $x^{o(1)}$. For each boundary, let $\partial_{\mathrm{tr}} E$ denote its derivative in s, t_0 or β used above. Its neighborhood volume is $\ll \varepsilon \max|\nabla E|/|\partial_{\mathrm{tr}} E| = x^{-c\delta/8+o(1)}$ at the chosen cube size. Sum the per-cube power errors before the moment limit; their product with the coefficient mass gives $O_A(L^{-2AR})$. These are the changes to the stated LWW argument. □

Proposition 5.3 (Block Type II estimate)**.** *Assume* H2′. *Let $M, N \ge x^\delta$, $MN \asymp x$ and let the coefficients satisfy the sifting and bound hypotheses above. After choosing A_0 and then K sufficiently large, independently of the fixed band exponents a_i,*

$$|Q_Y^{\mathrm{all}} - Q_Y^{\mathrm{maj},\mathcal{N}}| \ll_A \mathcal{N} Y L^{-A}. \tag{5.6}$$

Proof. We apply the arguments of LWW Sections 4.4–4.7 as written, with the replacements in Table 3. Section 4.4 uses relative coordinates, the linear form τ and the projection χ; use $\tau_{\mathcal{N}}$ and $\chi_{\mathcal{N}} = \mathbf{1}_{g_{\mathcal{N}} z \in \Omega}$. Its auxiliary operators have the same relative determinant, shared-prime lines and intrinsic tests. In (4.40)–(4.41), [12, pp. 27–28],

$$z' = j_1 w_z + h z, \qquad \frac{\tau_{\mathcal{N}}(z')}{\tau_{\mathcal{N}}(z)} = h + j_1 \frac{\tau_{\mathcal{N}}(w_z)}{\tau_{\mathcal{N}}(z)}. \tag{5.7}$$

The ratio strip leaves $O(1)$ integers h, uniformly in j_1. The clean and dirty bounds (4.42)–(4.50) use this geometry and the same bounded endpoint multipliers. The combinatorial signed-memory expansion of Section 4.5, pp. 32–35, uses these operator bounds, root averaging and bounded measure mass; $c_{\mathcal{N}} \le 1$ suffices.

For the endpoint pairing in Section 4.6, use the inner product conjugate-linear in its initial entry and set

$$f(Dbm, s) = \overline{\alpha_m} \beta_s. \tag{5.8}$$

At $P = (Dbm, s)$, $Q = (Dar, n)$ this gives exactly

$$\overline{f(P)} f(Q) = \alpha_m \overline{\alpha_r} \beta_n \overline{\beta_s}. \tag{5.9}$$

All support, list-invariance and norm bounds are unchanged by this conjugation. Primitivity must hold for comparison endpoints as well as raw endpoints. Every prime of Db is below Z, whereas s is Z-rough. Thus a common prime p' of Dbm, s exceeds Z and avoids $\mathcal{N}$. Write $t = bmn - ars$. Since $(b, ars) = 1$, $t \ne 0$. Both kernels impose $\mathcal{N} \mid t$ and $|t/\mathcal{N}| \ll Y$. The common prime would divide the nonzero integer $t/\mathcal{N}$, which is impossible because $Y = o(Z)$. The other endpoint is identical.

Use Lemma 3.4 for the partition in Section 4.6, and the positive matrix and ambient spectral Hölder argument in Lemma 3.7 for (4.64)–(4.68), pp. 36–37. The trace loss is $(d_0 Y)^{1/(2R)} = O(1)$.

Section 4.7 has $O(1)$ exact-edge and $O(Y)$ comparison-edge counts in the reduced determinant, as in (4.70), p. 37. For (4.75), p. 38, fix $k_0 = mn$, $\ell_0 = rs$. Then

$$\ell_0 \equiv ba^{-1}k_0 \pmod{\mathcal{N}}, \qquad |bk_0 - a\ell_0| \ll \mathcal{N}Y. \tag{5.10}$$

The interval for ℓ_0 has length $O(\mathcal{N})$ and one residue class, so there are $O(1)$ choices. Raw absolute estimates use the all-integer cofactor range of Proposition 2.6; the extra $\mathcal{N}/X$ costs a fixed logarithmic power.

The moment/root normalization itself remains d_0YMN, giving the pairing scale XY of LWW. The right side $\mathcal{N}Y$ in (5.6) is a permitted relaxation by $\mathcal{N}/X \ll L^B$, absorbed by increasing the fixed precision. It is independent of the Cauchy cofactor factor $\mathcal{N}/Y$. □

Remark 5.4 (Determinant scale). Using arcs for the unreduced determinant without the block congruence would connect a position to $\asymp \mathcal{N}Y$ positions. The trace loss would be $(d_0Y\mathcal{N})^{1/(2R)} = \exp(L^{1-r_*} + o(L^{1-r_*}))$ rather than $O(1)$. The block congruence absorbs this determinant scale and reappears in the major arcs.

5.1 Changes for general h: signed transport

The covering identity for the signed determinant is

$$\int_{\{\theta:\Delta\theta\in\mathfrak{M}_1\}} e(\theta(t - \Delta(b-a)))\,d\theta = \mathbf{1}_{H|t}\int_{\mathfrak{M}_1} e(\phi(t/\Delta - b + a))\,d\phi. \tag{5.11}$$

The proof of Lemma 3.1 uses H inverse branches with absolute Jacobian $1/H$ for either orientation. The reduced determinant is $\det(g_\Delta z, g_\Delta z')/(\Delta D) = \det(z,z')/D$, so its raw-minus-comparison bracket is exactly (3.5) with $t' = \det(z,z')/D$.

Root averaging and sharp boundaries. Use (3.24) for each path and Lemma 4.2 for its arithmetic roots. With $s = u/(D_0M)$ and $t_0 = v/N$, the physical boundaries become

$$s\tau_\beta(z) \in \{1, 16\}, \qquad t_0\tau_\beta(z) + \frac{\Delta z_2}{D_0MNs} \in \{1, 2\}. \tag{5.12}$$

Their transverse derivatives in s, t_0, respectively, equal $\tau_\beta(z) \in [1/16, 16]$. All other derivatives and boundary counts are $x^{o(1)}$; for example $|\Delta z_2/(D_0MN)| \ll RH/X \ll L^{B+C+r_*}$. The good-state ratio has the transverse derivative (3.23). The odd-H parity projection depends only on z. Partition normalized root space into cubes of side $x^{-c\delta/8}$. Boundary volume is $x^{-c\delta/8+o(1)}$, and the summed interval-uniform root errors still save a fixed power after all retained residue and path choices. The independent-line moment has density $c_{2H} \le 1$. The same reasoning with one edge also applies when goodness fails at either endpoint. Section 5.2 proves the new model-tail bound; Section 5.3 checks its separate arithmetic discrepancy budget at the retuned parameters.

Signed operators. The relative primitive box now has radius $O(RD_0)$, and its physical cutoff is $\mathbf{1}_{g_\Delta z\in\Omega}$. The shared-prime lattice, intrinsic good-state tests, weighted adjoints, signed memory and collision-rank restoration are precisely the H2′ operations described above. For a prescribed determinant j_1, write $z' = j_1 w_z + nz$; the ratio strip leaves $O(1)$ integers n. In the shared-prime lattice the parameter is $D\tau_\beta(w_*)/\tau_\beta(z_*) = D\beta_{Q_*}$ modulo one, and the same balanced box has sides between $Y^{0.1}$ and $Y^{0.9}$ and area comparable with Y. The signed clean kernel contains no Δ. Physical and parity restrictions are endpoint multipliers of norm at most one. In the dirty raw

equation $\det(z,z') = D(b-a)$, use the intrinsic ratio at its source; the dirty comparison congruence is nonzero since all its marking primes avoid H and the shared product. The unchanged signed-memory identities are used only with the scope of H2′. Appendix A records each interface.

Endpoint pairing. The coefficient $f(Dbm, s) = \overline{\alpha_m}\beta_s$ gives (5.9). Its norm and supremum satisfy

$$\|f\| \ll (D_0X)^{1/2}L^{C_6}, \qquad \sup|f| \le L^{C_6},$$

where the exponent is fixed independently of K. The complete band-prime part of $P_1 = Dbm$ is exactly Db, because m is sifted. Thus this coefficient depends on the position and not on the ordering of its lists. For both kernels, endpoint primitivity follows from $H \mid t$, $0 < |t/\Delta| \ll Y$ and $Y = o(Z)$ by the argument above.

For a slope cell j, let f_j be its endpoint vector and g_j the vector on its boundedly many interacting cells. For positive $B_0 = \mathcal{A}\mathcal{A}^*$, the principal coefficient-space Gram matrix on that cell has largest eigenvalue at most its trace d_j. Cell occupancy gives $\langle f_j, B_0^R f_j\rangle \le CD_0 \sup|f|^2 d_j$. Apply spectral Hölder in the ambient space first, and then Cauchy and Hölder over the cells, to obtain

$$|\langle f, \mathcal{A}g\rangle| \ll \|g\|\|f\|^{1-1/R}\left(CD_0\sup|f|^2\sum_j d_j\right)^{1/(2R)}. \tag{5.13}$$

Only actual unit endpoint roots enter $\sum_j d_j$, so no factor for the number of labelled blocks occurs. Root averaging and H2′ give $\sum_j d_j \ll D_0XL^{-2RE_0}$. The trace loss is $D_0^{1/(2R)} = O(1)$ by Section 5.3. This gives the pairing scale D_0X with arbitrary fixed saving.

Goodness, padding and common lists. The exceptional ratio measure saves $O_K(\exp(-cL^{0.1}))$. The one-edge root count and $O(1)$ raw/$O(Y)$ comparison counts give the same error on removing either good-state projection. Harmonic padding draws are summed over all d_0 intervals exactly once; their distinctness/avoidance probability differs from one by $O_K((J+1)^2\exp(-cL^{0.1}))$.

Here are the absolute estimates required for those operations. For fixed $\ell \asymp Y$ coprime to H, put $k = \Delta + \ell j \asymp X$. Its j interval has length $O(X/Y)$, regardless of its center. If $d \mid k$ then $(d,\ell) \mid \Delta$, hence $(d,\ell) = 1$. Deleting a largest factor in the four-factor decomposition that majorizes $\tau(k)^2$, the remaining product is at most $O(X^{3/4})$. Each such divisor gives $O(X/(Yd)+1)$ points. Summing gives

$$\sum_{j:\Delta+\ell j\asymp X}\tau(\Delta+\ell j)^2 \ll (X/Y)L^3 + X^{3/4}L^2 \ll (X/Y)L^3. \tag{5.14}$$

Cauchy for each product pair therefore bounds the absolute raw form by $XYL^{O(1)}$. Common-list pairs gain the extra factor $\sum_{p\in\cup_i\mathcal{P}_i} p^{-2} \ll \exp(-cL^{0.1})$. For the comparison form fix $k_0 = mn$ and $\ell_0 = rs$. Then

$$\ell_0 \equiv ba^{-1}k_0 \pmod H, \qquad |bk_0 - a\ell_0| \ll HY. \tag{5.15}$$

For either fixed variable the other has $O(1)$ choices, in a single residue class in an interval of length $O(H)$. Use $2\tau(k_0)\tau(\ell_0) \le \tau(k_0)^2 + \tau(\ell_0)^2$, sum over the $O(Y^2)$ product pairs, and multiply by arc mass $O(L^{3A_0}/Y)$. This gives $XYL^{O(1)}$ and the same additional common-list saving.

Proposition 5.5 (Affine Type II transport)**.** *Under* H2′*, with the parameters of Section 5.3, the affine square satisfies*

$$|Q_Y - Q_Y^{\mathrm{maj}}| \ll_A XYL^{-A} \tag{5.16}$$

at every fixed precision, uniformly in the A range and with the rough cofactor multiplier of D. Here Q_Y^{maj} is obtained from the raw equation in (2.24) by the signed covering kernel (5.11) and cutoff $\psi_1(t/(\Delta Y))$.

Proof. The preceding root, operator, pairing and absolute estimates verify the changed interfaces of LWW Sections 4.2–4.7. The retuning below supplies the model-tail and arithmetic-root errors with the required fixed precision. The number of product and padding dyads is $O_K(L)$. These fixed logarithmic costs are absorbed before choosing the bands. The root normalization is D_0X; the Cauchy cofactor count $(X+H)/Y$ remains a separate factor and does not change that normalization. □

5.2 Changes in version 2: the positive active-prime remainder

The signed main moment still uses the memory and ghost operators in H2′. The following auxiliary positive estimate is used for the model remainder in root averaging.

Lemma 5.6 (Positive active-prime bound)**.** *Consider the relative-coordinate path expansion of a $2R$-edge root moment, with the same active lists, normalized list permutations, position restrictions and shared-prime restrictions as above. Keep the definition $V_i = \sum_{p\in\mathcal{P}_i} 1/p$ and assume these masses are bounded above and below by fixed positive constants, $2KR \le L^2$, and the absolute geometric kernels admit $O(1)$ raw endpoints and $O(Y)$ comparison endpoints, the latter carrying $O(L^{3A_0}/Y)$. Let $\mathcal{A}_{\mathrm{act}}$ be the sum of absolute geometric path coefficients, including the list normalization, times the independent-line factors for primes that occur in active lists, while all factors for primes that never occur in a list are replaced by one. Then, uniformly in the real root variables and the permitted projections,*

$$\mathcal{A}_{\mathrm{act}} \le L^{C_*R} \tag{5.17}$$

for a fixed $C_* = C_*(K, A_0, \rho)$ and sufficiently large L.

Proof. For a prime p in group i, let $A_p \subset \{0,\dots,2R\}$ be its active time set, and let κ_p count its consecutive active runs. Put $q_0 = q_{2R} = \sqrt{\rho}$, $q_j = \rho$ for $0 < j < 2R$. If A_p is nonempty, its line average vanishes unless all active positions have one common projective line ℓ_p. In the consistent case its factor is exactly

$$\frac{V_i^{-\kappa_p}}{p+1} \prod_{j\notin A_p} q_j^{1_{[z_j]_p=\ell_p}}. \tag{5.18}$$

Indeed the first occurrence charges one uniform-line probability, and the physical list normalization charges one $1/V_i$ at the beginning of each active run. Shared entries continue a run without a new charge. This is also obtained directly from the local factor (4.37) of LWW before introducing b'_p.

Write $a_p = \min A_p$, $c_p = \max A_p$. Dropping the factors at $j < a_p$ and $j > c_p$ increases (5.18), because each belongs to $[0,1]$. All remaining inactive positions are internal, so their q_j equals ρ. It remains to bound the model with prime factors

$$\frac{V_i^{-\kappa_p}}{p+1} \prod_{\substack{a_p<j<c_p\\ j\notin A_p}} \rho^{1_{[z_j]_p=\ell_p}}. \tag{5.19}$$

Define $\nu_i^+(p) = 1/(V_i(p+1))$ and $\lambda_i^+(p,\ell) = \nu_i^+(p)$. Active lists have product measure ν_i^+; symmetric memory tuples have measure $(\lambda_i^+)^{\otimes k}/k!$. Since $V_i = \sum_{p\in\mathcal{P}_i} 1/p$, one has $\sum_p \nu_i^+(p) \le 1$ exactly. Use the active-list leave/discard and recall/fresh operations with *no ghost operators*. A fresh target entry has measure ν_i^+; recalling a memory entry contributes $1/V_i$. At an edge multiply source memory hits and target memory hits by $\vartheta_+^{h_s+h_t}$, with $\vartheta_+ = \sqrt{\rho}$. Use a nonnegative geometric upper kernel consisting of the absolute raw term plus the absolute comparison upper bound. Retain list permutations and all required shared-prime and complete-list restrictions.

First require all initial/fresh prime births to be distinct and the initial and final memory to be empty. A prime appearing in two separated active runs must remain in memory throughout the gap: discarding it would require a second fresh birth, and retaining it after its last run would leave final memory nonempty. Thus its history is forced. Each gap position receives two factors $\sqrt{\rho}$, one from each incident edge. The first active run costs $\nu_i^+(p)$, and each later run costs $1/V_i$, reproducing (5.19). Consistency of active lines is enforced by shared-prime conditions and by the line recorded in a recalled memory entry. There is no contribution from a never-active prime. Multiplicity factorials cancel the orderings of memory coordinates, so this is an identity of positive path sums.

Remove distinctness of births. All weights are nonnegative, so the sum increases. Starting empty, at most K memory coordinates are appended per edge; therefore every canonical path has memory size at most $2KR \le L^2$. Restrict the enlarged space to this cap, which still contains every canonical path.

We now prove a two-sided absolute bound for each positive edge operator. With Schur weight $v_0^{k_1+\cdots+k_K}$, fixed $v_0 > 1$, an edge changes the exponent by at most K. There are at most L^{2K} choices of recalled memory coordinates, and sums of fresh prime measures are at most one per group. The normalized list permutations have row and column sum one. For a fixed determinant, Bézout and the bounded τ ratio give $O(1)$ positions. There are $O(Y)$ comparison determinants, canceling the $1/Y$ comparison factor. Thus the weighted row sum is at most

$$C_{K,\rho} L^{2K}(1 + L^{3A_0}). \tag{5.20}$$

The adjoint coefficients follow from the complete factorial state measure. In one group let $S, I \in \{0,1\}$ indicate appending the departing prime and recalling the arriving prime. In the unordered memory description, each token type has measure $\nu(p)^n/n!$. If the source active prime is p and the target is q, cross-edge exclusion gives $p \ne q$. Multiplication by source measure divided by target measure gives:

(S,I)	Forward coefficient	Adjoint coefficient	Reversed rule
$(0,0)$	$\nu(q)$	$\nu(p)$	$(0,0)$
$(1,0)$	$\nu(q)$	n'_p	$(0,1)$
$(0,1)$	n_q/V_i	$\nu(p)/V_i$	$(1,0)$
$(1,1)$	n_q/V_i	n'_p/V_i	$(1,1)$

Here n_q and n'_p are the relevant source/target anchored memory multiplicities. Equivalently the adjoint is the reversed rule times V_i^{S-I}. Shared active-list measures cancel. This calculation includes repeated token types and active/memory overlaps in the enlarged space. In every reverse creation the memory line is anchored at its position; there is no free factor $p+1$. Taking the product of these identities over groups, and using the reversed $O(1)/O(Y)$ geometric count, proves the same

polynomial column bound as (5.20), up to a fixed constant. The cap and endpoint projections only decrease absolute sums. Weighted Schur gives the polynomial edge norm.

The vector b at relative position e_1, with empty memory and arbitrary admissible active lists, has squared norm at most $\prod_i(\sum_p \nu_i^+(p))^{J+1} \le 1$. Its positive matrix element across $2R$ such operators is bounded by the product of their norms, giving (5.17). The main signed-memory construction retains its original parameter $\vartheta^2 > \sqrt{\rho}$; that condition concerns ghost operators, which are absent from this auxiliary positive bound. The geometric input was the elementary endpoint count; the arithmetic root average enters only in the later transfer. □

Lemma 5.7 (Bonferroni model remainder)**.** *For damping Bonferroni depth B_d, the total independent-line main mass of the absolute remainder is at most*

$$L^{C_* R}\frac{(C_K R)^{B_d}}{B_d!}. \tag{5.21}$$

Proof. Fix a path and its active prime values first. For a never-active prime define

$$d_p(\ell) = 1 - \prod_{j=0}^{2R} q_j^{1_{[z_j]p=\ell}} \in [0,1]. \tag{5.22}$$

Its mean is at most $(2R+1)/(p+1)$ by $1-\prod(1-u_j) \le \sum u_j$. Distinct inactive primes have independent lines, independently also of active-prime consistency. The mean of the elementary-symmetric Bonferroni error of degree B_d is therefore at most $(C_K R)^{B_d}/B_d!$, uniformly in the path and its active values. Multiply by the exact active factors and absolute geometric coefficients, sum, and apply Lemma 5.6. This uses independence only after the path and active values have been fixed. It does not assert a relative error for arithmetic roots; their discrepancy is treated separately. □

5.3 Retuned numerical parameters

Fix $0 < a < 1/2$ and choose

$$\theta = \frac{\max(a,1/10)+1/2}{2}, \qquad j_* = \frac{1-2\theta}{10}, \qquad r_* = \theta + 2j_*, \qquad b_* = \theta + 3j_*. \tag{5.23}$$

Replace the version 1 numerical parameters by

$$Z = \exp(L^{3/4}), \quad J = \lfloor L^{j_*} \rfloor, \quad R = \lceil L^{r_*}/2 \rceil, \quad B_d = \lceil L^{b_*} \rceil, \qquad \max(a,1/10) < a_1 < \cdots < a_K < \theta. \tag{5.24}$$

The positive margins are

$$r_* - j_* - \theta = j_*, \qquad b_* - r_* = j_*, \qquad 1 - (r_* + j_* + \theta) = 1 - (b_* + \theta) = 7j_* > 0. \tag{5.25}$$

The changes to every affected use of the old numerical exponents are as follows.

1. *Vaughan and Type I:* $Z = x^{o(1)}$, $Z^{O(\log L)} = x^{o(1)}$, and every marking prime and selected product $a \asymp Y$ is below Z eventually. The Type I small-divisor BV margin remains $19/40 - 2/5 > 0$. Finite target-presieve moduli are still $\exp(O(L^a \log L)) = x^{o(1)}$. All factorial and Rankin arguments in Section 7 retain only fixed logarithmic losses.

2. *Sifted Möbius:* the smooth-divisor reciprocal tail in Lemma 6.11 becomes $O(L^{3/4}\exp(-c_\delta L^{1/4}))$, still smaller than any fixed logarithmic power. No other part of the small-conductor estimate changes.

3. *Paths and root moduli:* $\log(d_0 Y) = O_K(L^{j_*+\theta})$; the crude absolute path/list/permutation coefficient count has logarithm $O_K(L^{r_*+j_*+\theta})$. Retained prime moduli have logarithm $O_K(L^{r_*+j_*+\theta} + L^{b_*+\theta}) = O_K(L^{1-7j_*})$. Both are subpower. The Appendix A discrepancy and the signed root lemma are valid for any fixed modulus exponent below one, since every extra fixed power of S is absorbed by the coordinate power saving. Boundary derivatives and counts remain $x^{o(1)}$.

4. *Root-model tail:* use (5.21), not the crude path count, for the model Bonferroni error. Stirling bounds its logarithm by $O(R\log L) - (b_* - r_* + o(1))L^{b_*}\log L$, which is smaller than $-2AR\log L$ for every fixed A eventually. The arithmetic discrepancy uses the crude subpower coefficient count in item 3.

5. *Prime measures and signed memory:* $(J+1)R/p_{\min} \to 0$, so $b'_p > 0$, $s_i^{J+1} = 1 + o(1)$ and the original finite primewise identities are unchanged. The main signed model's memory cap remains L^2. Its truncation error has factor $2^{-L^2+O(R)}$ times $\exp(O_K(L^{1-7j_*}))$, hence remains negligible. The address count is polynomial in L, and the collision-rank threshold $\lceil C_0 2R\log L/L^{0.1}\rceil$ is $o(R)$. The high-/low-rank restoration argument therefore keeps the same saving.

6. *Signed edge norms:* the clean bound has $L^{\theta-A_0}$ in place of $L^{0.2-A_0}$, so $A_0 > G+3$ still suffices. The dirty raw bound becomes $L^{1-j_*K/2} + L^{-A}$; choose fixed $K > 2(G+3)/j_*$. Dirty comparison bounds retain exponential minimum-prime savings. The good-state powers of Y, the balanced-box powers, and the Vaughan exponent $1/5$ are not changed.

7. *Moment to pairing:* the trace factor has logarithm $O_K(L^{j_*+\theta-r_*}) = O_K(L^{-j_*})$. Thus it is bounded. There are $O_K(L)$ auxiliary-product dyads as before. Goodness and padding collisions still have exponential savings, because J is only a fixed power of L.

8. *Major arcs:* the short-prime contour proof works uniformly up to $\exp(O(L^\theta))$. The sparse-frequency count has logarithm $O(L^{0.9}\log L + L^\theta)$ and still satisfies the Gram estimates. In the reflected H1* interface, $a_K < \theta < 1/2$ allows the $\log Y$ term to be absorbed into $L^{1-a_K}\log L$ in Lemma LS. Its Burgess/Pólya–Vinogradov energy bounds do not depend on J or the marking ceiling. Deleting prime powers with base at most Z costs $O(ZL/V)$ after the long-prime cofactor sum, negligible since V is a fixed positive power of the large variable.

The determinant estimates at (5.24) hold under H2′. The model-tail bound in item 4 follows from Lemmas 5.6–5.7. The resulting roughness range is subpower.

The strict endpoint $a = 1/2$ is not obtained. In this fixed-parameter scheme the trace condition requires $r_* > j_* + \theta$, while the active-prime modulus budget requires $r_* + j_* + \theta < 1$. These already imply $2\theta + 2j_* < 1$. This is an obstruction to the displayed parameter scheme, not a claim about all methods.

6 Major arcs

6.1 Reduction to the averaged energy

Proposition 6.1 (Major-arc reduction)**.** *For every fixed $D_* > 0$, (α^*) at a sufficiently large fixed precision $A = D_* + O(A_0 + B_1)$ implies*

$$|Q_Y^{\mathrm{maj},\mathcal{N}}| \ll_{D_*} \mathcal{N} Y L^{-D_*}. \tag{6.1}$$

Proof. By Lemma 3.1, on the six-variable unit support,

$$\begin{aligned} H_{\mathcal{N}}(bmn - ars; a, b) &= \psi_1(t'/Y)\mathbf{1}_{bmn\equiv ars\ (\mathcal{N})} \int_{\mathfrak{M}_1} e(\phi(t' - b + a))\, d\phi, \\ t' = (bmn - ars)/\mathcal{N}, &\qquad \mathbf{1}_{bmn\equiv ars\ (\mathcal{N})} = \frac{1}{\varphi(\mathcal{N})} \sum_{\chi \bmod \mathcal{N}} \chi(bmn)\overline{\chi(ars)}. \end{aligned} \tag{6.2}$$

Write $\phi = h_0/k + \lambda_0/Y$, $(h_0, k) = 1$, $k \le L^{A_0}$, $|\lambda_0| \le 2L^{A_0}$, and $d\phi = d\lambda_0/Y$. Its factor is

$$e(h_0 t'/k)e(-h_0(b-a)/k)e(\lambda_0 t'/Y)e(-\lambda_0(b-a)/Y). \tag{6.3}$$

The factors in a, b modulo k expand in characters, with mass at most k each, since a, b are units modulo k.

For $e(h_0 t'/k)$, put $u_1 = bmn$, $u_1' = ars$. Both are units modulo $k\mathcal{N}$, and

$$\mathbf{1}_{\mathcal{N}|u_1 - u_1'} e(h_0 t'/k) = \mathbf{1}_{\mathcal{N}|u_1 - u_1'} e(h_0 u_1/(k\mathcal{N}))e(-h_0 u_1'/(k\mathcal{N})).$$

Let

$$H = \ker((\mathbb{Z}/k\mathcal{N}\mathbb{Z})^\times \longrightarrow (\mathbb{Z}/\mathcal{N}\mathbb{Z})^\times), \qquad |H| = k \prod_{\substack{p\mid k \\ p\nmid \mathcal{N}}} (1 - 1/p) \le k. \tag{6.4}$$

The reduction is surjective. For each unit class $a' \bmod \mathcal{N}$, choose a unit representative $\widetilde{a}' \bmod k\mathcal{N}$; its fibre is $\widetilde{a}' H$. Expand $w \mapsto e(h_0 \widetilde{a}' w/(k\mathcal{N}))$ in characters $\eta \in \widehat{H}$, with coefficients $|c_{a',\eta}| \le 1$. Every such character extends to a character ψ_η of the finite abelian group $(\mathbb{Z}/k\mathcal{N}\mathbb{Z})^\times$, independently of a'. Hence on that fibre

$$e(h_0 u_1/(k\mathcal{N})) = \sum_{\eta \in \widehat{H}} c_{a',\eta} \psi_\eta(\widetilde{a}')^{-1} \psi_\eta(u_1). \tag{6.5}$$

The same representative works on the u_1' side. Cauchy gives

$$\sum_{a'} \left| \sum_{\eta} c_{a',\eta} \psi_\eta(\widetilde{a}')^{-1} X_{a',\eta} \right|^2 \le k \sum_{\eta} \sum_{a'} |X_{a',\eta}|^2. \tag{6.6}$$

Thus class-dependent coefficients disappear at fixed powers of k, including when $(k, \mathcal{N}) > 1$. Together with the a, b factors, the character mass is at most k^4 (LWW use k^6).

For the smooth factor put $w = u_1/(\mathcal{N}Y)$, $w' = u_1'/(\mathcal{N}Y)$ and $\kappa_X = MN/\mathcal{N}$. The dyadic variables satisfy

$$w, w' \in [\kappa_X, 16\kappa_X], \qquad \kappa_X \gg L^{-B}, \quad \kappa_X \le 1, \qquad \psi_1(t'/Y)e(\lambda_0 t'/Y) = \psi_1(w - w')e(\lambda_0(w - w')). \tag{6.7}$$

If $\kappa_X \le 1/4$, the difference has absolute value at most 4 and $\psi_1 = 1$. Otherwise it varies on a bounded interval. Set $z = \log(w'/\kappa_X)$ and $\zeta = \log(w/w')$. With a fixed smooth cutoff ϖ equal to 1 on $[1, 16]$, the kernel is

$$\varpi(e^z)\varpi(e^{z+\zeta})\psi_1(\kappa_X e^z(e^\zeta - 1))e(\lambda_0\kappa_X e^z(e^\zeta - 1)). \tag{6.8}$$

Its support lies in a fixed box of side $O(1)$. In the variables $w/\kappa_X, w'/\kappa_X$, derivatives are bounded by fixed powers of L^{A_0} because $\kappa_X \le 1$. Consequently for a fixed sufficiently large B_0 the transform obeys

$$F_{\lambda_0}(v, s) \ll_j (1 + |v|/L^{B_0})^{-j}(1 + |s|/L^{B_0})^{-j}. \tag{6.9}$$

Taking B_1 sufficiently large leaves frequencies $|t|, |s| \le L^{B_1}$. This is the Mellin separation of LWW Section 5, pp. 42–43, at scale 1 rather than X. The cutoff here does not force a narrow product difference; the arithmetic is in the character congruence.

With the normalized polynomials (2.6), each product sum has size $YMN \cdot BMN$ with its displayed normalization. The $1/Y$ arc measure therefore gives the scale

$$\frac{(YMN)^2}{Y}\frac{1}{\varphi(\mathcal{N})}\sum_\chi \int |BMN|^2 = Y(MN)^2\frac{1}{\varphi(\mathcal{N})}\sum_\chi \int |BMN|^2. \tag{6.10}$$

To obtain (6.1) it suffices that the energy be $\ll L^{-D_*}\mathcal{N}/(MN)^2$, which follows from $L^{-A}/\mathcal{N}$ because $MN \le \mathcal{N}$. Cauchy in t, χ, and the dyadic tail estimates of LWW Section 5, p. 43, leave (α^*) with $\xi = \chi\psi$ on m, n and $\xi_b = \chi\psi_2$ on b, where $\psi = \psi_\eta$, $\psi_2 = \psi\psi_b'$ and ψ_b' is modulo k. The primed product uses $-\lambda_0, \overline{\alpha}, \overline{\beta}$; the estimates below also hold after these conjugations and the corresponding sign change of the frequency twist. The arc count L^{2A_0}, character mass k^4, λ_0 integral L^{A_0} and s integral L^{B_1} are fixed logarithmic costs, absorbed by $A = D_* + O(A_0 + B_1)$.

For completeness, Parseval for a product polynomial gives

$$\frac{1}{\varphi(\mathcal{N})}\sum_\chi \left|\sum_u c_u\chi(u)\psi(u)u^{it}\right|^2 = \sum_{a'\in(\mathbb{Z}/\mathcal{N}\mathbb{Z})^\times}\left|\sum_{u\equiv a'\ (\mathcal{N})} c_u\psi(u)u^{it}\right|^2 \ll (YX/\mathcal{N}+1)\sum_u |c_u|^2 \ll \frac{L^{O(1)}}{\mathcal{N}}. \tag{6.11}$$

Here the product length is $\asymp XY$ and $|c_u| \ll L^C\tau_3(u)/(XY)$. For MN alone the squared coefficient norm is $L^{O(1)}/(MN) \le L^{O(1)+B}/\mathcal{N}$; this extra power is absorbed. □

6.2 Prime sums from a fixed zero-free region

Lemma 6.2 (Lemma P). *Let $2 \le q \le \mathcal{N}^3$, and let $\chi \bmod q$ be induced by a primitive χ^* of conductor $q^* > 1$. Fix positive $C_{\mathrm{zf},1}, C_{\mathrm{zf},2}$ and $C_3 \ge 1$ with $C_{\mathrm{zf},2} \ge C_3 + 1$. Suppose $L(s, \chi^*) \ne 0$ in (1.3)'s rectangle for these constants. For $|u| \le L^{C_3}$ and $\mathcal{N}^{1/6} \le y \le \mathcal{N}$,*

$$S(y) := \sum_{\ell\le y}\Lambda(\ell)\chi(\ell)\ell^{iu} \ll yL^3(L^{-C_{\mathrm{zf},1}/12} + L^{-C_{\mathrm{zf},2}}). \tag{6.12}$$

The implied constant can be absolute once $L \ge L_0(C_{\mathrm{zf},1}, C_{\mathrm{zf},2}, C_3)$.

Proof. The imprimitive difference is supported on prime powers at primes dividing q, and contributes at most $\omega_{\mathrm{all}}(q)\log y \le 5L^2$, absorbed since $y \ge \mathcal{N}^{1/6}$. For χ^* put

$$F(s) = -\frac{L'}{L}(s - iu, \chi^*), \quad c = 1 + 1/L, \quad T = L^{C_{\mathrm{zf},2}} - L^{C_3} - 1 \ge \tfrac{1}{2}L^{C_{\mathrm{zf},2}}. \tag{6.13}$$

Truncated Perron [3, Chapter 17] gives

$$\sum_{\ell\le y}\Lambda(\ell)\chi^*(\ell)\ell^{iu}=\frac{1}{2\pi i}\int_{c-iT}^{c+iT}F(s)\frac{y^s}{s}\,ds+O(y(\log y)^2/T+\log y). \tag{6.14}$$

The usual error is the sum of $\Lambda(\ell)(y/\ell)^c\min(1,(T|\log(y/\ell)|)^{-1})$. The choice $y\le\mathcal{N}$ ensures $y^c\le ey$ and $\sum\Lambda(\ell)\ell^{-c}\ll L$.

For $1/2\le\sigma\le 2$, the partial-fraction formula and local zero count [3, Chapter 16] read

$$\frac{L'}{L}(s,\chi^*)=\sum_{|\gamma-t|\le 1}\frac{1}{s-\rho_0}+O(\log(q^*(|t|+2))),\quad N(t+1,\chi^*)-N(t,\chi^*)\ll\log(q^*(|t|+2)). \tag{6.15}$$

Put $\sigma_0=1-C_{\mathrm{zf},1}\log L/(2L)$. For $\sigma\ge\sigma_0$ and $|t|+1\le L^{C_{\mathrm{zf},2}}$, all nearby zeros have real part at most $1-C_{\mathrm{zf},1}\log L/L$ and horizontal distance at least $C_{\mathrm{zf},1}\log L/(2L)$. Their number is $O(\log(q^*L^{C_{\mathrm{zf},2}}))=O(L)$ after a threshold depending on the fixed constants. Hence

$$|L'/L(s,\chi^*)|\ll L\frac{2L}{C_{\mathrm{zf},1}\log L}+L\ll L^2. \tag{6.16}$$

The shifted $F(s)$ has this bound on $\sigma\ge\sigma_0$, $|t|\le T$.

Move the contour to σ_0. There is no zero pole in the rectangle: a pole $s=\rho_0+iu$ would have $\Re\rho_0\ge\sigma_0$ and $|\Im\rho_0|\le T+|u|\le L^{C_{\mathrm{zf},2}}-1$, contradicting the region. Also $\sigma_0>1/2$, so $s=0$ is outside, and χ^* is nonprincipal. The vertical and horizontal integrals are respectively

$$\ll y^{\sigma_0}L^2\int_{-T}^{T}\frac{dt}{|\sigma_0+it|}\ll y^{\sigma_0}L^3,\qquad\ll(c-\sigma_0)y^cL^2/T\ll yL^2/T.$$

Since $L/6\le\log y\le L$, $y^{\sigma_0}\le yL^{-C_{\mathrm{zf},1}/12}$. Combining these estimates yields

$$S(y)\ll yL^{3-C_{\mathrm{zf},1}/12}+yL^{2-C_{\mathrm{zf},2}}+y(\log y)^2/T+\log y+L^2\ll yL^3(L^{-C_{\mathrm{zf},1}/12}+L^{-C_{\mathrm{zf},2}}). \tag{6.17}$$

□

Remark 6.3. The width $C\log L/L$ is what gives the arbitrary fixed logarithmic saving on $y\ge\mathcal{N}^{1/6}$. A width $c/\log q^*$ with $q^*\asymp\mathcal{N}$ gives only a factor of the size $e^{-c/6}$. Only the logarithmic derivative on the contour is used.

Corollary 6.4 (Pointwise prime-factor polynomial)**.** *Suppose $\xi=\chi\psi \bmod k\mathcal{N}$ is induced by χ^* of conductor $L^{A_1}<q^*\le k\mathcal{N}$ and Lemma 6.2 applies. If*

$$C_{\mathrm{zf},1}\ge 12(D+4),\quad C_{\mathrm{zf},2}\ge D+4,\quad C_{\mathrm{zf},2}\ge C_3+1,\quad C_3=\max(B_1,2B)+1, \tag{6.18}$$

then, uniformly for $|t|\le T_0+1$ and sufficiently large L depending on the fixed parameters,

$$|N_\xi(t)|\le 8L^{1-D}+\frac{4Z+2L}{V}\le 9L^{1-D}. \tag{6.19}$$

Proof. Put $u=t+v$, so $|u|\le L^{C_3}$. Both R_Z and the unit indicator are completely multiplicative. Thus

$$\begin{aligned}c_V(n)R_Z(n)\mathbf{1}_{(n,\mathcal{N})=1}&=\sum_{\substack{\ell k'=n\\ \ell>V}}\Lambda(\ell)R_Z(\ell)\mathbf{1}_{(\ell,\mathcal{N})=1}R_Z(k')\mathbf{1}_{(k',\mathcal{N})=1},\\ N_\xi(t)&=\frac{1}{N}\sum_{k'<2N/V}R_Z(k')\mathbf{1}_{(k',\mathcal{N})=1}\xi(k')(k')^{iu}\sum_{\ell\in I_{k'}}\Lambda(\ell)R_Z(\ell)\mathbf{1}_{(\ell,\mathcal{N})=1}\xi(\ell)\ell^{iu},\\ I_{k'}&=(\max(V,N/k'),2N/k'].\end{aligned} \tag{6.20}$$

In a ratio-two interval at most one power of each prime is present. Dropping the indicators in the inner sum therefore costs at most $\sum_{p\le Z}\log p+\sum_{p\mid\mathcal{N}}\log p\le 2Z+L$, and after the k' sum at most $(4Z+2L)/V$. For every nonempty $I_{k'}$, the two endpoints are between V and $2N\le\mathcal{N}$, and

$$\log V=(\log x)/5\ge(L-B\log L-\log 2)/5\ge L/6$$

for sufficiently large L. The modulus is at most $\mathcal{N}^2$. Lemma 6.2 and (6.18) give $|S(y)|\le 2C_PyL^{-D-1}\le yL^{-D}$ for $L\ge 2C_P$. Consequently

$$|N_\xi(t)|\le 4L^{-D}(\log(2N/V)+1)+(4Z+2L)/V\le 8L^{1-D}+(4Z+2L)/V.$$

Since $V\ge\mathcal{N}^{1/6}$ and $Z=\exp(L^{3/4})$, the last ratio is at most L^{1-D} for large L. Only endpoint sums with $y\ge V$ are used, even when $I_{k'}$ is short. □

The mean value theorem used below has the form

$$\int_{-T}^{T}\left|\sum_u b_u u^{it}\right|^2dt=\sum_u|b_u|^2\bigl(2T+O(\delta_u^{-1})\bigr),\tag{6.21}$$

where δ_u is the separation of $\log u$ from the other logarithmic frequencies [16].

6.3 Exceptional values of the short polynomial

For $0\le i<2T_0$ let $J_i=[-T_0+i,-T_0+i+1]$. For each $\chi \bmod \mathcal{N}$ choose a point $t_{\chi,i}$ maximizing $|B_{\xi_b}M_\xi N_\xi|$ on J_i. Let $\mathcal{P}_0$ use the even i and $\mathcal{P}_1$ the odd i. Within each family, points for a fixed character are 1-spaced, and

$$\sum_\chi\int_{|t|\le T_0}|BMN|^2\,dt\le\sum_{\varepsilon=0,1}\sum_{(\chi,t)\in\mathcal{P}_\varepsilon}|B_{\xi_b}(t)M_\xi(t)N_\xi(t)|^2.\tag{6.22}$$

Below $\mathcal{P}$ is either family, and each argument applies to both.

Lemma 6.5 (Sobolev estimate)**.** *For $F\in C^1([-T_0-1,T_0+1])$ and 1-spaced points $t_1<\cdots<t_r$ in $[-T_0,T_0]$,*

$$\sum_j|F(t_j)|^2\le\int_{|t|\le T_0+1}|F|^2\,dt+2\left(\int_{|t|\le T_0+1}|F|^2\,dt\right)^{1/2}\left(\int_{|t|\le T_0+1}|F'|^2\,dt\right)^{1/2}.\tag{6.23}$$

Proof. Average $|F(t_j)|^2=|F(s)|^2+\int_s^{t_j}(|F|^2)'$ over $s\in[t_j-1/2,t_j+1/2]$. These intervals have disjoint interiors; use $|(|F|^2)'|\le 2|F||F'|$ and Cauchy–Schwarz. □

Lemma 6.6 (Mean value over residue classes)**.** *If (c_u) is supported on positive $u\le 4\mathcal{N}$ and $|\psi_3|\le 1$, then for $T\ge 1$,*

$$\sum_{\chi \bmod \mathcal{N}}\int_{-T}^{T}\left|\sum_u c_u\chi(u)\psi_3(u)u^{it}\right|^2dt\le\varphi(\mathcal{N})(2T+C_0')\sum_u|c_u|^2\tag{6.24}$$

with absolute C_0'.

Proof. Character orthogonality gives

$$\sum_{\chi}\left|\sum_{u} c_u\chi(u)d_u\right|^2=\varphi(\mathcal{N})\sum_{a'\in(\mathbb{Z}/\mathcal{N}\mathbb{Z})^\times}\left|\sum_{u\equiv a'\ (\mathcal{N})} c_u d_u\right|^2.$$

Each class has at most four positive integers at most $4\mathcal{N}$. If $u<u'$ are in it, $u'\ge u+\mathcal{N}\ge (4/3)u$ since $u\le 3\mathcal{N}$. The Montgomery–Vaughan mean value theorem [16] therefore gives $C_0'=O(1/\log(4/3))$. For length at most $\mathcal{N}$, a unit class has at most one term and the integral in that class is exactly $2T$ times its squared coefficient. □

Lemma 6.7 (Lemma LS)**.** *Fix $A_s\ge 1$ and put $E=\{(\chi,t)\in\mathcal{P}:|B_{\xi_b}(t)|>L^{-A_s}\}$. For $L\ge L_0(K,a_K,A_s,B_1,C_B)$,*

$$|E|\le \exp\bigl(L^{1-a_K}(2A_s+K+10)\log L\bigr)\le \exp(L^{0.9}). \tag{6.25}$$

Proof. Let $j=\lfloor L/\log(16Y)\rfloor\ge 1$; then $(16Y)^j\le\mathcal{N}<(16Y)^{j+1}$. Write $B_{\xi_b}(t)^j=\sum_u c_u^{(j)}\xi_b(u)u^{it}$, where

$$c_u^{(j)}=\sum_{\substack{b_1\cdots b_j=u\\ b_i\in\mathcal{A}}}\prod_i c_{b_i},\qquad c_b=\frac{\eta(b/Y)e(-\lambda_0 b/Y)}{Y\prod_i V_i},\qquad |c_b|\le C_B/(16Y). \tag{6.26}$$

This polynomial is supported on $[Y^j,(16Y)^j]\subset[1,\mathcal{N}]$. For each disjoint band the j prime factors of u, with multiplicity, can be assigned to the j slots in at most $j!$ ways. Hence there are at most $(j!)^K$ representations and

$$|c_u^{(j)}|^2\le (j!)^K\sum_{\text{reps}}\prod_i|c_{b_i}|^2,\qquad \sum_u|c_u^{(j)}|^2\le (j!)^K\left(\sum_b|c_b|^2\right)^j\le (j!)^K\left(\frac{C_B^2}{16Y}\right)^j. \tag{6.27}$$

Use Lemmas 6.5 and 6.6 on B^j; its derivative inserts a coefficient $i\log u$, with $|\log u|\le L$. Since $\xi_b=\chi\psi_2$ and $|\psi_2|\le 1$,

$$\sum_{(\chi,t)\in\mathcal{P}}|B_{\xi_b}(t)|^{2j}\le 3L\varphi(\mathcal{N})(2T_0+2+C_0')(j!)^K(C_B^2/(16Y))^j. \tag{6.28}$$

The left side is at least $|E|L^{-2jA_s}$. Since $Y^{-j}<16^{j+1}Y/\mathcal{N}$ and $\varphi(\mathcal{N})\le\mathcal{N}$,

$$|E|\le L^{2jA_s}3L(2T_0+2+C_0')(j!)^K C_B^{2j}16Y. \tag{6.29}$$

Every $b\in\mathcal{A}$ exceeds $\exp(L^{a_K})$, so $j\le L^{1-a_K}$, $\log j\le\log L$, and $\log Y\le 2\sum_i L^{a_i}+\log 16\le 3KL^{a_K}$. It follows that

$$\log|E|\le 2A_sj\log L+Kj\log j+2j\log C_B+\log Y+O(\log L)\le L^{1-a_K}(2A_s+K+10)\log L.$$

For fixed $a_K>1/10$, this is eventually at most $L^{0.9}/2$. □

Corollary 6.8 (Bulk estimate)**.** *For each family $\mathcal{P}$,*

$$\begin{aligned}\sum_{\mathcal{P}\setminus E}|BMN|^2&\le L^{-2A_s}\sum_{\mathcal{P}}|M_\xi N_\xi|^2\\ &\le L^{-2A_s}3L\varphi(\mathcal{N})(2T_0+O(1))\frac{L^2\sum_{u\le 4MN}\tau(u)^2}{(MN)^2}\\ &\ll L^{-2A_s+B_1+B+8}\frac{\varphi(\mathcal{N})}{\mathcal{N}}.\end{aligned} \tag{6.30}$$

Proof. The product MN polynomial is supported in $(MN,4MN]$, with $4MN\le 8x\le 4\mathcal{N}$ and coefficients at most $L\tau(u)/(MN)$. Apply Lemmas 6.5 and 6.6, and $\sum_{u\le y}\tau(u)^2\ll y(\log y)^3$. □

6.4 Energy on the exceptional set

Lemma 6.9 (Lemma HM)**.** *Let $(\xi_r, t_r)_{r\le R'}$ be character-frequency pairs modulo q, 1-spaced in t whenever their characters agree. Let a_m be supported on $(X', X'']$, $X' < X'' \le 8X'$, and let $w \ge 0$ be at least 1 on that interval. Then*

$$\sum_r \left|\sum_m a_m \xi_r(m) m^{it_r}\right|^2 \le \|a\|_2^2 \max_r \sum_s |G_{rs}|, \qquad G_{rs} = \sum_m w(m)\xi_r(m)\overline{\xi_s(m)} m^{i(t_r - t_s)}. \tag{6.31}$$

Proof. Put $S_r = \sum_m a_m \xi_r(m) m^{it_r}$ and $b_r = \overline{S_r}/(\sum_s |S_s|^2)^{1/2}$; the zero case is immediate. Cauchy gives

$$(\sum_r |S_r|^2)^{1/2} = \sum_r b_r S_r \le \|a\|_2 \left(\sum_m w(m)\left|\sum_r b_r \xi_r(m) m^{it_r}\right|^2\right)^{1/2}.$$

The squared last factor is $\sum_{r,s} b_r \bar{b}_s G_{rs}$. The Hermitian Schur bound and $\sum |b_r|^2 = 1$ prove the claim. □

Choose $q = k\mathcal{N}$ and a smooth w_0 with $\mathbf{1}_{(X',X'']} \le w_0 \le \mathbf{1}_{[X'/2,16X']}$ and $|w_0^{(i)}| \le C_i (X')^{-i}$. There are three kinds of Gram entry.

1. On the diagonal, $G_{rr} \le 16X'$.

2. If $\xi_r = \xi_s$, $r \ne s$, and $\tau = t_r - t_s$, then $|\tau| \ge 1$ and

$$G_{rs} = \sum_{d|q} \mu(d) d^{i\tau} \sum_{m'} w_0(dm')(m')^{i\tau}. \tag{6.32}$$

   For $d \le 16X'$, Euler summation replaces the inner sum by its integral with error $O(1 + |\tau|)$: for $f(y) = w_0(dy)y^{i\tau}$,

$$\int |f'| \le \int |w_0'| + |\tau| \int_{X'/(2d)}^{16X'/d} dy/y \le C(1 + |\tau|).$$

   Two integrations by parts give integral size $O((X'/d)(1 + |\tau|)^{-2})$. For $d > 16X'$ the sum is empty. Thus

$$|G_{rs}| \le C' X' (1 + |\tau|)^{-2} \log L + \tau(k\mathcal{N})(1 + |\tau|), \tag{6.33}$$

   using $\sum_{d|q} 1/d \ll \log\log q$. The 1-spacing gives

$$\sum_{\substack{s\ne r \\ \xi_s = \xi_r}} |G_{rs}| \le C'' X' \log L + (2T_0 + 1)^2 \tau(k\mathcal{N}). \tag{6.34}$$

3. If $\xi_r \ne \xi_s$, their quotient is nonprincipal. On units it is $\chi_r \overline{\chi_s}$, and these are distinct characters modulo $\mathcal{N}$. Partial summation gives

$$|G_{rs}| \le C(1 + |\tau|)\Delta_\xi(16X'), \qquad \Delta_\xi(Y') = \max_{y \le Y'} \left|\sum_{m \le y} \xi(m)\right|, \quad |\tau| \le 2T_0 + 1. \tag{6.35}$$

If ξ is induced by primitive $\xi^* \bmod q^* > 1$, put $q_1 = \prod_{p \mid k\mathcal{N}, p \nmid q^*} p$. Then

$$\sum_{m \le y} \xi(m) = \sum_{d \mid q_1} \mu(d)\xi^*(d) \sum_{m' \le y/d} \xi^*(m'), \qquad \Delta_\xi(Y') \le 2^{\omega_{\mathrm{all}}(k\mathcal{N})} \Delta_{\xi^*}(Y'). \tag{6.36}$$

The two inputs are Pólya–Vinogradov and Burgess at $r = 2$:

$$\Delta_{\xi^*}(Y') \le C(q^*)^{1/2} \log q^* \le C(L^{A_0}\mathcal{N})^{1/2}\, 2L, \tag{6.37}$$

$$\left| \sum_{H_0 < n \le H_0 + H} \xi^*(n) \right| \ll_\varepsilon H^{1/2}(q^*)^{3/16+\varepsilon} \quad (H \ge 1), \tag{6.38}$$

where (6.38) holds for every modulus, as in [7, introduction, (1)] (arbitrary moduli are allowed for $r \le 3$). Also

$$2^{\omega_{\mathrm{all}}(k\mathcal{N})} \le 2^{A_0 \log L} 2^{\omega_{\mathrm{all}}(\mathcal{N})} \le L^{A_0} \exp(O(L/\log L)) = \mathcal{N}^{o(1)}, \qquad \tau(k\mathcal{N}) = \mathcal{N}^{o(1)}.$$

Corollary 6.10 (Exceptional energy)**.** *Let E be as in Lemma 6.7 and fix $\eta_0 = 1/40$. Then*

$$M \ge \mathcal{N}^{3/8+\eta_0} \implies \sum_E |M_\xi(t)|^2 \ll \log L, \tag{6.39}$$

$$\sum_E |M_\xi(t)|^6 \ll L^9, \tag{6.40}$$

$$N \ge \mathcal{N}^{1/2+2\eta_0} \implies \sum_E |N_\xi(t)|^2 \ll L^2 \log L. \tag{6.41}$$

Proof. For (6.39) use $a_m = \alpha_m/M$, $\|a\|_2^2 \le 2/M$, $X' = M$ and Burgess with $\varepsilon = \eta_0/8$. The distinct-character row contribution is at most

$$\begin{aligned} &|E| C(2T_0+2) 2^{\omega_{\mathrm{all}}(k\mathcal{N})} C_\varepsilon (16M)^{1/2} (L^{A_0}\mathcal{N})^{3/16+\varepsilon} \\ &\quad \le M\mathcal{N}^{-\eta_0/2+\varepsilon+o(1)} \exp(L^{0.9}) = M\mathcal{N}^{-3/320+o(1)} \exp(L^{0.9}) = o(M). \end{aligned} \tag{6.42}$$

Together with $16M + C''M\log L + (2T_0+1)^2\tau(k\mathcal{N})$ this proves (6.39). For (6.40), $M^3 \ge (x^{1/5}/2)^3 \ge \mathcal{N}^{3/5-o(1)}$. The coefficients of M_ξ^3 are at most $M^{-3}\tau_3(u)$ on $(M^3, 8M^3]$, with squared norm $\ll L^8/M^3$. Pólya–Vinogradov makes the distinct-character row at most

$$|E| L^{O(1)} \mathcal{N}^{1/2+o(1)} \le M^3 \mathcal{N}^{1/2-3/5+o(1)} \exp(L^{0.9}) = o(M^3),$$

yielding L^9. For (6.41), take $a_n = \beta_n/N$, $\|a\|_2^2 \le 2L^2/N$; Pólya–Vinogradov gives row $\le N\mathcal{N}^{-2\eta_0+o(1)} \exp(L^{0.9}) = o(N)$. In all cases the same-character row is $O(X' \log L)$, because $X' \ge \mathcal{N}^{1/5-o(1)}$ absorbs $(2T_0+1)^2\tau(k\mathcal{N})$. □

Lemma 6.11 (Twisted sifted Möbius sums)**.** *For a primitive character of conductor at most L^D, every fixed $C_t, A, \delta > 0$ admits*

$$\sup_{|t| \le x^{C_t}} \left| \sum_{m \in I} \mu(m) R_Z(m) \xi^*(m) m^{it} \right| \ll ML^{-A}, \qquad I \subset [M, 2M], \quad M \ge x^\delta. \tag{6.43}$$

Proof. First use [14, Lemma 3.9(ii)] at any dyadic scale $D_0 \ge x^{\delta/2}$ with auxiliary variable $X_* = D_0^{R_*}$, fixed $R_* > 2(C_t+1)/\delta$, and exponent $1/R_*$. Its freely chosen logarithmic precision covers both L^D and the desired saving, uniformly over these scales. To obtain arbitrary subintervals, decompose a prefix ending at any $u \in [M, 2M]$ into dyads down to scale ML^{-E}, for a large fixed E. The sum of their square-root lengths is $O(\sqrt{M})$, while the remaining initial segment contributes at most $O(\sqrt{M}L^{-E/2})$ to the weighted sum with coefficient $m^{-1/2}$. All retained scales exceed $x^{\delta/2}$ for sufficiently large x. Thus weighted prefix sums, and hence their differences on every subinterval of $[M, 2M]$, are $O(\sqrt{M}L^{-A})$ after selecting the precision and E. Partial summation against the *real* factor $m^{1/2}$ now gives $O(ML^{-A})$ for the untapered interval sum; the phase is not differentiated. This derives the interval estimate from the full-dyad consequence of the cited lemma.

Choose the fixed rescaling exponent once large enough for the same argument with $\delta/2$ as well as δ; for example $R_* > 8(C_t+1)/\delta$ suffices for the frequency requirement. After the convolution below, all retained prefix dyads then have length at least $x^{\delta/4}$ for sufficiently large x. The identity

$$\mu(m)R_Z(m) = \sum_{\substack{d \mid m \\ P^+(d) \le Z}} \mu(m/d) \tag{6.44}$$

is checked prime by prime. For $d \le M^{1/2}$ apply the preceding untapered estimate at scale $M/d \ge x^{\delta/2}$; summing $1/d$ over Z-smooth d costs $O(\log Z)$. For $d > M^{1/2}$, Rankin with exponent $1/\log Z$ bounds its reciprocal mass by $O(\log Z \exp(-c_\delta L^{1/4}))$. This proves (6.43). If the coefficient also has the unit restriction $(m,h)=1$, remove it at cost $O(M\omega_{\mathrm{all}}(h)/Z)$, uniformly in the phase, since any excluded sifted factor exceeds Z. Consequently (6.43) still holds. Fixed Mellin phases and smooth weights of logarithmic variation are absorbed by increasing the fixed precision. □

Remark 6.12 (The published interval estimate). Lemma 3.9(ii) of [14, p. 46] is stated with a supremum over subintervals of its dyadic range. The proof above uses its full-dyad consequence and recovers the interval estimate by prefix decomposition and partial summation against a real weight. For a bounded-variation weight the estimate is multiplied by its supremum plus variation.

6.5 The energy theorem

Theorem 6.13 (Reflected energy estimate)**.** *Fix A and $A_1 \ge 1$, and put*

$$\begin{aligned} D &= \left\lceil \frac{3}{2}(A + B_1 + B + 20) \right\rceil, \qquad C_3 = \max(B_1, 2B) + 1, \\ C_{\mathrm{zf},2} &= \max(D+4, C_3+1), \qquad C_{\mathrm{zf},1} = 12(D+4), \\ A_s &= A + B_1 + B + 10. \end{aligned} \tag{6.45}$$

In the setting of Section 6.1, $\mathrm{H1}^\star(C_{\mathrm{zf},1}, C_{\mathrm{zf},2}; A_1)$ *at* $\mathcal{N}$ *implies* (α^*)*, for a threshold depending on all fixed parameters and* A_1*.*

Proof. Split the characters into $\mathcal{X}_s = \{\chi : \mathrm{cond}(\chi\psi) \le L^{A_1}\}$ and its complement $\mathcal{X}_b$. For $\chi \in \mathcal{X}_s$, let ξ^* induce $\xi = \chi\psi$. On the coefficient support,

$$M_\xi(t) = \frac{1}{M}\sum_m \mu(m)R_Z(m)\mathbf{1}_{(m,\mathcal{N})=1}\xi^*(m)m^{i(t+v)}. \tag{6.46}$$

Removing the unit indicator changes this by at most

$$\sum_{\substack{p\mid\mathcal{N}\\ p>Z}}\frac{2M/p}{M}\le 2\omega_{\mathrm{all}}(\mathcal{N})/Z\le 2Le^{-L^{3/4}}.$$

Apply Lemma 6.11 and its bounded-variation extension to the sum without the indicator: the modulus is at most L^{A_1}, $M\ge x^{\delta}$ and $|t+v|\le x$ for large x. For every fixed A' this gives $\sup_{|t|\le T_0}|M_\xi(t)|\ll L^{-A'}$. This is the use of the sifted Möbius input, whose proof cites [14, Lemma 3.9(ii)]. For fixed ψ, the primitive inducing character determines χ, so $|\mathcal{X}_s|\le L^{2A_1}$. The polynomial $B_{\xi_b}N_\xi$ is supported on $u\le 32YN$ with coefficients at most $C_BL\tau(u)/(16YN)$ and squared coefficient norm $\ll L^5/(YN)$. The ordinary mean value theorem gives $\int_{|t|\le T_0}|BN|^2\ll (T_0+YN)L^5/(YN)\ll L^5$. Thus with $A'=A+A_1+4$,

$$\frac{1}{\varphi(\mathcal{N})}\sum_{\chi\in\mathcal{X}_s}\int|BMN|^2\ll\frac{L^{2A_1-2A'+5}}{\varphi(\mathcal{N})}\ll L^{-A}/\mathcal{N},\tag{6.47}$$

using $\mathcal{N}/\varphi(\mathcal{N})\ll\log L$.

For $\chi\in\mathcal{X}_b$, $q^*\mid k\mathcal{N}$ and $L^{A_1}<q^*\le k\mathcal{N}$. The assumed region and Corollary 6.4 yield $|N_\xi(t)|\le 9L^{1-D}$ for $|t|\le T_0+1$. Discretize on this subset as above; restricting characters only reduces the bounds for $\mathcal{P}$ and E. The bulk is at most

$$L^{-2A_s+B_1+B+8}\varphi(\mathcal{N})/\mathcal{N}\le L^{-A-1}\varphi(\mathcal{N})/\mathcal{N}.$$

If $M\ge\mathcal{N}^{3/8+\eta_0}$, (6.39) gives

$$\sum_E|BMN|^2\le C_B^2\sup_E|N|^2\sum_E|M|^2\ll L^{2-2D}\log L.\tag{6.48}$$

If $M<\mathcal{N}^{3/8+\eta_0}$, then

$$N=\frac{MN}{M}\gg\frac{x}{M}\gg\frac{\mathcal{N}^{5/8-\eta_0}}{L^B},\qquad N\ge\mathcal{N}^{1/2+2\eta_0}\quad\text{for sufficiently large }\mathcal{N}.\tag{6.49}$$

Hölder with exponents $3,3/2$, followed by (6.40) and (6.41), gives

$$\begin{aligned}\sum_E|B|^2|M|^2|N|^2&\le C_B^2\sup_E|N|^{2/3}\sum_E|M|^2|N|^{4/3}\\&\le C_B^2(9L^{1-D})^{2/3}\Big(\sum_E|M|^6\Big)^{1/3}\Big(\sum_E|N|^2\Big)^{2/3}\ll L^{(2/3)(1-D)+3+2}.\end{aligned}\tag{6.50}$$

In either case the exceptional contribution is

$$\ll L^{6-2D/3}\le L^{-A-B_1-B-14}\le L^{-A-1}.\tag{6.51}$$

One may also use the looser exponent $(2/3)(1-D)+6$ in the second case and $3-2D$ with a factor $\log L$ in the initial case; both leave the same required margin. Sum the two families and divide by $\varphi(\mathcal{N})$:

$$\frac{1}{\varphi(\mathcal{N})}\sum_{\chi\in\mathcal{X}_b}\int|BMN|^2\ll L^{-A-1}/\mathcal{N}+L^{-A-1}/\varphi(\mathcal{N})\ll L^{-A}/\mathcal{N}.$$

□

The inputs to Theorem 6.13 are the ordinary mean value theorem, Pólya–Vinogradov, Burgess at $r = 2$, partial fractions and zero counting, Perron, and Lemma 6.11. The half-plane (1.2) enters only by supplying (1.3). For one fixed region the theorem supplies its selected fixed precision A; it does not assert all precisions from one fixed width.

6.6 Changes for general h: the hybrid analytic interface

In this subsection $L = \log x$, $X = MN \asymp y$, $y \geq x/L^B$, $\Delta = -\varepsilon h$ and $H = |h|$. Normalize B, M, N as in (2.6), replacing the unit modulus $\mathcal{N}$ by H. The same fibre expansion used in Section 6.1 applies to the units modulo kH over those modulo H: the kernel of reduction has size at most k, for either sign of Δ. On the arc $\phi = c/k + \lambda_0/Y$, expand $e(\mathrm{sgn}(\Delta)cu/(kH))$ on each fibre. Its Fourier coefficients have modulus at most one, and each character of the fibre group extends to the full unit group. The same representative at both endpoints and Cauchy as in (6.6) remove the fibre-dependent coefficients before orthogonality modulo H. Together with the a, b phases modulo k, a bound k^8 for all character losses suffices and is independent of H. The characters occur in the two locations

$$\xi = \chi\psi \text{ on } m, n, \qquad \xi_b = \chi\psi\psi'_b \text{ on } b, \qquad \chi \pmod H, \quad \psi'_b \pmod k. \tag{6.52}$$

Choose $A_1 > A_0$. If $\mathrm{cond}(\xi) > L^{A_1}$, then ξ_b is nonprincipal; otherwise ξ would be induced from modulus k. For each fixed extension ψ, at most L^{2A_1+1} characters χ lie in the complementary range, since the primitive inducing character determines χ.

6.6.1 Uniform Mellin scale

Let $\mathcal{T} = \max(1, X/H)$, $w = u/(YX)$, $w' = u'/(YX)$, so w, w' belong to a fixed compact subset of $(0, \infty)$. The smooth factor is

$$\psi_1((X/\Delta)(w - w'))e(\lambda_0(X/\Delta)(w - w')). \tag{6.53}$$

With $z_0 = \log w'$ and $\zeta = \mathcal{T}\log(w/w')$, insert fixed cutoffs in w, w' equal to one on their ranges and call the resulting function $K_{\lambda_0}(z_0, \zeta)$. If $H \leq X$, the compact support of ψ_1 makes ζ lie in a fixed interval. If $H > X$, $\mathcal{T} = 1$ and the two fixed cutoffs give that conclusion. All mixed derivatives have bounds by fixed powers of L^{A_0}, uniformly in H: in the former case $(X/H)/\mathcal{T} = 1$, while in the latter $X/H \leq 1$.

Its Fourier transform satisfies, for every fixed integer j,

$$|\widehat{K}_{\lambda_0}(v, s)| \ll_j (1 + |v|/L^{B_0})^{-j}(1 + |s|/L^{B_0})^{-j}, \tag{6.54}$$

with fixed B_0. Fourier inversion with $t = \mathcal{T}v$ yields an *exact factor* $1/\mathcal{T}$ and factors $w^{it}(w')^{i(s-t)}$. Including normalized endpoint polynomials and arc measure $d\phi = d\lambda_0/Y$ gives

$$\frac{YX^2}{\mathcal{T}} \cdot \frac{1}{\varphi(H)} \sum_{\chi} \iint \widehat{K}_{\lambda_0}(t/\mathcal{T}, s) Q_\chi(t) Q'_\chi(s - t)\, dt\, ds. \tag{6.55}$$

Here $Q = BMN$, with M^{-1}, N^{-1}, Y^{-1} normalizations as in the reflected proof; the b-phase is $e(-\lambda_0 b/Y)$ and the primed a-phase is its inverse. The precise sign has no effect on the estimates below. Frequency cutoffs can be chosen as

$$|s| \leq L^{B_1}, \qquad |t| \leq \mathcal{T}L^{B_1}, \qquad B_1 = B_0 + 2, \tag{6.56}$$

independently of the requested saving. Higher integration-by-parts order, not a growing B_1, supplies arbitrary fixed precision in the tails.

6.6.2 Hybrid mean value

For arbitrary bounded coefficient polynomials of lengths M, N, their product has coefficients c_u on $u \asymp X$ with

$$\sum_u |c_u|^2 \ll L^{C_2}/X. \tag{6.57}$$

For a fixed character extension ψ and any real interval I of length $T \geq 1$,

$$\frac{1}{\varphi(H)} \sum_{\chi \bmod H} \int_I \left| \sum_u c_u \psi(u)\chi(u) u^{it} \right|^2 dt \ll (T + X/H) \sum_u |c_u|^2. \tag{6.58}$$

Orthogonality leaves a sum of time integrals over residue classes modulo H. Distinct integers in one class, inside a fixed multiple of X, have logarithms spaced at least cH/X. Montgomery–Vaughan's Hilbert-inequality form of the mean value bounds each class by $(T + O(X/H))$ times its coefficient square sum. If a class has only one term its integral is exactly T times that square. Summing proves (6.58). Translation of I multiplies coefficients by phases and changes none of the estimates.

At $T \asymp \mathcal{T} L^{B_1}$, (2.21) and $\mathcal{T} = \max(1, X/H)$ imply the bound L^{C_3}/H. This is also a uniform bound for polynomially weighted tails after multiplication by (6.54). Dyadic decomposition of t and then s, using (6.58) and Cauchy on the primed and unprimed sides, shows that the part outside (6.56) contributes at most L^{-A}/H for any fixed A, by taking j large. The possible factor $H/X \ll L^{B+C}$ is included in the fixed precision budget. The trivial bound $|B| \ll_{K,\rho} 1$ is enough for this tail argument.

6.6.3 Short-prime cancellation under H1

Suppose ξ is nonprincipal modulo $q \leq x^{C_q}$, and $|t| \leq x^{C_t}$ for fixed C_q, C_t. Under H1, uniformly for $P \geq \exp(cL^{a_1})$ and $P \leq \exp(CL^{\theta})$,

$$\sum_{P < p \leq P'} \frac{\xi(p) p^{it}}{p} \ll P^{-1/16} L^{C_4}, \qquad P \leq P' \leq 2P. \tag{6.59}$$

To prove this, pass first to the primitive inducing character. On $\sigma = 15/16$ the partial-fraction formula for L'/L and the zero count in a unit-height interval give $|L'/L(\sigma + iu)| \ll \log(q(|u| + 3))$: every nontrivial zero is at horizontal distance at least $1/16$, and there is no pole for a nonprincipal character. The same bound holds on the contour between $15/16$ and $1 + 1/\log P$. Apply truncated Perron to $-L'/L(s - it, \xi)$ at height $x^{C_t + C_q + 20}$ and move to $15/16$. Taking half-integer endpoints, then restoring an endpoint term if necessary, bounds the truncation error by $O(PL^{O(1)} x^{-C_t - C_q - 20} + L^{O(1)})$. The vertical integral is $O(P^{15/16} L^{O(1)})$, and the horizontal integrals are negligible. This proves the twisted von Mangoldt endpoint bound. Prime powers contribute $O(P^{1/2} L^{O(1)})$. Partial summation with the real weight $1/(u \log u)$ gives (6.59); the phase u^{it} is part of the sum and is never differentiated. Imprimitive Euler factors and puncturing at $p \mid h$ cost $O(L/P)$ on these bands, absorbed by the bound.

To estimate B_{ξ_b}, put $g(y) = y\eta(y)e(-\lambda_0 y)$ and use its Mellin transform. Isolating one prime from the first band gives

$$B_{\xi_b}(t)=\frac{1}{V_1}\int_{\mathbb{R}}\widehat{g}(v)Y^{-iv}\left(\sum_{p\in\mathcal{P}_1}\frac{\xi_b(p)p^{i(t+v)}}{p}\right)\prod_{i=2}^{K}\left(\frac{1}{V_i}\sum_{p\in\mathcal{P}_i}\frac{\xi_b(p)p^{i(t+v)}}{p}\right)dv. \tag{6.60}$$

The product is at most one in modulus, including the empty product when $K=1$. The transform has L^1 norm at most a fixed logarithmic power. Truncate $|v|\le x^2$ using three or more integrations by parts; its tail is $O(x^{-1})$ times a fixed logarithmic factor. On (6.56), $|t+v|\le x^3$ for large x. Decompose $\mathcal{P}_1$ into ordinary dyadic intervals and apply (6.59). Thus for each fixed D,

$$\sup_{|t|\le 2\mathcal{T}L^{B_1}}|B_{\xi_b}(t)|\ll_D L^{-D} \tag{6.61}$$

whenever ξ_b is nonprincipal. Combining (6.61) with (6.58) proves the large-conductor contribution $\ll_A L^{-A}/H$.

6.6.4 Small conductors and sparse-frequency energy

Let $\mathcal{P}\subset[P,2P)$ with $cL^{0.1}\le\log P\le CL^{\theta}$, and

$$P_s(t)=P^{-1}\sum_{p\in\mathcal{P}}v_p\xi_b(p)p^{i(t+v)},\quad |v_p|\le 1, \tag{6.62}$$

where v is any real number. For fixed A_s,B_1, let $\mathcal{I}$ be the unit intervals with integer endpoints meeting $|t|\le 2XL^{B_1}$ and $|P_s(t)|>L^{-A_s}$. Then

$$|\mathcal{I}|\le\exp(O(L^{0.9}\log L)),\qquad \sum_{I\in\mathcal{I}}\sup_{t\in I}|N_\xi(t)|^2\ll L^{C_5}. \tag{6.63}$$

For the first assertion take $T_0=4XL^{B_1}$ and $j=\lfloor\log T_0/\log(2P)\rfloor$. Unique factorization gives coefficient-square mass of P_s^j at most $j!(2/P)^j$, and its length is at most T_0. Choose a large-value point in each interval and divide the intervals into three classes modulo 3, making the points separated. On disjoint unit neighborhoods the inequality

$$|F(t_r)|^2\ll\int_{t_r-1/2}^{t_r+1/2}(|F(u)|^2+|F(u)F'(u)|)du \tag{6.64}$$

follows by integrating the derivative of $|F|^2$. Apply the ordinary Dirichlet-polynomial mean value to $F=P_s^j$ and F', whose coefficients acquire factors $\log n\ll L$. It follows that

$$|\mathcal{I}|\ll T_0L^{O(1)}L^{2A_sj}j!(2/P)^j. \tag{6.65}$$

Here $j\ll L^{0.9}$ and $\log T_0-j\log P\ll\log P+j$, so the logarithm of this bound is $O(L^{0.9}\log L)$. The growing-order factorial has been retained.

For the second assertion choose a maximizing point for $|N_\xi|$ in each closed interval and again separate by three classes. The unweighted evaluation Gram entries are $\sum_{N<n\le 2N}n^{i(t_r-t_s)}$. Put $T_*=\exp(L/(\log L)^2)$. For $1\le|u|\le T_*$, Euler summation and one integration by parts give the normalized bound $O(1/|u|+x^{-c_\delta})$; the discretization error is $O((1+|u|)/N)$. For $T_*<|u|\le x^{C_0}$, the second-derivative estimate when $|u|\le N$ gives $O(\sqrt{|u|}/N+|u|^{-1/2})$. When $|u|\ge N$, choose a fixed integer $k>C_0/\delta$ and the classical exponent pair

$$\kappa_k=(2^{k+2}-2)^{-1},\qquad \lambda_k=1-(k+1)\kappa_k. \tag{6.66}$$

It bounds the normalized logarithmic-phase sum by $O(N^{-(k+2-C_0/\delta)\kappa_k}+N^{-1})$. Thus in the high range the normalized Gram entry is $O(\exp(-cL/(\log L)^2))$. The number of intervals in (6.63) times either of the two error terms tends to zero. Separation gives row sum of $1/|t_r-t_s|$ at most $O(\log(2+|\mathcal{I}|))=O(L)$. The Gram row sums are therefore $O(NL)$. The Hermitian Schur bound and $\sum_n|\beta_n\xi(n)/N|^2\ll L^{O(1)}/N$ prove the second assertion of (6.63), with no dependence on the character conductor or the shift v.

For each small-conductor ξ, split the integral of $|P_sMN|^2$ into $|P_s|\le L^{-A_s}$ and its complement. The first part is $\ll L^{-2A_s+C_6}$ by the ordinary mean value of MN over length $O(XL^{B_1})$. On the second part use (6.43) for M and the second assertion of (6.63) for N, giving $\ll L^{-2A'+C_5}$. Choosing fixed A_s,A' sufficiently large gives every fixed saving. Minkowski applied to (6.60) and the ordinary dyadic prime decomposition then gives

$$\int_{|t|\le 2XL^{B_1}}|B_{\xi_b}(t)M_\xi(t)N_\xi(t)|^2dt\ll_A L^{-A}. \tag{6.67}$$

The argument is uniform in the bounded b-character and in the Mellin shift in (6.60). In (6.43), the extra frequency is the original t plus the Vaughan Mellin twist, not that prime-polynomial shift. Summing (6.67) over at most L^{2A_1+1} small-conductor characters and dividing by $\varphi(H)$ yields $\ll_A L^{-A}/H$, since $H/\varphi(H)\ll\log\log(3H)\ll\log L$ with a harmless constant in the bounded-H cases.

Proposition 6.14 (General affine major arcs)**.** *Under* H1*, with the preceding fixed choices, the affine comparison square has bound* $|Q_Y^{\mathrm{maj}}|\ll_A XYL^{-A}$ *at every fixed precision.*

Proof. Combining both conductor ranges gives

$$\frac{1}{\varphi(H)}\sum_{\chi \bmod H}\int_{|t|\le 2\mathcal{T}L^{B_1}}|B_{\chi\psi\psi'_b}(t)M_{\chi\psi}(t)N_{\chi\psi}(t)|^2dt\ll_A L^{-A}/H. \tag{6.68}$$

The primed endpoint, coefficient conjugations, translated intervals, and reversed signs of the frequency twists satisfy the same estimates. Cauchy in t,χ, then the s integration and the bounded-denominator character/arc sums in Section 6.1, cost only fixed logarithmic powers. Equations (6.55) and (6.68) give

$$|Q_Y^{\mathrm{maj}}|\ll_D\frac{YX^2}{\mathcal{T}H}L^{-D}\le XYL^{-D}, \tag{6.69}$$

because $\mathcal{T}H=\max(X,H)$. The tail estimate in Section 6.6 supplies the same precision. This is the signed comparison form defined in Proposition 5.5, so the bound follows under H1. The principal and low-conductor contribution is controlled by (6.67).

□

6.7 The small-radical input

Input 6.15 (Iwaniec, 1974, Theorem 2)**.** *For* $q\ge 3$*, put* $d=\mathrm{rad}(q)$*,* $\ell=\log(q(|t|+3))$ *and*

$$\vartheta^{-1}=4\cdot 10^4\{\log d+(\ell\log(2\ell))^{3/4}\}. \tag{6.70}$$

There is at most one nonprincipal character modulo q *and one zero* ρ_0 *in* $\Re\rho_0>1-\vartheta$*; if present, the character is real and the zero real and simple [8, Theorem 2, p. 97].*

Lemma 6.16 (Lemma Q). *If* $\operatorname{rad}(\mathcal{N}) \le L^C$ *and* $k \le L^{A_0}$, *every real primitive character of conductor* $q^* \mid k\mathcal{N}$ *has*

$$q^* \le 8\operatorname{rad}(k\mathcal{N}) \le 8L^{A_0+C} \le L^{A_1}, \qquad A_1 = A_0 + C + 1, \quad L \ge 8. \tag{6.71}$$

Proof. A real primitive character is a Kronecker symbol with fundamental discriminant of absolute value m, $4m$ or $8m$, with m odd and squarefree [3, Chapter 5]. Hence $q^* \mid 8\operatorname{rad}(q^*) \mid 8\operatorname{rad}(k\mathcal{N})$. □

Corollary 6.17 (Small-radical region). *For* $\operatorname{rad}(\mathcal{N}) \le L^C$ *and* $A_1 = A_0 + C + 1$, (1.3) *holds for every fixed* $C_{\mathrm{zf},1}, C_{\mathrm{zf},2}$ *once* $L \ge L_0(C_{\mathrm{zf},1}, C_{\mathrm{zf},2}, C, A_0)$.

Proof. For $q^* \mid k\mathcal{N}$, $q^* > L^{A_1}$, apply Input 6.15 with $q = q^*$. Here $\operatorname{rad}(q^*) \le L^{A_0+C}$ and $\ell = \log(q^*(|t|+3)) \le 2L$ at the fixed height $L^{C_{\mathrm{zf},2}}$. Thus

$$\begin{aligned} \vartheta^{-1} &\le 4 \cdot 10^4 \{(A_0 + C)\log L + (2L\log(4L))^{3/4}\}, \\ \vartheta &\ge c_I L^{-3/4}(\log L)^{-3/4} > C_{\mathrm{zf},1}\log L/L \end{aligned} \tag{6.72}$$

for sufficiently large L, with absolute $c_I > 0$ of size 10^{-5}. A real exceptional character would contradict Lemma 6.16, so the exceptional case is absent. The strict width comparison also gives the closed boundary in (1.3). □

The explicit radical dependence is needed because k contributes radical up to L^{A_0} even when $\mathcal{N} = 2^m$; a prime-power region alone does not cover these conductors.

Corollary 6.18 (The analytic alternatives). *Theorem 6.13 supplies* (α^*) *from the selected* H1*, *and therefore from* H1 *for every even* $\mathcal{N}$. *In the class* $\operatorname{rad}(\mathcal{N}) \le L^C$ *it follows from Corollary 6.17. For* $0 < |h| \le L^C$, *all characters have conductor at most* L^{A_0+C}, *but the scale-*1 *Mellin reduction is replaced in Section 11.1 by the reconstructed hybrid small-conductor proof.*

6.8 Changes in version 2: radical criteria

For use on the larger radical classes, the elementary real-conductor bound in Lemma 6.16 can be seen directly. At an odd p^e, the units congruent to one modulo p have odd order, so a real character is trivial on them and factors modulo p. At powers of two, units congruent to one modulo eight are squares; the real conductor has 2-part at most eight. Thus $q^* \le 8\operatorname{rad}(q^*) \le 8\operatorname{rad}(k\mathcal{N})$.

In the remainder of this subsection write N for the even integer $\mathcal{N}$ (rather than a Vaughan polynomial length) and $L = \log N$.

The LS(c) **criterion.** Assume LS(c) as in Theorem F. Fix the desired H1* width and height constants and the arc-denominator exponent A_0, and fix a nonnegative envelope $\eta(L) \to 0$. Uniformly for even N with $L = \log N$ and

$$\log\operatorname{rad}(N) \le \eta(L)\frac{L}{\log L}, \tag{6.73}$$

LS(c) and Iwaniec's 1974 Theorem 2 imply the selected H1*, once N exceeds a threshold depending on the envelope and fixed parameters.

Proof. For a primitive conductor $q^* \mid kN$, $k \le L^{A_0}$,

$$\log\operatorname{rad}(q^*) \le \log\operatorname{rad}(N) + A_0\log L = o(L/\log L) \tag{6.74}$$

uniformly in the class. At the fixed height $|t| \le L^{C_2}$, $\ell = \log(q^*(|t|+3)) \le 2L$ for sufficiently large L. The inverse of Iwaniec's width is

$$4 \cdot 10^4 \{\log \operatorname{rad}(q^*) + (\ell \log(2\ell))^{3/4}\} = o(L/\log L). \tag{6.75}$$

Thus the Iwaniec region strictly contains the desired closed strip of width $C_1 \log L/L$. Its only possible exception is a real simple zero of a real character. For a primitive real character, Lemma 6.16 gives

$$\log q^* \le \log 8 + \log \operatorname{rad}(q^*) = o(L/\log L). \tag{6.76}$$

LS(c) therefore puts its real zero strictly farther from one than $C_1 \log L/L$, uniformly. For H1* only conductors above the fixed threshold L^{A_1} need this argument; they eventually exceed q_0. This removes the exception and proves the closed zero-free strip. □

An explicit envelope is $\eta(L) = 1/\log L$, giving

$$\operatorname{rad}(N) \le \exp\left(\frac{\log N}{(\log\log N)^2}\right). \tag{6.77}$$

For every fixed $a_0 < 1$, the class $\operatorname{rad}(N) \le \exp((\log N)^{a_0})$ is another example. These classes can include $N = 2^m r_m$ with odd primes $r_m \le \exp(m^\delta)$, fixed $\delta < 1$, whose radicals need not be bounded by a fixed power of $\log N$.

For a fixed requested width C_1, a stronger explicit boundary is also permitted: let

$$\kappa = \frac{\min(c, 1/(4 \cdot 10^4))}{8C_1}. \tag{6.78}$$

If $\log \operatorname{rad}(N) \le \kappa L/\log L$, then for large L the extra $A_0 \log L$ and the $3/4$ term can each be bounded by a fixed fraction of $\kappa L/\log L$. Both the Iwaniec width and the real-zero gap exceed the requested width by a fixed strict margin. The exact threshold may depend on κ, A_0, C_1, C_2. This assertion provides one fixed selected precision; it does not supply arbitrary growing precision from one fixed κ.

The Landau–Page criterion. We use the classical theorem in the form recalled in [4, Lemma 7.1]: for an absolute $c_{\mathrm{LP}} > 0$, among primitive conductors at most Q at most one character has a real zero $\beta > 1 - c_{\mathrm{LP}}/\log Q$; such a character is real. This is the only part of that paper used here. Fix $A_0, A_1, C_1, C_2 > 0$, the parameters of the selected H1* interface, and choose

$$0 < \kappa \le \frac{\min(c_{\mathrm{LP}}, 1/(4 \cdot 10^4))}{32C_1}. \tag{6.79}$$

For a dyadic scale X, put $\ell = \log X$ and

$$R_X = \exp(\kappa\ell/\log\ell), \qquad Q_X = 8(2\ell)^{A_0} R_X. \tag{6.80}$$

Let q_X be the conductor of the possible Landau–Page character at Q_X, leaving q_X undefined if there is no such character. For every sufficiently large X, the selected H1* holds uniformly for even $X \le N \le 2X$ with $\operatorname{rad}(N) \le R_X$, provided either the exceptional character does not exist, or $q_X \le \ell^{A_1}$, or

$$\frac{q_X}{\gcd(q_X, N)} > (\log N)^{A_0}. \tag{6.81}$$

Proof. Write $L_N = \log N$, so $\ell \le L_N \le 2\ell$ for large X. For $q^* \mid kN$, $k \le L_N^{A_0}$, one has

$$\log \operatorname{rad}(q^*) \le \kappa \ell / \log \ell + A_0 \log(2\ell). \tag{6.82}$$

At $|t| \le L_N^{C_2}$, the logarithm in Iwaniec's theorem is at most 2ℓ eventually. Since $(2\ell \log(4\ell))^{3/4} = o(\ell/\log \ell)$, the denominator of Iwaniec's width is at most $8 \cdot 10^4 \kappa \ell / \log \ell$ eventually. The resulting width is at least $16 C_1 \log \ell / \ell$, strictly greater than the requested $C_1 \log L_N / L_N$. Thus only a real zero of a primitive real character can remain in the requested closed strip.

Such a character has $q^* \le 8 \operatorname{rad}(kN) \le Q_X$. Also $\log Q_X \le 2\kappa \ell / \log \ell$ eventually. Every character other than the possible Landau–Page character therefore has real-zero distance at least

$$\frac{c_{\mathrm{LP}}}{\log Q_X} \ge 16 C_1 \frac{\log \ell}{\ell}, \tag{6.83}$$

again a strict margin. If the exceptional character does not exist there is nothing to remove. If $q_X \le \ell^{A_1}$, its conductor is at most $L_N^{A_1}$, so it is outside the conductor range required by H1*. In the remaining case, for positive integers q, N,

$$q \mid kN \quad \Longleftrightarrow \quad q/\gcd(q, N) \mid k. \tag{6.84}$$

This follows prime by prime by comparing valuations. Consequently there is a positive integer $k \le L_N^{A_0}$ with $q_X \mid kN$ if and only if $q_X / \gcd(q_X, N) \le L_N^{A_0}$. Condition (6.81) excludes precisely this possibility. Every required conductor is now covered, proving H1*. All margins are uniform on the stated dyadic class. □

Reflected consequence on H2′. The A and D conclusions and their functional rate follow on this dyadic class from Section 11, with Iwaniec, Landau–Page, Burgess and the other named classical inputs. Neither H1 nor LS(c) is required here. This is an explicitly specified exceptional-conductor avoidance class, not an exception-free numerical list: the identity of q_X may be unknown. The scale and the fixed width must be selected before κ.

For reflected sign consequences the requested C_1, C_2 are the fixed energy precision required by Theorem 6.13 and Section 10; κ is chosen afterwards. This proves both analytic criteria of Theorem F. The LS criterion supplies every fixed requested precision for each prescribed envelope; its threshold may depend on that envelope. At conductor comparable with N, a gap $c/\log q$ alone does not give a width $C_1 \log\log N / \log N$ or control complex zeros. The reflected consequences are on the displayed classes only.

6.9 The classical-region obstruction

Remark 6.19 (A separate contour estimate). Fix $C_t \ge 2$. Under (1.2), formula (5.15) of LWW, p. 44, separates a prime in $\mathcal{P}_1$ from B_{ξ_b}. Lemma 3.5 of [5], followed by dyadic summation and partial summation, gives

$$\sum_{p \in \mathcal{P}_1} \frac{\xi_b(p) p^{it'}}{p} \ll L^{C_{\mathrm{prime}}+1} \exp(-L^{a_1}/16) \tag{6.85}$$

uniformly for $|t'| \le \mathcal{N}^{C_t}$ and nonprincipal ξ_b of modulus $k\mathcal{N} \le \mathcal{N}^2$. Here C_{prime} is fixed, depending on C_t and the fixed exponent ranges. The cited Perron argument uses $\Re s = 15/16$ and $L'/L \ll \log^2(q(|t|+3))$.

For the classical real-zero lower bound $1 - \beta_1 \gg_\varepsilon q^{-\varepsilon}$, even at $q \le \exp(L^{a_1/2})$ the guaranteed lower bound for $(1-\beta_1)L^{a_1}$ can tend to zero. It gives no divergence to infinity, and leaves a possible exceptional conductor in $(L^C, \exp(L^{a_1/2})]$ outside this argument.

Proposition 6.20 (Obstruction (EC)). *Let $\chi \bmod q$ be real primitive, with $q \mid \mathcal{N}$. For $p \nmid \mathcal{N}$ one has*

$$\mu(\mathcal{N}-p) = \chi(-1)\chi(p)\varepsilon(\mathcal{N}-p)\mu^2(\mathcal{N}-p), \qquad \varepsilon(n) = \prod_{p'\mid n}(-\chi(p')). \tag{6.86}$$

Suppose every split prime ($\chi(p')=1$) at most $\log q$ divides $\mathcal{N}$, and at most $\pi(\mathcal{N})/20$ primes $p<\mathcal{N}$ have either $\chi(p)=1$ or a split prime factor exceeding $\log q$ in $\mathcal{N}-p$. Then, for sufficiently large such $\mathcal{N}$,

$$\left|\sum_{p<\mathcal{N}} \mu(\mathcal{N}-p)\right| \geq 0.27\pi(\mathcal{N}). \tag{6.87}$$

Proof. For squarefree n coprime to q, $\varepsilon(n)=\mu(n)\chi(n)$; for nonsquarefree n both sides of (6.86) vanish after multiplication by $\mu^2(n)$. Also $\chi(\mathcal{N}-p)=\chi(-1)\chi(p)$ when $p\nmid\mathcal{N}$, proving the identity. Outside the stated exceptional set, $\chi(p)=-1$. No split prime can divide $\mathcal{N}-p$: a small split prime divides $\mathcal{N}$ and would then divide p, contrary to $p\nmid\mathcal{N}$; a larger split prime is excluded by hypothesis. Thus $\varepsilon(\mathcal{N}-p)=1$, and each squarefree such value has $\mu(\mathcal{N}-p)=-\chi(-1)$. Section 9, whose proof uses only Siegel–Walfisz, gives $(A_{\mathcal{N}}+o(1))\pi(\mathcal{N})$ squarefree values uniformly. Removing the exceptional set loses at most $\pi(\mathcal{N})/20$ of them, and their possible opposite contribution loses another $\pi(\mathcal{N})/20$. The $O(\omega_{\mathrm{all}}(\mathcal{N}))$ nonunit primes are negligible. Hence the left side of (6.87) is at least $(0.3739-2/20+o(1))\pi(\mathcal{N}) > 0.27\pi(\mathcal{N})$. □

This proposition establishes an obstruction under its split-prime hypotheses. The classical lower bound $1-\beta \gg_\varepsilon q^{-\varepsilon}$ does not by itself exclude that scenario for a large conductor dividing $\mathcal{N}$. The possible zero-size condition $(1-\beta)\log\mathcal{N}\to 0$ does not replace the separate split-prime hypotheses in this proposition. It identifies why the present uniform argument uses (1.3). In the small-radical class all the real conductors dividing $k\mathcal{N}$ are small by Lemma 6.16.

Remark 6.21 (Changes in version 2: the scope of EC). The split-prime assumptions in Proposition 6.20 remain separate from a zero assumption. Under H1, the Perron argument at 15/16 in Section 6.6.3, applied up to $\mathcal{N}$, gives $\sum_{p<\mathcal{N}}\chi(p) = O(\mathcal{N}^{15/16}(\log\mathcal{N})^{O(1)})$ uniformly for primitive nonprincipal conductors at most $\mathcal{N}$. For a real such character, half the primes are therefore split, up to $o(\pi(\mathcal{N}))$; this already excludes EC's bound $\pi(\mathcal{N})/20$. A sufficiently strong selected H1* gives the corresponding conclusion through Lemma 6.2, with small conductors treated by Siegel–Walfisz. The weaker-input statements apply only on their specified classes. The LS criterion supplies the required wider strip on its prescribed radical envelope; the avoidance criterion excludes $q_X \mid \mathcal{N}$ by (1.14). For D at $z=2$, $V_{\mathcal{N}}(2)=1$ and $U_{\mathcal{N}}(2)=A_{\mathcal{N}}$, apart from the possible prime 2 in passing to full roughness.

7 Type I and the common divisor

7.1 Finite presieving of the target

Remark 7.1 (Changes in version 2: Type I). This section replaces the version 1 reflected Type I proof by a proof for both orientations. Its common-divisor extraction is the same; the finite presieve and the endpoint/Rankin bounds below also cover the rough target. The retained exponents 2/5 and 1/20 refer to the large variable, not to the marking-band ceiling.

Assume h is even and $2\leq z\leq \exp(L^a)$, $a<1/2$. Let $\mathcal{Q}=\{r\in\mathbb{P}: r\leq z, r\nmid h\}$; it consists of odd primes. For an integer $R_z\geq 0$ put

$$S_{R_z}(n) = \sum_{\substack{e|n,\ e|P_h(z)\\ \omega_{\mathrm{all}}(e)\le R_z}} \mu(e), \qquad E_{R_z}(n) = \sum_{\substack{e|n,\ e|P_h(z)\\ \omega_{\mathrm{all}}(e)= R_z+1}} 1. \tag{7.1}$$

The binomial identity $\sum_{j=0}^{R}(-1)^j\binom{u}{j} = (-1)^R\binom{u-1}{R}$ for $u>0$ gives

$$|\mathcal{R}_{z,h}(n) - S_{R_z}(n)| \le E_{R_z}(n). \tag{7.2}$$

For $u \le R$ the error is zero; for $u > R$, $\binom{u-1}{R} \le \binom{u}{R+1}$. This proves the pointwise inequality, including $u = 0$. The same bound works in the independent divisor model with local probabilities $1/(r-1)$, because h is coprime to every sieve prime. In particular

$$\sum_{\substack{e|P_h(z)\\ \omega_{\mathrm{all}}(e)=R_z+1}} \frac{1}{\varphi(e)} \le \frac{(\sum_{r\in\mathcal{Q}} 1/(r-1))^{R_z+1}}{(R_z+1)!}. \tag{7.3}$$

The numerator's base is at most $a\log L + O(1)$. Given a fixed desired precision A, choose a sufficiently large fixed C_z and $R_z = \lceil C_z \log L\rceil$. Stirling's elementary lower bound $R! \ge (R/e)^R$ makes (7.3) $O_A(L^{-A})$. Every retained e obeys

$$e \le z^{R_z+1} \le \exp(O_A(L^a \log L)) = x^{o(1)}. \tag{7.4}$$

This is uniform for all z in the stated range, including fixed z. Also

$$c/\log z \le V_h(z) \le 1, \qquad V_h(z) \gg_a L^{-a}. \tag{7.5}$$

Indeed omitting factors increases the product, and for odd primes

$$\frac{r-2}{r-1} = (1-1/r)\left(1 - \frac{1}{(r-1)^2}\right). \tag{7.6}$$

Mertens' product estimate and convergence of the second positive product prove (7.5). Thus any error $O_A(xL^{-A})$ with $A > a$ is negligible relative to $xV_h(z)$.

7.2 Type I for both orientations

Proposition 7.2 (Uniform Type I)**.** *For the function F below, both Type I terms of Proposition 2.3 have bound $O_A(yL^{-A})$ at every fixed precision, in the A range and, for the rough target, in the D range. The same is true for the Liouville target.*

Proof. Let $f = \mu$ or λ, and use the bounded function

$$F(n) = f(\varepsilon n + h)\mathcal{R}_{z,h}(\varepsilon n + h)W(\varepsilon n + h)\,1_{\varepsilon n + h \ge 1}. \tag{7.7}$$

For the unrough version omit $\mathcal{R}_{z,h}$. On a prime dyad of scale $y \ge x/L^B$, the assertion to prove is $T_1(F), T_2(F) \ll_A yL^{-A}$ at each fixed precision.

Write the outer Type I variable as $q \le y^{2/5}$ with $R_Z(q) = 1$ and coefficient $|\alpha_q| \le O(L)$. Discard q having a common prime with h. Such a prime is greater than Z, and summing reciprocal multiples costs $O(yL^3\omega_{\mathrm{all}}(h)/Z)$; the harmless endpoint counts are of smaller power size. This is negligible at every fixed logarithmic precision. Separate k odd first, and expand the factors at odd primes in the inner roughness $R_Z(k)$ at depth $R_I = \lceil C_I \log L\rceil$, obtaining odd squarefree $r \le Z^{R_I+1} = x^{o(1)}$ with $r \mid k$. Keep k odd. Expand W at depth $R_W = \lceil C_W \log L\rceil$ as

$$W(n) = \sum_{d|n} c_d + \text{remainder}, \tag{7.8}$$

where d is squarefree, uses only selected band primes, $d = \exp(O_K(R_W L^\theta)) = x^{o(1)}$, $\sum_d |c_d|/d \ll_{K,\rho} 1$, and the remainder is bounded by a positive finite divisor sum with reciprocal coefficient mass $O_{K,\rho}(1/R_W!)$. Here is the finite expansion used at this interface. In a single band, writing $u = \omega_i(n)$ and $V = V_i$, Taylor's formula for $(1-t)^{u-1}$ at $t = 0$ gives

$$\frac{u\rho^{u-1}}{V} = \frac{1}{V}\sum_{j=0}^{R_W-1} (j+1)(\rho-1)^j \sum_{\substack{d|n,\ p|d\Rightarrow p\in\mathcal{P}_i \\ d \text{ squarefree},\ \omega_{\mathrm{all}}(d)=j+1}} 1 + \mathcal{E}_i(n). \tag{7.9}$$

Indeed $u\binom{u-1}{j} = (j+1)\binom{u}{j+1}$, and

$$|\mathcal{E}_i(n)| \le \frac{(R_W+1)(1-\rho)^{R_W}}{V} \sum_{\substack{d|n,\ p|d\Rightarrow p\in\mathcal{P}_i \\ d \text{ squarefree},\ \omega_{\mathrm{all}}(d)=R_W+1}} 1. \tag{7.10}$$

The reciprocal coefficient mass of this positive remainder is $O(1/R_W!)$, also with $1/\varphi(d)$ in place of $1/d$, because both band sums are eventually bounded above by one. The absolute reciprocal mass of a retained single-band polynomial is bounded by a fixed constant. Multiplying the K expansions and telescoping their errors proves the assertions for W. This supplies directly the finite identity corresponding to LWW (2.17) and its Type I truncation.

Insert (7.2) for the target roughness. The sieve divisor e and list divisor d are coprime, since every band prime exceeds z, and $D = de$ is squarefree and coprime to h. All of its primes are below Z. A term with $(r, D) > 1$ is impossible: a common prime divides both qk and $\varepsilon qk + h$, hence divides h, contrary to $(D, h) = 1$. For the other terms the conditions $r \mid k$, $D \mid \varepsilon qk + h$, and k odd give one class modulo $2rD$, because $(q, D) = 1$. Their count is

$$O(y/(qrD) + 1). \tag{7.11}$$

Use (7.11) for every finite Taylor or Bonferroni error. The reciprocal mass in r is $O(\log Z)$, in e is $O(\log z)$, and in d is bounded. A tail of depth $C \log L$ has any requested fixed logarithmic saving by (7.3) or the Taylor factorial bound. All $+1$ terms total $x^{2/5+o(1)}$, since the total absolute coefficient mass and the number of sieve divisors are $x^{o(1)}$. Choose the depths to absorb the fixed logarithmic factors. This proves the required truncation error estimate without summing over divisors of the full primorial.

Now put $k = rj$ and $g = (r, h)$. The integer g is odd and squarefree, $g \mid \varepsilon qrj + h$, and $(D, g) = 1$. For Möbius, the exact local-factor identity gives

$$\mu(Dgv) = \mu(Dg) \sum_{\substack{b|v \\ p|b\Rightarrow p|Dg}} \mu(v/b). \tag{7.12}$$

Here the displayed support means every prime dividing b divides Dg; arbitrary prime powers in b are allowed. At a prime dividing Dg, the two potentially nonzero terms cancel when that prime divides v, proving the identity prime by prime. Thus write

$$\varepsilon qrj + h = Dgbt, \qquad t \equiv (h/g)(Db)^{-1} \pmod{qr/g}. \tag{7.13}$$

The inverse exists: D is disjoint from qr, while the primes of g have been removed from the squarefree r and are below every prime dividing q. Also $(h/g, qr/g) = 1$, since q was made coprime to h and $(r/g, h) = 1$. Hence (7.13) is reduced. There is no extra requirement $(g, Db) = 1$. When h is even, j odd is equivalent to t odd, giving a reduced class modulo $2qr/g$. The positive interval for t is contained in

$$[1, Y_0], \qquad Y_0 = (|h| + 3y)/(Dgb). \tag{7.14}$$

The sign ε changes the order of the interval endpoints, not its modulus or reducedness.

For $Dgb \le y^{1/20}$, one has $Y_0 \ge y^{19/20}$ and $\log Y_0 \asymp L$. Moreover $2qr/g \le y^{2/5+o(1)}$, below the Bombieri–Vinogradov level $Y_0^{1/2} L^{-B_{\mathrm{BV}}}$ for every fixed B_{BV}, once x is sufficiently large. For fixed g, D, b, group by qr/g. The coefficient is at most $O(L\tau(qr/g))$; in fact the rough/small-prime factorization is unique. The weighted reduced-class Möbius estimate follows from ordinary BV by

$$\Big(\sum_{m \le Q} \tau(m) E(Y_0; m) \Big)^2 \le \Big(\sum E(Y_0; m) \Big) \Big(\sum \tau(m)^2 E(Y_0; m) \Big), \tag{7.15}$$

using $E(Y_0; m) \ll Y_0/m + 1$ and $\sum \tau(m)^2/m \ll L^4$. At sufficiently large fixed BV precision this gives $\ll Y_0 L^{-A}$ for the weighted sum. Sum *separately for each* g, since the level depends on it, and then use

$$\sum_{g \mid \mathrm{rad}(h)} 1/g \ll \log L, \qquad \sum_{b:\ p \mid b \Rightarrow p \mid Dg} 1/b \le \prod_{p \le Z} (1 - 1/p)^{-1} \ll \log Z. \tag{7.16}$$

Together with $\sum_e 1/e \ll \log z$, $\sum_d |c_d|/d \ll 1$, and $(|h| + 3y)/y \ll L^{B+C}$, these are only fixed logarithmic losses.

For $Dgb > y^{1/20}$, the factors $Dg = x^{o(1)}$ imply $b > y^{1/25}$ for sufficiently large x. Rankin at exponents $1/2$ and $1/20$ gives

$$\sum_{b > y^{1/25}} \frac{1}{b} \ll y^{-1/50} L^{O(1)}, \qquad \#\{b \le |h| + 3y : p \mid b \Rightarrow p \mid Dg\} \ll y^{1/20} L^{O(1)}. \tag{7.17}$$

To justify the products, Dg has $O_{K,A}(\log L)$ distinct prime divisors, all odd; each factor $(1 - p^{-\eta})^{-1}$ is bounded by a constant depending only on fixed $\eta > 0$. Thus the whole factor is $L^{O_{K,A}(1)}$. Counting (7.13) absolutely gives $O(y/(Dbqr) + 1)$. The reciprocal term has the displayed power saving; the endpoint term totals $y^{9/20+o(1)}$. Both absorb every fixed logarithmic loss. This completes the Möbius Type I bound. Partial summation for the factor $\log k$ supplies T_1 with one additional fixed power of L.

For Liouville use $\lambda(Dgt) = \lambda(Dg)\lambda(t)$, so there is no b-sum and $Dg = x^{o(1)}$ always lies in the small-divisor branch. The needed Liouville BV follows from

$$\lambda(n) = \sum_{b^2 \mid n} \mu(n/b^2). \tag{7.18}$$

For a reduced class modulo m, $(b, m) = 1$ and the inner Möbius class is $c\bar{b}^2 \pmod m$. Truncate at $b \le L^{A+2}$. Use Möbius BV at scale Y_0/b^2 with a larger fixed logarithmic level exponent. For the tail summed over $m \le Q$ use $Y_0/(b^2 m)+1$, obtaining $O(Y_0 L/L^{A+2} + Q\sqrt{Y_0})$; choose $Q \le \sqrt{Y_0} L^{-A-2}$ in addition to the head's BV restriction. The same Cauchy/divisor-square estimate gives the τ-weighted version. This proves the Liouville Type I bound.

For the unrough A statement with *odd* h, take $e = 1$. Equation (7.13) forces even t. Write $t = 2t'$; for Möbius use $\mu(2t') = -\mu(t')1_{t' \text{ odd}}$ and a reduced class modulo $2qr/g$. For Liouville use $\lambda(2t') = -\lambda(t')$, with no parity restriction, and modulus qr/g. All inverses exist because these moduli apart from the explicit 2 are odd. This supplies the odd-shift Type I case.

□

8 Removal of the auxiliary weights

8.1 Positive moments on the eligible prime interval

In this section $L = \log x$ and I is the eligible interval in $[1, x]$. For D retain its even-shift and fixed-$a < 1/2$ conditions. For A omit every low-prime sieve factor and put $V = 1$; no parity assumption on h is then needed.

Remark 8.1 (Changes in version 2: relative measures). The version 1 reduced-class mean and fixed-number variance are used below with a finite positive presieve. The variance retains the factor $V_h(z)$, and the later Cauchy inequality uses the rough prime measure. The last subsection replaces the fixed-b limit by a uniform construction with growing b_x.

Let I be the interval of eligible primes in $[1, x]$; endpoints depend on ε, h. Put $\Theta_I = \sum_{p\in I} \log p$. For $F = 1$, W_j, or W_jW_k with $j \neq k$, the assertion is

$$\sum_{p\in I} (\log p)\mathcal{R}_{z,h}(\varepsilon p + h)F(\varepsilon p + h) = V_h(z)m_F\Theta_I + O_A(xL^{-A}), \tag{8.1}$$

where $m_1 = 1$, $m_{W_j} = m_j$, and $m_{W_jW_k} = m_jm_k$. The bound is uniform in z, h and in both endpoints, for each fixed band array.

For completeness, the independent local divisor model assigns $r \mid \varepsilon p + h$ probability $1/(r-1)$ at any odd $r \nmid h$. The CRT proves independence at distinct primes, because the affine coefficient ε is a unit and the required residue is reduced. Every band avoids h, and different bands are disjoint. Consequently

$$m_j = \prod_{i=1}^{K} \left[\frac{1}{V_i} \sum_{r\in\mathcal{P}_i} \frac{1}{r-1} \prod_{\substack{q\in\mathcal{P}_i \\ q\neq r}} \left(1 - \frac{1-\rho}{q-1}\right) \right] = 2^{K(\rho-1)} + o(1). \tag{8.2}$$

The rough primes and all band primes are disjoint, so the local-model average of $\mathcal{R}_{z,h}F$ is exactly $V_h(z)m_F$.

To pass to primes, use the finite presieve (7.2) and the finite Taylor divisor expansions from Section 7.2, now with at most $2K$ bands. A retained product modulus is $ed = x^{o(1)}$ and coprime to h; its prime class is $p \equiv -\varepsilon h \pmod{ed}$. The modulus uniquely determines its low-prime factor e and band factor d. The Taylor coefficient is bounded by a fixed power of $\log L$, so BV with a larger fixed logarithmic precision applies to the sum over all retained moduli, and is uniform in the two interval endpoints. Prime powers and the few primes dividing h cost $x^{1/2+o(1)}$ and $x^{o(1)}$, respectively, after all finite coefficient sums.

The Taylor remainder has reciprocal coefficient mass $O(1/R_W!)$ also with $1/\varphi(d)$ instead of $1/d$, since each band's sum $\sum 1/(r-1)$ tends to $\log 2 < 1$. The roughness remainder multiplied by the absolute truncated Taylor polynomial has local-model mass bounded by (7.3) times $\sum_d |c_d|/\varphi(d) \ll_{K,\rho} 1$. Apply BV to these positive finite error sums as well. Choosing the truncation

depths and BV precision proves errors $O_A(xL^{-A})$. Replacing the interval length by Θ_I uses the endpoint-uniform PNT error, again with enough logarithmic precision. This establishes (8.1).

Each W_j/m_j is bounded by a constant depending on K, ρ but not on b, after increasing the threshold for the fixed array. Thus its square is at most a fixed constant times itself. With

$$\mathcal{W}_b(n) = b^{-1} \sum_{j=1}^{b} W_j(n)/m_j, \tag{8.3}$$

expand the square and apply (8.1). The cross terms for $j \neq k$ are $V_h(z)\Theta_I$ after normalization. The first moments and constant term have the same mean. The diagonal terms are bounded as just described. Hence

$$\sum_{p\in I} (\log p)\mathcal{R}_{z,h}(\varepsilon p + h)|\mathcal{W}_b(\varepsilon p + h) - 1|^2 \leq \frac{C_{K,\rho}}{b} V_h(z)\Theta_I + O_{A,b}(xL^{-A}). \tag{8.4}$$

The constant $C_{K,\rho}$ is independent of b and of the band locations. This is the relative variance needed for D; the factor $V_h(z)$ has not been discarded.

Proposition 8.2 (Normalized fixed-number weight removal)**.** *For any fixed b, normalized weights $\mathcal{W}_b = b^{-1}\sum_j W_j/m_j$ satisfy (8.4). For a bounded target, Cauchy–Schwarz with measure $(\log p)\mathcal{R}_{z,h}(\varepsilon p + h)$ gives*

$$\left| \sum_{p\in I} (\log p) f(\varepsilon p + h)\mathcal{R}_{z,h}(\varepsilon p + h)(1 - \mathcal{W}_b(\varepsilon p + h)) \right| \ll xV_h(z)b^{-1/2} + o_b(xV_h(z)). \tag{8.5}$$

For the unrough statement replace $V_h(z)$ by one. This includes the version 1 fixed-b conclusion when $x = h = \mathcal{N}$ and $\varepsilon = -1$.

Proof. The total mass of this measure is $V_h(z)\Theta_I + O_A(xL^{-A})$. Multiply its square root by the square root of (8.4); use $\Theta_I \ll x$ and $V_h(z) \gg L^{-a}$ with a sufficiently large fixed precision. The constant is independent of b. □

8.2 A common threshold for separated band families

Fix K, $\alpha > 1/10$, an admissible ceiling θ, and $\delta_b > 0$. Permit the individual exponents to vary with x subject to $a_1 \geq \alpha$, $a_K \leq \theta$ and $a_{i+1} - a_i \geq \delta_b$. All weighted estimates above have a common threshold and implied constant on this family.

The dependence on band locations is controlled as follows. Band masses converge uniformly because every lower endpoint exceeds $\exp(L^\alpha)$; puncturing costs $O(L\exp(-L^\alpha))$. Maximum product sizes, root moduli, coefficient masses, and boundary derivatives use only the fixed ceiling. Minimum-prime and overlap bounds use only α. The sole hierarchy comparison in the second good-state test is bounded explicitly by

$$\log \#\{DZ\} \leq C_K \log L + C_K L^{a_{i_*} - \delta_b}, \tag{8.6}$$

so its ratio to $L^{a_{i_*}}$ tends to zero uniformly. All finite memory identities and the positive bound in Section 5.3 have constants depending only on the fixed normalization bounds and operator parameters. Prime-polynomial contours and mean values are uniform in their endpoints. In Type I and the auxiliary moments, at most K or $2K$ groups occur in any one estimate; their coefficient bounds and factorial tails are uniform. Cross moments between two different copies require disjoint

bands, not a fixed gap between those copies. The reflected weak-input interface has the same uniform large-value margin, as checked in the last item in Section 5.3. This accounts for every use of band locations in the cited arguments.

8.3 Packing and the explicit rate

Use $a = 0$ for the unrough statement. After fixing K, choose an admissible $\theta > \max(a, 1/10)$ and put

$$g = \frac{\theta - \max(a, 1/10)}{4K + 4}, \quad \alpha_i = \max(a, 1/10) + (2i - 1)g, \quad b_x = \left\lfloor \frac{g \log L}{\log 4} \right\rfloor. \tag{8.7}$$

For $1 \le j \le b_x$ choose the ith band by logarithmic endpoints

$$t_{i,j} = 4^{j-1} L^{\alpha_i}, \qquad \exp(t_{i,j}) < p \le \exp(2t_{i,j}), \quad p \nmid h. \tag{8.8}$$

Its exponent $a_{i,j} = \alpha_i + (j - 1) \log 4 / \log L$ lies in $[\alpha_i, \alpha_i + g]$. Each weight has consecutive exponent gaps exactly $2g$. Across different i, the neighborhoods are separated by g; within one i, adjacent logarithmic lower endpoints have ratio four, exceeding the previous band's upper/lower ratio two. Thus all $b_x K$ bands are disjoint for sufficiently large x, and each weight belongs to one fixed separated family. The marking primes exceed the rough threshold. Only the number of copies grows; K and every analytic precision are fixed.

Apply (8.4) with uniform errors to these copies. The b_x^2 mixed errors are divided by b_x^2; averaging the uniformly small marked sums also loses no factor b_x. Since $b_x \asymp \log L$, the positive-measure Cauchy argument gives

$$\left| \sum_{p \in I} (\log p) f(\varepsilon p + h) \mathcal{R}(p) \right| \ll \frac{xV}{\sqrt{\log L}} + xL^{-A} + \sqrt{Vx \cdot xL^{-A}}, \tag{8.9}$$

where $V = 1$ or $V_h(z)$. With $V \gg L^{-a}$, a sufficiently large *fixed* A absorbs the final terms. Low primes and removal of the prime logarithm contribute relative error $O(L^{a-1} \log L + L^{a+1-B})$, smaller than $(\log L)^{-1/2}$ for every fixed $a < 1/2$ and sufficiently large fixed B.

Consequently the final errors are

$$O_C\left(\frac{\pi(x)}{\sqrt{\log\log x}} \right) \quad \text{in A}, \qquad O_{a,C}\left(\frac{V_h(z)\pi(x)}{\sqrt{\log\log x}} \right) \quad \text{in D}. \tag{8.10}$$

The sign formulas use the exact main terms already computed, and have the same errors. For Möbius relative to the whole rough population, replacing $U_h(z)$ by $V_h(z)$ adds $O(1/z)$ to the relative error. Constants and thresholds remain ineffective through the named classical inputs. This is an explicit functional rate, not a numerical effective estimate.

The original fixed-b argument alone did not prove (8.10). The additional uniformity check and explicit packing are essential. No prime distribution estimate is applied at the full product of all marking primes; each analytic moment uses only its finite truncated divisors from at most two weight copies.

9 Squarefree values

Proposition 9.1 (Uniform squarefree density)**.** *For even $\mathcal{N} \to \infty$,*

$$\sum_{\substack{p<\mathcal{N}\\ p\nmid\mathcal{N}}} \mu^2(\mathcal{N}-p)\log p = A_{\mathcal{N}}\mathcal{N} + O_A(\mathcal{N}L^{-A}), \qquad A_{\mathcal{N}} \geq 0.3739. \tag{9.1}$$

Consequently

$$\#\{p<\mathcal{N} : \mu^2(\mathcal{N}-p)=1\} = (A_{\mathcal{N}}+o(1))\pi(\mathcal{N}), \tag{9.2}$$

uniformly in even $\mathcal{N}$.

Proof. Use $\mu^2(n)=\sum_{d^2\mid n}\mu(d)$. If $p\nmid\mathcal{N}$ and $d^2\mid\mathcal{N}-p$, then $(d,\mathcal{N})=1$. For $d\leq L^{A+2}$, Siegel–Walfisz in the reduced class $p\equiv\mathcal{N} \pmod{d^2}$, at a sufficiently large fixed precision, gives main term

$$\mathcal{N}\sum_{(d,\mathcal{N})=1}\frac{\mu(d)}{\varphi(d^2)} = A_{\mathcal{N}}\mathcal{N}, \tag{9.3}$$

with extension error $O(\mathcal{N}L^{-A-1})$. The tail has the finite range required by $\mathcal{N}-p>0$:

$$\sum_{L^{A+2}<d\leq\sqrt{\mathcal{N}}}(\mathcal{N}/d^2+1)L \ll \mathcal{N}L^{-A-1}+\sqrt{\mathcal{N}}L. \tag{9.4}$$

The last term is absorbed in $O_A(\mathcal{N}L^{-A})$. This is LWW Proposition 6.1 and (6.3), pp. 46–47, with the size and reduced-class checks for the varying shift supplied here. Primes dividing $\mathcal{N}$ cost $O(\omega_{\mathrm{all}}(\mathcal{N})L)=O(L^2)$. To pass to ordinary counts, use Lemma 9.2 below with the bounded target $\mu^2(\mathcal{N}-p)$. The same progression proof permits any endpoint $y\leq\mathcal{N}$. □

Lemma 9.2 (Removing the prime logarithm)**.** *For any bounded target $|f(p)|\leq 1$ on primes below $\mathcal{N}$, possibly depending on $\mathcal{N}$, and fixed $B>2$,*

$$\sum_{p<\mathcal{N}} f(p) = \frac{1}{L}\sum_{p<\mathcal{N}} f(p)\log p + O_B\left(\frac{\mathcal{N}\log L}{L^2}+\frac{\mathcal{N}}{L^B}\right). \tag{9.5}$$

Proof. For $p>\mathcal{N}/L^B$, $\log p/L = 1+O_B(\log L/L)$. Chebyshev's bound gives $O(\mathcal{N}/L)$ such primes. For the remaining primes bound both sums by their number, at most $\mathcal{N}/L^B$; this proves the identity without varying the target sequence in a partial-summation limit. □

9.1 Changes for general h: positive eligible counts

This assertion uses only the classical inputs. Uniformly for the entire A range,

$$\#\{p\in\mathcal{P}_{\varepsilon,h}(x) : \mu^2(\varepsilon p+h)=1\} = A_h\Pi_{\varepsilon,h}(x)+o(\pi(x)). \tag{9.6}$$

Remove primes dividing h, at most $O(L)$ terms. For the others, $d^2\mid\varepsilon p+h$ forces $(d,h)=1$. Truncate the square-divisor expansion at $d\leq L^D$; the unweighted tail is at most

$$\sum_{L^D<d\leq\sqrt{x+|h|}}(x/d^2+1) \ll xL^{-D}+\sqrt{xL^C}, \tag{9.7}$$

which is $o(\pi(x))$ for fixed $D > 2$. For the head, apply Siegel–Walfisz to $p \equiv -\varepsilon h \pmod{d^2}$, a reduced class, at both endpoints of the eligible interval. The modulus is a fixed power of $\log x$; arbitrarily large fixed saving absorbs the L^D divisors. Replace the interval's logarithmic-integral mass by its actual prime count using the same endpoint-uniform PNT estimate. Extending the absolutely convergent series gives

$$\sum_{(d,h)=1} \frac{\mu(d)}{\varphi(d^2)} = \prod_{r\nmid h}(1 - 1/(r(r-1))) = A_h, \tag{9.8}$$

with an admissible tail. Restoring the removed primes changes only $O(L)$ terms. This proves (9.6), including odd h and empty eligible intervals.

The same argument with a larger fixed truncation exponent gives any requested error $O_A(xL^{-A})$ in the unweighted count. In particular its error is smaller than $\pi(x)E(x)$.

9.2 Rough and squarefree-rough populations

The positive population formula follows from (8.1) with $F = 1$ and the same removal of $\log p$:

$$\#\{p \in \mathcal{P}_{\varepsilon,h}(x) : P^-(\varepsilon p + h) > z\} = V_h(z)\Pi_{\varepsilon,h}(x) + o(V_h(z)\pi(x)). \tag{9.9}$$

For squarefree rough values use $\mu^2(n) = \sum_{d^2\mid n}\mu(d)$. Remove the $O(\omega_{\mathrm{all}}(h))$ primes dividing h. On the remainder any contributing d is coprime to h, and roughness forces $(d, P_h(z)) = 1$. Truncate at $d \le L^D$ with D a sufficiently large fixed number. Since $1 \le n \le x + |h| \ll xL^C$, the weighted discarded tail is

$$\ll L \sum_{L^D < d \le \sqrt{x+|h|}} (x/d^2 + 1) \ll xL^{1-D} + \sqrt{xL^C}\, L. \tag{9.10}$$

It is smaller than any prescribed fixed logarithmic error by choosing D suitably. For the head expand roughness using (7.2). Moduli are $d^2e = x^{o(1)}$, with $(d, e) = 1$, and the reduced prime residue is $-\varepsilon h$. The modulus has the unique decomposition into its square part d^2 and its squarefree part e; thus the sum of BV discrepancies has no uncontrolled multiplicity. Positive Bonferroni tails are bounded by (7.3) times $\sum_d 1/\varphi(d^2) \ll 1$. Restore the finitely many omitted primes only after bounding the total finite divisor coefficient mass by $x^{o(1)}$; their error is negligible.

The main series factors as

$$V_h(z) \sum_{(d,hP_h(z))=1} \frac{\mu(d)}{\varphi(d^2)} = U_h(z). \tag{9.11}$$

Extending the d-series has error $O(L^{-D+1})$, since $d/\varphi(d) \ll \log(2d)$ suffices. Consequently

$$\#\{p \in \mathcal{P}_{\varepsilon,h}(x) : P^-(\varepsilon p + h) > z,\ \mu^2(\varepsilon p + h) = 1\} = U_h(z)\Pi_{\varepsilon,h}(x) + o(V_h(z)\pi(x)). \tag{9.12}$$

The proofs of (9.9) and (9.12) give arbitrarily strong fixed-logarithmic weighted remainders before removing the prime logarithm. After that removal their relative errors are bounded by $O(L^{a-1}\log L)$ plus arbitrarily small fixed powers of L^{-1}. They are therefore smaller than $E(x)$ throughout the final fixed-$a < 1/2$ range, not merely unspecified little-o errors.

For $\kappa = \pm 1$, the identities $1_{\mu(n)=\kappa} = (\mu^2(n) + \kappa\mu(n))/2$ and $1_{\lambda(n)=\kappa} = (1 + \kappa\lambda(n))/2$, multiplied by the roughness indicator, prove the sign formulas from (1.11), (9.9), and (9.12), at the precision of Section 8.3. Finally

$$0 \le 1 - U_h(z)/V_h(z) \le \sum_{r>z} \frac{1}{r(r-1)} \ll 1/z, \tag{9.13}$$

uniformly in h. This is the exact reason both signs have density $1/2$ within the rough population when $z \to \infty$ and the eligible interval has a fixed positive proportion of the primes.

9.3 The progression local factors

Fix $C, C_q, A > 0$. Uniformly for $0 < |h| \le x(\log x)^C$, $q \le (\log x)^{C_q}$, $(a_0, q) = 1$, and $\varepsilon = \pm 1$, let $\Pi_{q,a_0;\varepsilon,h}(x)$ count the eligible primes additionally satisfying $p \equiv a_0 \pmod q$. Then

$$\#\{p \in \mathcal{P}_{\varepsilon,h}(x) : p \equiv a_0 \ (q),\ \mu^2(\varepsilon p + h) = 1\} = \mathfrak{A}(q, a_0; \varepsilon, h)\Pi_{q,a_0;\varepsilon,h}(x) + O_{A,C,C_q}(x(\log x)^{-A}), \tag{9.14}$$

where the product, interpreted as zero if any forced-square factor occurs, is

$$\mathfrak{A}(q, a_0; \varepsilon, h) = \prod_{r \nmid qh} \left(1 - \frac{1}{r(r-1)}\right) \prod_{\substack{r \| q \\ r \mid \varepsilon a_0 + h}} \left(1 - \frac{1}{r}\right) \prod_{\substack{r^2 \mid q \\ r^2 \mid \varepsilon a_0 + h}} 0. \tag{9.15}$$

Proof. Write $L = \log x$ and delete the $O(\log(2|h|)) = O_C(L)$ primes dividing h. On the remaining primes, a square divisor $d^2 \mid \varepsilon p + h$ necessarily has $(d, h) = 1$. Truncate $\mu^2(n) = \sum_{d^2 \mid n} \mu(d)$ at $d \le L^D$, with a fixed $D > A + C_q + 3$. The absolute discarded sum is at most

$$\sum_{L^D < d \le \sqrt{x+|h|}} (x/d^2 + 1) \ll xL^{-D} + \sqrt{xL^C} = O_{A,C,C_q}(xL^{-A}). \tag{9.16}$$

For each retained squarefree d, the simultaneous conditions $p \equiv a_0 \pmod q$ and $p \equiv -\varepsilon h \pmod{d^2}$ are either incompatible or specify one class modulo $m_d = \operatorname{lcm}(q, d^2)$. Compatibility means $a_0 \equiv -\varepsilon h \pmod{\gcd(q, d^2)}$. When compatible the class is reduced: it is a unit at primes dividing q by $(a_0, q) = 1$, and at primes dividing d by $(d, h) = 1$. Since $m_d \le L^{C_q + 2D}$, endpoint-uniform Siegel–Walfisz applies at arbitrarily high fixed precision. Summing its errors over $d \le L^D$ gives the required error after increasing that precision. A small interval endpoint causes no difficulty: use the trivial bound below xL^{-B} and the usual Siegel–Walfisz estimate above it, with fixed B chosen at the requested precision.

Relative to the prime progression's logarithmic-integral mass, the main coefficient is

$$\sum_{\substack{(d,h)=1 \\ a_0 \equiv -\varepsilon h \ (\gcd(q,d^2))}} \mu(d) \frac{\varphi(q)}{\varphi(\operatorname{lcm}(q, d^2))}. \tag{9.17}$$

The series is absolutely convergent for fixed q. Its tail beyond L^D is at most

$$\varphi(q) \sum_{d > L^D} \frac{1}{\varphi(d^2)} \ll L^{C_q - D + 1},$$

, uniformly in the displayed range. Even multiplying this by x is within the requested error. Compatibility and the totient ratio factor prime by prime. If $r \nmid qh$, selecting $r \mid d$ costs $1/\varphi(r^2) = 1/(r(r-1))$. If $r \mid h$, it cannot be selected. If $r \| q$, selection is compatible exactly when $r \mid \varepsilon a_0 + h$,

and then costs $\varphi(r)/\varphi(r^2) = 1/r$. If $r^2 \mid q$, selection is compatible exactly when $r^2 \mid \varepsilon a_0 + h$, and then costs one. These are precisely (9.15); in particular the coefficient lies in $[0,1]$.

Siegel–Walfisz for the original progression replaces its logarithmic-integral mass by the actual eligible-prime count at the same precision. Restoring the removed primes costs $O_C(L)$ in the original count; their contributions in the finite expansion are at most $O_C(L^{D+1})$, also admissible. This proves (9.14). □

For $q = 9$, $a_0 = 1$, $\varepsilon = 1$, $h = 8$, the last product is zero and every eligible shifted value is divisible by 9. Thus keeping A_h as a progression sign-density constant would be false. Formula (9.14) supplies the correct positive population but does not split it into two signs.

10 Assembly of the Goldbach statements

Proofs of Theorems G and S. Choose ρ, $A^* > 0$ and then $B > A^* + 3$. Let C_{coef} be the corresponding fixed coefficient and smoothing-loss exponent for this choice of B. Complete the parameter choices in the following order:

$$\begin{gathered}\rho\ ;\ A^*\ ;\ B\ ;\ C_{\mathrm{coef}}\ ;\\ D^* > 2(A^* + B + C_{\mathrm{coef}} + c_*)\ ;\ E_0\ ;\ G\ ;\\ A_0\ ;\ K\ ;\ C^*, C^\dagger, C_0\ ;\ B_0, B_1\ ;\\ A_{\mathrm{energy}} = D^* + O(A_0 + B_1)\ ;\\ D, C_{\mathrm{zf},1}, C_{\mathrm{zf},2}, C_3, A_s, A_1\ ;\\ \text{the fixed precision of Lemma 6.11}\ ;\ a_i\ ;\ L_0.\end{gathered} \tag{10.1}$$

Here c_* is a fixed reserve for scale and dyadic losses, chosen large enough before the later parameters. The choices for the moment estimate are those in LWW Section 4.7, p. 39; A_0, K are independent of the band locations. The zero-free constants are subsequently defined by (6.45) with $A = A_{\mathrm{energy}}$ and do not enter C_{coef} or the inequality for D^*. For any fixed b the bK band exponents can be selected after these saving exponents; their positions affect thresholds, not these choices.

On each of the $O(\log L)$ dyadic prime intervals, the Vaughan reduction and Proposition 7.2 reduce the weighted prime sum to Type II sums T_Y on $O_K(L^\theta)$ product scales. Propositions 5.3 and 6.1, followed by Theorem 6.13, give

$$Q_Y^{\mathrm{all}} = Q_Y^{\mathrm{maj},\mathcal{N}} + O(\mathcal{N}YL^{-D^*}), \quad |Q_Y^{\mathrm{maj},\mathcal{N}}| \ll \mathcal{N}YL^{-D^*}, \quad |T_Y|^2 \ll (\mathcal{N}/Y)Q_Y^{\mathrm{all}} \ll \mathcal{N}^2L^{-D^*}. \tag{10.2}$$

Thus $|T_Y| \ll \mathcal{N}L^{-D^*/2}$, and the larger fixed budget in (10.1), using $\mathcal{N}/X \ll 2L^B$, gives $T_3(F) \ll xL^{-A^*}$. For Theorem G, use the selected H1*; (1.2) supplies it for every fixed selection of constants. For Theorem S, use Corollary 6.17 with $A_1 = A_0 + C + 1$ and the Burgess input in Theorem 6.13. In either case

$$\sum_{x<p\le 2x} (\log p)\mu(\mathcal{N}-p)W(\mathcal{N}-p) = O(xL^{-A^*}), \qquad \sum_{p<\mathcal{N}} (\log p)\mu(\mathcal{N}-p)W(\mathcal{N}-p) = o(\mathcal{N}). \tag{10.3}$$

The omitted small primes have the negligible bounded-weight cost stated in Remark 2.2.

For each fixed b, every normalized weighted limit in (8.5) is zero. Then let $b \to \infty$ to get $\sum_{p<\mathcal{N}} \mu(\mathcal{N}-p)\log p = o(\mathcal{N})$. Lemma 9.2 and $\pi(\mathcal{N}) \sim \mathcal{N}/L$ give the initial conclusion of (1.4). Finally, for $\kappa = \pm 1$,

$$\#\{p < \mathcal{N} : \mu(\mathcal{N}-p) = \kappa\} = \tfrac{1}{2}\left(\#\{p < \mathcal{N} : \mu^2(\mathcal{N}-p) = 1\} + \kappa \sum_{p<\mathcal{N}} \mu(\mathcal{N}-p)\right). \tag{10.4}$$

Proposition 9.1 proves the sign densities.

All comparisons use $\log x = L + O_B(\log L)$ on the stated range. Puncturing, the large common-prime discard, the power errors in Lemma 4.2 and the $O(\log L)$ dyadic summation are uniform in even $\mathcal{N}$. Constants depend on the fixed bands, K, ρ, b, and requested precisions, and in the small-radical case on C. The limits are ineffective through Siegel–Walfisz and Lemma 6.11. A fixed H1* supplies the selected fixed energy precision; the half-plane and the small-radical region supply every fixed precision needed in this order. □

10.1 Changes in version 2: assembly of A, C and D

Proofs of Theorems A, C and D. Use $L = \log x$, $\Delta = -\varepsilon h$, $H = |h|$. For D fix $a < 1/2$ and the retuned exponents of Section 5.3; for A one may use the same exponents with any fixed auxiliary $a \in (0, 1/10]$ and omit the rough sieve. Choose a sufficiently large fixed lower-prime cutoff B and smoothing precision, and then the required saving D_* so that it absorbs the coefficient, Mellin, dyadic and L^{B+C} losses. Choose E_0, G, A_0, K in the order of (10.1), with $A_0 > G + 3$ and $K > 2(G+3)/j_*$. These choices precede the band locations. Next choose the fixed transform exponents, conductor cutoff and small-conductor, Type I and moment precisions. Finally choose the separated band family and its packing. For the reflected branch choose the energy precision and zero-free constants by (6.45) at this stage; they do not enter the earlier coefficient-loss exponent.

The sifted Vaughan identity, Proposition 7.2, Proposition 5.5 and Proposition 6.14 give, under H1 and H2′,

$$Q_Y \ll XYL^{-D_*}, \qquad |T_Y|^2 \ll \frac{X+H}{Y} Q_Y \ll X(X+H)L^{-D_*} \ll X^2 L^{B+C-D_*}. \tag{10.5}$$

The rough indicator transfers to the bounded common cofactor in Proposition 2.8; it gives precisely the same upper square. A larger fixed D_* absorbs all subsequent losses. The number of product scales is $O_K(L^\theta)$ and the number of retained prime dyads is $O_B(\log L)$. For each fixed requested A, and with $B > A + 2$, this proves uniformly for every permitted single weight

$$\sum_{p \in \mathcal{P}_{\varepsilon,h}(x)} (\log p) f(\varepsilon p + h) \mathcal{R}(\varepsilon p + h) W_j(\varepsilon p + h) \ll_A xL^{-A}, \tag{10.6}$$

where $\mathcal{R} = 1$ for A and $\mathcal{R} = \mathcal{R}_{z,h}$ for D. The omitted low primes cost $O(xL^{1-B})$.

Section 8.2 supplies one threshold for these estimates, and Section 8.3 gives $b_x \asymp \log L$ disjoint copies. Apply its Cauchy bound with the rough prime measure when relevant. A sufficiently large fixed A, B yields

$$\left| \sum_{p \in \mathcal{P}_{\varepsilon,h}(x)} (\log p) f(\varepsilon p + h) \mathcal{R}(\varepsilon p + h) \right| \ll xV/\sqrt{\log L}, \quad V = 1 \text{ or } V_h(z). \tag{10.7}$$

For $p > x/L^B$, replace $\log p$ by L with relative error $O_B(\log L/L)$ using Chebyshev. Low primes contribute $O(x/L^B)$. Relative to $V\pi(x)$ their total error is $O(L^{a-1} \log L + L^{a+1-B})$ in D and the same expression with $a = 0$ in A. It is smaller than $(\log L)^{-1/2}$. The $O(\log x)$ primes dividing h can be restored on this scale; on the others punctured and full roughness agree. This proves A and D and the rate C.

The positive counts in Section 9 and the exact identities

$$\mathbf{1}_{\mu(n)=\kappa} = \frac{\mu^2(n) + \kappa\mu(n)}{2}, \qquad \mathbf{1}_{\lambda(n)=\kappa} = \frac{1 + \kappa\lambda(n)}{2} \quad (n \geq 1) \tag{10.8}$$

give their sign formulas, with the actual eligible counts. Section 13 proves B with its own relative measure and uses the same packing for its instance of C. □

Corollary 10.1 (Quantitative reflected branches)**.** *For $\varepsilon = -1$, $x = h = \mathcal{N}$ even, the A and D conclusions and their explicit rate hold under* H2′*, Burgess and the sufficiently strong fixed* H1⋆ *selected above. They hold on the small-radical class of S with Iwaniec and Burgess, and on the classes in F with its respective analytic inputs.*

Proof. In this specialization $X \le \mathcal{N}$ and the Mellin scale is one. The reflected energy theorem supplies the major-arc bound at the selected precision. Its large-value count remains uniform for $a_K \le \theta < 1/2$, because $L^{a_K} = o(L^{1-a_K} \log L)$, and the Burgess and Pólya–Vinogradov power margins are unchanged. Replacing Z by $\exp(L^{3/4})$ costs at most $O(ZL/V)$ in the long-prime deletion, still negligible since V is a fixed positive power of the large variable. The reconstructed sifted Möbius lemma and the Vaughan identity supply the other changed analytic interfaces. The common signed transport, positive moments and packing then give (10.7) and its unweighted consequences. The small-radical region and the criteria in F supply precisely this selected rectangle. This also completes the reflected consequences of Theorem F. □

11 Small shifts

Here $L = \log x$ and $0 < |h| \le L^{C_0}$, with fixed C_0.

11.1 Changes to the version 1 small-shift proof

The unnormalized basis used there was $g_h(P) = \left(\begin{smallmatrix} u & -hc \\ v & -hd \end{smallmatrix}\right)$, where $ud - vc = 1$. It has determinant $-h$, but can have a relative first-coordinate bound proportional to $|h|$. We now use the normalized signed basis (3.19) with $\Delta = -\varepsilon h$. Lemma 3.9 bounds both cell occupancy and path coordinates independently of $|h|$. The parity conditions are (3.22); after the normalization the even-h condition is not simply that z_1 be odd. Lemma 4.2 counts the normalized signed complementary ratio directly. These replacements reuse the block, root and transport sections instead of requiring a separate small-shift geometric proof.

Proposition 11.1 (Small shifts without H1)**.** *For fixed C_0, A and D restricted to $0 < |h| \le L^{C_0}$ hold with their displayed rates under* H2′ *and the listed classical inputs alone. For D retain nonzero even h and every fixed $a < 1/2$.*

Proof. In the hybrid major arcs all characters are modulo kH, $k \le L^{A_0}$, and have conductor at most $L^{A_0+C_0}$. Choose the fixed cutoff $A_1 > A_0 + C_0 + 1$. The small-conductor proof in Section 6.6.4 then estimates the whole character average, including its principal character, at the same Mellin scale $\max(1, X/H)$. The short-prime contour under H1 is unused. The reconstructed Type I and positive moment proofs, common signed transport, retuning and uniform packing are unchanged. Thus the A/D assembly applies with H2′ as its sole nonclassical input. □

Proof of Theorem H. Use Proposition 11.1 with $\varepsilon = 1$. The stated hypotheses of H include H2′. Its cancellation follows from A with the explicit error above; the sign formula follows from the positive count and (10.8). In this range $\Pi_{1,h}(x) = \pi(x) + O(L^{C_0})$, so replacing the eligible count by $\pi(x)$ costs less than the error. This proves H with its original hypotheses and gives the separately stated weaker-input branch. Positivity is retained throughout. □

The reflected weaker-input branch is Corollary 10.1. Neither branch asserts that one selected polylogarithmic-height rectangle suffices for all intermediate affine shifts, where frequencies can be of size X/H.

12 Liouville versions

The function $\lambda(n) = (-1)^{\Omega(n)}$ is completely multiplicative and takes values in $\{-1, 1\}$ for every $n \geq 1$. We retain the Vaughan coefficient $\mu(m)R_Z(m)$, since it comes from decomposing the prime variable. For the shifted target define explicitly

$$\begin{aligned} F_{\lambda,G}(n) &= \begin{cases} \lambda(\mathcal{N}-n)W(\mathcal{N}-n), & 1 \leq n < \mathcal{N}, \\ 0, & \text{otherwise}, \end{cases} \\ F_{\lambda,h}(n) &= \begin{cases} \lambda(n+h)W(n+h), & n \geq 1,\ n+h \geq 1, \\ 0, & \text{otherwise}. \end{cases} \end{aligned} \tag{12.1}$$

For $\kappa = \pm 1$, the three statements of Theorem 1.1 are explicitly

$$\sum_{p<\mathcal{N}} \lambda(\mathcal{N}-p) = o(\pi(\mathcal{N})), \qquad \#\{p < \mathcal{N} : \lambda(\mathcal{N}-p) = \kappa\} = \left(\tfrac{1}{2} + o(1)\right)\pi(\mathcal{N}), \tag{12.2}$$

$$\sum_{\substack{p\leq x \\ p+h\geq 1}} \lambda(p+h) = o(\pi(x)), \qquad \#\{p \leq x : p+h \geq 1,\ \lambda(p+h) = \kappa\} = \left(\tfrac{1}{2} + o(1)\right)\pi(x). \tag{12.3}$$

The former holds for all even $\mathcal{N}$ under the hypotheses of Theorem G, and for the fixed-C small-radical class under those of Theorem S; the latter is uniform for $0 < |h| \leq (\log x)^C$ under those of Theorem H. LWW Section 1, pp. 1–3, mentions the Liouville counterpart; we supply the expansion and Type I substitutions needed for it.

12.1 Expansion of the weight

The exact identities are

$$\begin{aligned} \omega_i(n)\rho^{\omega_i(n)-1} &= \sum_{\substack{p\in\mathcal{P}_i \\ p\mid n}} \rho^{\omega_i(n)-1}, \\ W(n) &= \frac{1}{\prod_i V_i} \sum_{\substack{a\in\mathcal{A} \\ a\mid n}} \rho^{\omega(n)-K}, \\ \omega(ah) - K &= \omega(h) - s(a,h), \qquad s(a,h) = \#\{i : p_i \mid h\} \in \{0,\dots,K\}, \end{aligned} \tag{12.4}$$

where $\omega_i(ah) = \omega_i(h) + \mathbf{1}_{p_i\nmid h}$ and $s(a,h) = 0$ precisely when $(a,h) = 1$.

Lemma 12.1 (Liouville expansion)**.** *Let w denote $mn+2$, $\mathcal{N}-mn$ or $mn+h$, with the positive argument convention used above and the corresponding coefficients. Then*

$$\begin{aligned} \sum_{\substack{m,n \\ w\geq 1}} \alpha_m\beta_n\lambda(w)W(w) = \frac{(-1)^K}{\prod_i V_i} \sum_{\ell\geq 1} \lambda(\ell)\rho^{\omega(\ell)} \sum_{\substack{a\in\mathcal{A},m,n \\ w=a\ell,(a,\ell)=1}} \alpha_m\beta_n \\ + O_{K,\rho}(XLe^{-L^{0.1}}). \end{aligned} \tag{12.5}$$

For a fixed positive w, the discrepancy before multiplication by $\alpha_m\beta_n$ is exactly

$$\frac{\lambda(w)}{\prod_i V_i} \sum_{\substack{a\in\mathcal{A},a\mid w\\ s(a,w/a)\geq 1}} \rho^{\omega(w)-K}. \tag{12.6}$$

It lies in $\lambda(w)[0,W(w)]$; every summand weight is at most 1.

Proof. For $a \in \mathcal{A}$, $\lambda(a) = (-1)^K$, and complete multiplicativity gives $\lambda(w) = (-1)^K\lambda(w/a)$ for every $a \mid w$, with no coprimality assumption. On $(a, w/a) = 1$, (12.4) gives exactly the main term. The omitted summands are precisely (12.6). They have some $p_i^2 \mid w$, and $\omega(w) \geq K$ makes their weights at most 1. Their absolute value is bounded, with constant 1, by

$$\frac{1}{\prod_i V_i} \sum_{\substack{a\in\mathcal{A},a\mid w\\ (a,w/a)>1}} 1.$$

Since $W(w) \ll_{K,\rho} 1$, the absolute discrepancy is at most

$$C_{K,\rho} \sum_{p\in\cup_i\mathcal{P}_i} \mathbf{1}_{p^2\mid w}.$$

For fixed m on the coefficient support, $(m, p) = 1$ and the condition $p^2 \mid w$ selects one class of n modulo p^2. Since $N \gg p^2$, its count is at most $N/p^2 + 1 \ll N/p^2$. Summing with $|\alpha_m| \leq 1$, $|\beta_n| \ll L$ and $\sum_{p\in\cup_i\mathcal{P}_i} p^{-2} \ll e^{-L^{0.1}}$ gives the claimed error for all three positive shifted arguments. This is the bounded-weight square-divisibility argument of LWW (4.3), p. 15.

If the restriction $(a, \ell) = 1$ is then removed before Cauchy, the added terms have $s \geq 1$ and $\rho^{\omega(\ell)} \leq \rho^{\omega(w)-K}$, by (12.4). They are bounded by the same count, at an additional fixed constant. Cauchy uses only $|\lambda(\ell)\rho^{\omega(\ell)}| \leq 1$. For the Goldbach shift use $Q_Y^+ \leq Q_Y^{\mathrm{all}}$ in (2.18); for small shifts use (2.23). In each case the resulting square contains no target function. □

12.2 Bombieri–Vinogradov for Liouville

The Möbius input is LWW (3.1), p. 10: for every $A > 0$ there exists $B = B(A)$ such that

$$\sum_{q\leq x^{1/2}L^{-B}} \max_{(c,q)=1} \max_{y\leq x} \left| \sum_{\substack{n\leq y\\ n\equiv c\ (q)}} \mu(n) \right| \ll_A xL^{-A}. \tag{12.7}$$

Lemma 12.2 (Liouville distribution)**.** *For every $A > 0$ there is $B'(A)$ such that*

$$\sum_{q\leq x^{1/2}L^{-B'(A)}} \max_{(c,q)=1} \max_{y\leq x} \left| \sum_{\substack{n\leq y\\ n\equiv c\ (q)}} \lambda(n) \right| \ll_A xL^{-A}. \tag{12.8}$$

One may take $B'(A) = \max(B(A), A) + A + 1$.

Proof. The multiplicative identity

$$\lambda(n) = \sum_{d^2\mid n} \mu(n/d^2) \tag{12.9}$$

holds prime by prime: at p^k the sum is 1 for even k and -1 for odd k. For $(c,q)=1$,

$$\sum_{\substack{n\le y\\ n\equiv c\ (q)}}\lambda(n)=\sum_{d\le\sqrt{y}}\sum_{\substack{m\le y/d^2\\ d^2m\equiv c\ (q)}}\mu(m).$$

There is no solution when $(d,q)>1$; otherwise the class is $m\equiv c\bar{d}^2 \pmod q$, which is reduced. Put $D_0=L^{A+1}$ and $Q=x^{1/2}L^{-B'}$. For $d\le D_0$, use the discrepancy

$$E_\mu(x/d^2;q)=\max_{(c',q)=1}\max_{z\le x/d^2}\left|\sum_{\substack{m\le z\\ m\equiv c'\ (q)}}\mu(m)\right|.$$

For $d>D_0$ use $y/(d^2q)+1$. The tail summed over $q\le Q$ is

$$\ll xL/D_0+Q\sqrt{x}\le xL^{-A}+xL^{-B'}. \tag{12.10}$$

For the head, $x_d=x/d^2\ge xL^{-2A-2}$ and $\log x_d\ge L/2$. Condition $B'\ge B(A)+A+1$ ensures $Q\le\sqrt{x_d}(\log x_d)^{-B(A)}$. Apply (12.7) at x_d and sum $\sum_{d\le D_0}xd^{-2}L^{-A}\ll_A xL^{-A}$. The stated $B'(A)$ also absorbs the tail. □

Lemma 12.3 (Weighted Liouville distribution)**.** *Put*

$$E_\lambda(Y;m)=\max_{(c,m)=1}\max_{z\le Y}\left|\sum_{\substack{t\le z\\ t\equiv c\ (m)}}\lambda(t)\right|.$$

For fixed $A_{\mathrm{BV}}>2A+4$ and $Q\le Y^{1/2}L^{-B'(A_{\mathrm{BV}})}$,

$$\sum_{m\le Q}\tau(m)E_\lambda(Y;m)\ll_A YL^{-A}. \tag{12.11}$$

Here $\log Y\asymp L$ in the applications, with fixed comparison constants and a corresponding fixed precision margin.

Proof. The trivial bound is $E_\lambda(Y;m)\ll Y/m+1$. As in the Cauchy step of LWW Section 3, p. 13,

$$\Big(\sum\tau E\Big)^2\le\Big(\sum E\Big)\Big(\sum\tau^2E\Big),\quad \sum E\ll YL^{-A_{\mathrm{BV}}},\quad \sum\tau^2E\ll YL^4+QL^3. \tag{12.12}$$

Use Lemma 12.2 at precision A_{BV}, rather than A, and take square roots. □

12.3 Type I for Liouville

Proposition 12.4 (Liouville Type I)**.** *The Type I estimates of Proposition 7.2 hold for the Liouville target in both orientations, with or without the D roughness condition, on their respective stated ranges.*

Proof. Use (7.13) with $\lambda(Dgt)=\lambda(Dg)\lambda(t)$; there is no auxiliary smooth b-sum. The retained $Dg=x^{o(1)}$ lies in the small-divisor case. For even h use the reduced odd class modulo $2qr/g$. For odd h put $t=2t'$ and use $\lambda(2t')=-\lambda(t')$, with reduced modulus qr/g and no parity condition on t'. Apply Lemma 12.3 at the g-dependent level before summing $\sum_{g\mid\mathrm{rad}(h)}1/g$. All finite presieve/Taylor errors use boundedness only and were proved in Section 7.2. □

Remark 12.5 (Changes in version 2: the Liouville target). The version 1 expansion and distribution lemmas above are retained. The square-overlap proof applies to $w = \varepsilon mn + h \geq 1$ because every marking prime avoids h and is coprime to m. Its positive cofactor has exactly the range in Proposition 2.8; the rough indicator is constant under removal of the selected product. The coefficient $\mu(m)R_Z(m)$ in the prime decomposition remains Möbius. The progression proof uses the same Liouville distribution lemma after its explicit nonunit-class factorization.

12.4 Uses of the target function

Table 2: Target-function substitutions.

Item	LWW input or operation	Liouville substitution
1	(4.3), p. 15; weight expansion	Lemma 12.1, including the square-overlap count.
2	(4.4)–(4.5), p. 15; Cauchy	$\lvert\lambda(\ell)\rho^{\omega(\ell)}\rvert \leq 1$; the Goldbach square is (2.18).
3	Sections 4.2–4.7, pp. 17–39; Appendix A, pp. 47–51	The square has no target function; the same moment and root arguments apply.
4	Section 5, pp. 39–45	The Möbius coefficient in M remains $\mu(m)R_Z(m)$ from Vaughan's identity.
5	(3.1), p. 10	Lemma 12.2.
6	(3.4), p. 12, and (3.5), p. 13	$\lambda(dv) = \lambda(d)\lambda(v)$; no a-sum or Rankin branch.
7	Weighted grouping, p. 13	Lemma 12.3, using the higher fixed precision.
8–10	(3.3) truncations; Proposition 2.1; Proposition 2.3 and weight removal	Only boundedness of the target and residue counts; use the same positive means m_j.
11	Proposition 6.1, pp. 46–47	No squarefree-density input is required.
12	Final sign assembly	Use (12.13).

Other Möbius and squarefree factors in LWW occur in the divisor combinatorics of Proposition 2.3 (pp. 7–9), Bombieri–Vinogradov for primes (p. 9), squarefree r, c_d in the Type I expansion (pp. 11–14), the relative-coordinate/list structure (pp. 22–26, 35), the Vaughan coefficient, and the appendix sieve (p. 50). They do not refer to the target sign and are retained. The additional unit discards in Sections 2 and 7 use only boundedness.

12.5 Sign densities and assembly

Proof of Theorem 1.1. For any finite prime set $\mathcal{P}$ and positive $n(p)$,

$$\#\{p \in \mathcal{P} : \lambda(n(p)) = \kappa\} = \tfrac{1}{2}\left(|\mathcal{P}| + \kappa \sum_{p\in\mathcal{P}} \lambda(n(p))\right), \quad \kappa = \pm 1. \tag{12.13}$$

Use $\mathcal{P} = \{p < \mathcal{N}\}$, of size $\pi(\mathcal{N})$ for even $\mathcal{N} > 2$, or $\mathcal{P} = \{p \leq x : p + h \geq 1\}$, of size $\pi(x) + O(L^C)$.

In the Goldbach assembly, use Lemma 12.1 and Proposition 12.4 for the expansion and Type I sums. The block estimate, major arcs and Theorem 6.13 are independent of the target. The same normalized weight removal gives the unweighted cancellation. For small shifts use the identical substitutions in Section 11. Lemma 12.2 changes the fixed distribution exponent $B(A)$ to $B'(A)$; all choices still precede the large parameter. Uniformity in $\mathcal{N}$ or $|h| \leq L^C$ and the respective hypotheses are unchanged. Identity (12.13) proves the sign densities without Proposition 9.1.

On squarefree values $\lambda = \mu$. Thus the nonsquarefree values in the Goldbach setting have density $(1 - A_{\mathcal{N}})/2 + o(1)$ for each Liouville sign, consistent with (1.4) and (12.2). The errors remain ineffective through the cancellation argument. □

13 Progressions

Remark 13.1 (Changes in version 2: progression restriction). This section adds the prime-variable character to the existing determinant and analytic arguments. The local density was proved in Section 9.3. Restriction to a progression also requires the relative moment and logarithm-removal calculation below.

Proof of Theorem B. **The prime-variable character and Type I**

Let υ be any Dirichlet character modulo q. We first prove the marked estimate with the factor $\upsilon(p)$ at arbitrary fixed logarithmic precision, uniformly over q, υ. Fix a sufficiently large low-prime cutoff exponent B. At the Vaughan level, $\upsilon(mn) = \upsilon(m)\upsilon(n)$, so replace the two coefficient sequences by $\alpha_m \upsilon(m)$ and $\beta_n \upsilon(n)$. Their absolute coefficient bounds are unchanged. Since $q < Z$ eventually, every sifted variable is already coprime to q.

The affine cofactor identity and Cauchy square are unchanged: the extra character factors belong to the coefficients, while the bounded target on the cofactor is still $f(\ell)$ times its damping. The signed root counting, labelled intermediate space, endpoint primitivity, comparison kernel, memory moments, goodness removal and minor-arc error in Section 5.1 apply to these coefficients with the same bounds. In particular no congruence modulo q is added to the root-counting modulus.

For Type I the outer character factor has modulus at most one. To make the inner character completely explicit, in the notation of Section 7.2 one has $\varepsilon q_1 rj + h = Dgbt$ (rename its outer variable q_1 to avoid the present modulus q). Thus the prime-variable character is

$$\upsilon(q_1 rj) = \upsilon\big(\varepsilon(Dgbt - h)\big). \tag{13.1}$$

Partition t into its residue classes modulo q. On each it is constant and bounded. The reduced Möbius/Liouville class modulo $2q_1 r/g$ is intersected with one residue modulo q. These latter classes need not be reduced in t, so one must not invoke reduced-class BV on them directly. Instead, an equivalent simpler treatment is to partition the original inner variable k modulo q before the transformation. Since the sieve already gives $(q_1 k, q) = 1$, all these are unit classes for k, but the resulting target class can still be nonunit. To handle that case, absorb the fixed prime-power part forced by the modulus into the target using the same local-factor identity (7.12), as detailed next.

For a residue $t \equiv c \pmod q$, put $v = (c, q)$, $q' = q/v$, and write $t = vt_1$. Then $t_1 \equiv c/v \pmod{q'}$ is reduced, but (v, t_1) may be nontrivial. For Möbius use the general identity

$$\mu(vt_1) = \begin{cases} \mu(v) \displaystyle\sum_{\substack{b_1 \mid t_1 \\ p \mid b_1 \Rightarrow p \mid v}} \mu(t_1/b_1), & v \text{ squarefree}, \\ 0, & v \text{ not squarefree}. \end{cases} \tag{13.2}$$

If a prime divides both b_1 and q', the original reduced class of t_1 is impossible, so only $(b_1, q') = 1$ survives. The other original modulus $m = 2q_1 r/g$ is already reduced in t, hence $(v, m) = 1$ is necessary, and any prime of b_1 dividing m likewise gives an impossible term. On the surviving terms the new variable $t_2 = t_1/b_1$ satisfies two compatible reduced classes, hence one reduced class

modulo $\operatorname{lcm}(m,q') \le qm$. The numerator changes by inverses of v, b_1, which are units wherever needed. For Liouville simply use $\lambda(vt_1) = \lambda(v)\lambda(t_1)$, with no b_1 expansion.

All extra finite choices and multiplicities cost only fixed powers of L: there are $q \le L^{C_q}$ residue classes, $v \le q$, and at most q preimages of a modulus under $m \mapsto \operatorname{lcm}(m,q')$. To justify the latter bound prime by prime, let e be the exponent in q'. If the fixed lcm has larger exponent, the exponent in m is forced; if it equals e, the exponent in m has at most e+1 choices. Thus there are at most $\tau(q') \le q'$ possible m. The small-divisor BV margin in Section 7.2 remains fixed after these logarithmic factors and replacing its divisor cutoff by $Dgbvb_1 \le y^{1/20}$. The extra reciprocal sum over b_1 is at most $\prod_{p\mid q}(1-1/p)^{-1} \ll \log L$.

For the complementary smooth tail, $v \le L^{C_q}$ and all primes of b_1 divide q; the set of primes supporting bb_1 has $O_{K,A,C_q}(\log L)$ elements. Apply the same Rankin argument as in Section 7.2 to the combined smooth factor bb_1. Its convolution multiplicity contributes the square of the corresponding Euler product, still a fixed power of L at each fixed Rankin exponent. Thus the power-saving tail and endpoint estimates survive. Odd h uses the already separated factor 2 from Section 7.2 before this residue decomposition. This proves the uniformly twisted Type I estimate. The additional residue and divisor costs are absorbed by raising its fixed BV and truncation precisions, without changing the Type II coefficient budget.

The two character locations in the major arcs

The finite fibre expansion and Mellin kernel are unchanged. Its characters now have the form

$$\xi = \chi\psi\upsilon \quad \text{on } m,n, \qquad \xi_b = \chi\psi\psi_b' \quad \text{on the marking product } b. \tag{13.3}$$

All are regarded on the common modulus $\operatorname{lcm}(kH,q)$. The prime-variable character υ occurs only in the two long polynomials. Choose a fixed $A_1 > A_0 + C_q + 2$. If ξ_b were principal, then $\chi\psi = (\psi_b')^{-1}$ and $\xi = \upsilon(\psi_b')^{-1}$ would have conductor dividing $\operatorname{lcm}(q,k)$, at most $qk \le L^{A_0+C_q}$. Therefore every ξ above the chosen cutoff has nonprincipal ξ_b, and H1 gives the same short-prime saving.

The hybrid mean value still averages over $\chi \pmod H$; the fixed additional character υ is absorbed into the coefficients and has modulus at most one. Orthogonality and the logarithmic spacing H/X are unchanged. In the small-conductor range the primitive inducer of ξ determines χ for fixed ψ, υ, so there are still at most L^{2A_1+1} characters. The small-conductor Möbius bound applies at the raised fixed cutoff. Primes dividing q cause no extra loss since they are below Z; primes dividing H are removed exactly as in Lemma 6.11. The sparse-frequency estimate accepts arbitrary bounded coefficients in the long polynomial, including υ, and is uniform in the marking character. Thus (6.68)–(6.69) retain their bounds.

Combining Type I and Type II gives, at every fixed precision A,

$$\sum_{p\in\mathcal{P}_{\varepsilon,h}(x)} (\log p) f(\varepsilon p+h) W_j(\varepsilon p+h)\upsilon(p) \ll_{A,C,C_q} xL^{-A}. \tag{13.4}$$

Character orthogonality modulo q and the factor $1/\varphi(q)$ then give the same bound for each marked progression sum. The uniform band-family proof applies unchanged after the fixed cutoff and precisions are raised; no parameter grows with q beyond its fixed logarithmic exponent.

Progression-relative weight removal

Let I be the eligible prime interval and put $\Theta_{I;q,a_0} = \sum_{p\in I,\, p\equiv a_0(q)} \log p$. For $F = 1, W_j, W_jW_k$ with $j \ne k$, the finite Taylor expansion and CRT give

$$\sum_{\substack{p\in I\\ p\equiv a_0\ (q)}} (\log p)F(\varepsilon p+h) = m_F\Theta_{I;q,a_0} + O_A(xL^{-A}). \tag{13.5}$$

Indeed every band prime exceeds q and is coprime to h, so each retained band divisor d is coprime to q. Combining its reduced shifted class with $a_0 \pmod q$ gives one reduced class modulo qd. Different d give different qd, all $x^{o(1)}$, so the summed BV discrepancies retain arbitrary fixed precision. The same positive Taylor remainder bound applies. The local probabilities at band primes remain $1/(r-1)$; hence the means m_F are exactly those in (8.2) and factor for disjoint copies. Endpoint-uniform Siegel–Walfisz gives $\Theta_{I;q,a_0} \ll x/\varphi(q)$.

Using the b_x copies in Section 8.3, (13.5) yields variance at most $C\Theta_{I;q,a_0}/b_x + O_A(xL^{-A})$. Cauchy with this progression measure and the uniformly small marked progression sums bounds the unmarked weighted sum by

$$\frac{Cx}{\varphi(q)\sqrt{\log L}} + O_A(xL^{-A}) + O_A\left(\sqrt{\frac{x}{\varphi(q)}\,xL^{-A}}\right). \tag{13.6}$$

Choose fixed $A > C_q + 4$. Both errors are absorbed by the first term. To remove $\log p$, use the progression bound $\pi(x;q,a_0) \ll x/(\varphi(q)L)$ for the high primes; this gives a relative error $O_B(\log L/L)$. The low-prime segment contributes at most x/L^B, a relative error $O(L^{C_q+1-B})$, so take fixed $B > C_q + 4$. Using only the ambient Chebyshev bound here would fail when C_q is large. This proves (1.9). Finally choose the precision in (9.14) larger than $C_q + 3$ and use the exact sign identities, proving the stated sign counts.

□

The modulus range is $q \le (\log x)^{C_q}$ with fixed C_q.

A Changes relative to LWW

All LWW references in Table 3 are to version 1, 14 October 2026. Page numbers are the numbered pages of that paper. Its arguments are used as written with these replacements. The square being represented is the full all-integer nonnegative square (2.18), whose expansion is

$$Q_Y^{\mathrm{all}} = \prod_i V_i^{-2} \sum_{\substack{a,b,m,n,r,s\\ (\mathcal{N}-mn)/a=(\mathcal{N}-rs)/b\in\mathbb{Z}\setminus\{0\}}} \eta(a/Y)\eta(b/Y)\alpha_m\overline{\alpha_r}\beta_n\overline{\beta_s}. \tag{A.1}$$

After the common-list removal, (2.19) is exactly its determinant condition. Endpoint pairing uses (5.8) with a conjugate-linear initial entry. The moment normalization remains d_0YMN, and Cauchy uses $\mathcal{N}/Y$ separately.

Table 3: The block dictionary.

Row	LWW location	Replacement or retained input
1	Section 4.1, p. 14; (4.3)–(4.5), p. 15	Separate the product cutoff as there, with $F = 0$ off positive shifted arguments. Use the nonnegative enlargement (2.18) and its exact integer-cofactor expansion (A.1). Its common range is $\lvert\ell_0\rvert \ll \mathcal{N}/Y$, and $\lvert T_Y\rvert^2 \ll (\mathcal{N}/Y)Q_Y^{\mathrm{all}}$.
2	(4.8)–(4.9), p. 16; (4.24)–(4.25), p. 21	Set $t' = t/\mathcal{N}$, use $\psi_1(t/(\mathcal{N}Y)) = \psi_1(t'/Y)$ and $\mathfrak{M}_{\mathcal{N}}$. The $\mathcal{N}$ branches cancel the Jacobian $1/\mathcal{N}$ in Lemma 3.1.

Row	LWW location	Replacement or retained input
3	Proposition 4.3, pp. 20–21; Section 4.6, p. 35	Use labelled index-$\mathcal{N}$ blocks and $P_j = g_{\mathcal{N}}(P)z_j$ with $g_{\mathcal{N}} = gE_{\mathcal{N}}U_0$. At endpoints use $f(Dbm, s) = \overline{\alpha_m}\beta_s$, so $\overline{f(P)}f(Q) = \alpha_m\overline{\alpha_r}\beta_n\overline{\beta_s}$.
4	Continuous extension, p. 22	Keep the entry $\mathcal{N}/u$ exactly in (3.14); $\tau_{\mathcal{N}}(z) = z_1 + \beta z_2$ replaces $z_1 - 2rz_2$.
5	Path bound, p. 23	Lemma 3.5 directly gives $\lvert z_2\rvert \le 64d_0YMN/\mathcal{N}$, $\lvert z_1\rvert \le 16 + \lvert z_2\rvert$. It implies the needed $\lvert z\rvert \ll Rd_0Y$ and coefficient mass $\exp(O_K(L^{r_*+j_*+\theta}))$.
6	(4.30), (4.32), pp. 23–24	The second boundary is $E_2 = t_0\tau_{\mathcal{N}} + z_2\mathcal{N}/(d_0YMNs)$. The extra term is $O(1)$, as is its s derivative; $\partial_{t_0}E_2 = \tau_{\mathcal{N}} \in [1/16, 16]$ and $\partial_\beta E_2 = t_0z_2$ and $\lvert\partial_\beta E_2\rvert \ll \exp(O_K(L^{j_*+\theta}))$. The neighborhood volume is $x^{-c\delta/8+o(1)}$.
7	(4.31), p. 24	$\partial_\beta(\tau_{\mathcal{N}}(w)/\tau_{\mathcal{N}}(z)) = 1/\tau_{\mathcal{N}}(z)^2$ for $\det(z, w) = 1$, replacing $-2/\tau(z)^2$.
8	Boundary cubes, p. 24	Use (s, t_0, β), the same transverse derivatives and $x^{o(1)}$ choices. Sum the interval-uniform errors of Lemma 4.2 over cubes before the moment limit.
9	(4.33)–(4.34), pp. 24–25	Use the primitive relative box of radius $2C_{\mathrm{pos}}Rd_0Y$ and the same τ window. The multiplier is $\chi_{\mathcal{N}} = \mathbf{1}_{g_{\mathcal{N}}z\in\Omega}$, of norm at most 1. Endpoint unit support replaces LWW's z_1-odd test; that test is not implied for the normalized coordinates used here.
10	(4.40)–(4.41), pp. 27–28	$z' = j_1w_z + hz$ and $\tau_{\mathcal{N}}(z')/\tau_{\mathcal{N}}(z) = h + j_1\tau_{\mathcal{N}}(w_z)/\tau_{\mathcal{N}}(z)$. The allowed ratio strip leaves $O(1)$ integers h per j_1.
11	Slope-cell identity, p. 28	$z_2'/\tau_{\mathcal{N}}(z') - z_2/\tau_{\mathcal{N}}(z) = \det(z, z')/(\tau_{\mathcal{N}}(z)\tau_{\mathcal{N}}(z'))$. An edge has reduced determinant $O(d_0Y)$; cells of width d_0Y have bounded interaction overlap. A Goldbach block meets $O(1)$ such cells.
12	(4.42)–(4.46), pp. 28–30	Use the shared-prime lattice, Minkowski box $A'B' \asymp Y$ and the same signed kernel $Y^{-1}\psi_1(\det(k, \ell)/Y)\int_{\mathbb{T}\setminus\mathfrak{M}_1} e(\theta\det(k, \ell))\,d\theta$. Its $\alpha = D\tau_{\mathcal{N}}(w_*)/\tau_{\mathcal{N}}(z_*)$ is $D\beta_{Q_*}$ modulo 1. The enclosing region has $\lvert\ell\rvert \ll Y$ and bounded τ ratios.
13	(4.45), p. 29	$\chi(B_Dk)$ is a bounded endpoint multiplier. Sharp or nonrectangular position restrictions preserve its contraction property.
14	(4.47)–(4.48), pp. 30–31	$j_1 = P_{\mathrm{full}} - Dp_*ZT_f = \det(z, z') = D(b - a)$; $\tau_{\mathcal{N}}(z')/\tau_{\mathcal{N}}(z) \in [1/16, 16]$. The second good-state test is applied to the intrinsic ratio at the source $g_{\mathcal{N}}(P)z$.
15	(4.49)–(4.50), p. 31	The reduced integer box has the same nonzero linear congruence modulo a list prime. List primes are coprime to $\mathcal{N}$ and to the chosen shared D.
16	Endpoint primitivity, p. 35	A common prime of Dbm, s is $> Z$ and coprime to $\mathcal{N}$. For both kernels, $\mathcal{N} \mid t$, $t \neq 0$ by $(b, ars) = 1$, and $\lvert t/\mathcal{N}\rvert \ll Y = o(Z)$. Such a prime would divide $t/\mathcal{N}$, which is impossible.
17	Section 4.6 partition, p. 36	Use labelled blocks with $O(d_0Y)$ lattice-primitive positions in the application of Lemma 3.4, with no interaction between blocks. The disjoint endpoint union is restricted to units as in (3.7).
18	(4.63)–(4.68), pp. 36–37	Use the positive principal submatrix on $\mathcal{C}_{\mathcal{N}} \cap \Lambda_\varrho$ in the enlarged ambient space. Spectral Hölder precedes root restriction; $\lambda_{\max} \le d_\varrho$ and $(d_0Y)^{1/(2R)} = O(1)$. Only the restricted-root trace is summed.
19	(4.70), p. 37	Exact-edge $O(1)$ and comparison-edge $O(Y)$ counts are in t'. The bad-state ratio measure is unchanged.
20	(4.74)–(4.76), p. 38; (4.6), p. 16	For fixed $k_0 = mn$, $\ell_0 = rs$ is in the class $ba^{-1}k_0 \bmod \mathcal{N}$ in an interval of length $O(\mathcal{N})$, giving $O(1)$ choices. Raw absolute divisor-square counts use the translated cofactor interval or $\lvert\ell_0\rvert \ll \mathcal{N}/Y$, at a cost $\mathcal{N}/X \ll L^B$.

Row	LWW location	Replacement or retained input
21	Appendix A, swapped case, p. 50	Retain $\mathcal{N}/(uv)$, with $\mathcal{N}/(uv) \ll L^B/(d_0Y)$. Use both $k = \lfloor \beta + \mathcal{N}/(uv) \rfloor \in \{0,1\}$ branches, the separate matrix residues and the interval envelopes in the proof of Lemma 4.2.
22	Lemma A.1 and (A.5), pp. 47–51	Use Lemma 4.2, the determinant-$\mathcal{N}$ matrix coset, $c_{\mathcal{N}} = \zeta(2)^{-1} \prod_{p\mid\mathcal{N}} (p-1)/(p+1)$ and line masses $1/(p+1)$. Its factor at 2 is already included. Summing matrix errors costs $S^3 = x^{o(1)}$ and decreases the final fixed line-error exponent.
23	Arc measure, p. 16	Haar measure under $\theta \mapsto \mathcal{N}\theta$ gives $\lvert\mathfrak{M}_{\mathcal{N}}\rvert = \lvert\mathfrak{M}_1\rvert \ll L^{3A_0}/Y$.

A.1 Additional version 2 changes and reconstructed inputs

The preceding table retains the reflected dictionary, with the common numerical retuning. The following entries give the further changes for both orientations, the quantitative rate, rough values and progressions. They also identify arguments outside H2′ that are now supplied in the paper.

LWW location	Version 2 replacement or retained input
Proposition 2.1, p. 4	The exact convolution identity (2.13), both prime-power corrections and the boundary-strip/Mellin proof of Proposition 2.3.
Proposition 2.2, pp. 5–6	Lemma 6.11, proved from MSTT at full dyads, followed by prefix dyads, real-weight partial summation and a smooth-divisor convolution. The sifted tail has $\exp(-cL^{1/4})$ decay with $Z = \exp(L^{3/4})$.
(2.13)–(2.19), pp. 7–10	The same weight. Finite Taylor divisors, reduced-class probabilities $1/(r-1)$ and positive presieve errors give the relative moments in Section 8.1.
Section 3, pp. 10–14	Extract $g = (r,h)$ before applying (7.12); use the reduced class modulo $2qr/g$, or the explicit odd-h parity rule, with each g-dependent BV level treated before the reciprocal g-sum. Large smooth divisors use the fixed-exponent Rankin bounds in Section 7.2.
(4.3)–(4.8), pp. 15–16	$w = \varepsilon(mn - \Delta) > 0$, $\Delta = -\varepsilon h$, $H = \lvert h\rvert$. Cauchy uses $(X+H)/Y$ and the all-nonzero-integer square (2.23); its raw equation is (2.24). Roughness passes to the common cofactor by (2.25).
(4.24)–(4.32), pp. 21–24; Appendix A	The normalized signed basis (3.19), index-H labelled blocks, parity (3.22), intrinsic derivative (3.23) and signed root Lemma 4.2. The swap has all $k = -1, 0, 1$ branches and intervals are enlarged before clipping.
(4.29)–(4.32), pp. 23–24	Root density c_{2H} and probability $1/(p+1)$ per prescribed line. Every fixed $S \le \exp(C_0 L^{\sigma})$, $\sigma < 1$, is allowed. Boundary transversality uses (5.12) with $x^{o(1)}$ gradients.
(4.35)–(4.38), pp. 25–27	Keep the signed prime measures, ghosts and factorial identities. For the separate positive remainder use (5.18), remove only exterior damping, and use the exact adjoint table of Lemma 5.6.
Bonferroni step, p. 23	The model tail is $L^{C_*R}(C_K R)^{B_d}/B_d!$. Arithmetic discrepancy still uses the crude subpower residue/coefficient count. Sections 5.2–5.3.
(4.42)–(4.50), pp. 29–31	The relative signed kernels and balanced powers of Y remain. The clean norm uses $L^{\theta - A_0}$ and the dirty raw norm $L^{1 - j_* K/2}$. Choose $A_0 > G + 3$, $K > 2(G+3)/j_*$.
(4.53)–(4.62), pp. 32–35	Retain cap L^2, polynomial address counts and collision threshold $o(R)$. The new subpower cost exponent is $1 - 7j_* < 1$. The signed ghost operators are unchanged.
(4.63)–(4.68), pp. 36–37	Within each block use cells of width $H/(d_0 Y M^2)$ and occupancy $O(d_0 Y)$. Spectral Hölder precedes restriction to actual unit roots. The trace factor $(d_0 Y)^{1/(2R)}$ is independent of H.

LWW location	Version 2 replacement or retained input
(4.70)–(4.76), pp. 37–39	The absolute raw and comparison bounds (5.14) and (5.15) are uniform in signed Δ. Endpoint primitivity uses $H \mid t$ for comparison edges too.
Section 5, pp. 39–45	Fibre character mass $k^{O(1)}$, Mellin scale $\mathcal{T} = \max(1, X/H)$ and exact prefactor $YX^2/\mathcal{T}$. Hybrid spacing is H/X. Large conductors use the 15/16 contour under H1; small conductors use the reconstructed sparse-frequency Gram argument.
Section 5, reflected branch	Retain the long-prime energy theorem under one sufficiently strong fixed H1*. The comparison $a_K < 1/2$ keeps its large-value bound; Burgess and Pólya–Vinogradov retain fixed power margins.
Section 5.3, p. 46	Uniformity is proved on separated single-weight families. Pack $b_x \asymp \log\log x$ disjoint copies by (8.8); each mixed moment uses at most $2K$ bands, and its averaged error has no b_x loss.
Proposition 6.1, pp. 46–47	Positive counts use the actual eligible interval. Rough/squarefree-rough factors are V_h, U_h; progression squarefreeness uses (1.8), including forced zeros.
New progression interfaces	The prime-variable twist is on m, n, not on the marking product. Nonunit target classes are factored by (13.2); progression-relative variance and prime bounds remove logarithms at scale $\pi(x)/\varphi(q)$.

B Finite checks

The finite checks cover block algebra, unit charts, parity, signed normalized bases, all three complementary-basis branches, cofactor equations, presieve and local-factor identities, and the sifted Vaughan convolution with both prime-power corrections. The positive-memory models include factorial-measure adjoints, shared lists, repeated memory types, active/memory overlaps, projective infinity, longer nonalternating histories and never-active primes. Rational parameter checks include the earlier one-third range and the half-exponent choice. These checks concern finite identities and models. Root discrepancy, analytic estimates and signed operator bounds require the arguments and inputs stated in the paper.

References

[1] William D. Banks. On the OpenAI proof that Landau–Siegel zeros do not exist, 2026. URL https://arxiv.org/abs/2610.15483v1. Version 1, 11 October 2026.

[2] N. A. Carella. Result on the Mobius function over shifted primes, 2022. URL https://arxiv.org/abs/2206.12956v1. Version 1, 26 June 2022; Theorems 1.1 and 1.2.

[3] Harold Davenport. *Multiplicative Number Theory*, volume 74 of *Graduate Texts in Mathematics*. Springer, 2 edition, 1980. doi: 10.1007/978-1-4757-5927-3. URL https://link.springer.com/book/10.1007/978-1-4757-5927-3.

[4] Kevin Ford, Ben Green, Sergei Konyagin, James Maynard, and Terence Tao. Long gaps between primes. *Journal of the American Mathematical Society*, 31:65–105, 2018. doi: 10.1090/jams/876. URL https://www.ford126.web.illinois.edu/wwwpapers/primegaps2.pdf. Lemma 7.1.

[5] Jizhou Guo. Sums of two rough numbers with prescribed Liouville signs, October 2026. URL

https://github.com/aster2024/openai-math-followups/tree/9fc56815121a461cd79dbe7c30f2c59fde7aa938/rough-goldbach-parity. Preprint, 9 October 2026.

[6] Jizhou Guo. Shifted primes with a prescribed parity of the number of prime factors, October 2026. URL https://github.com/aster2024/openai-math-followups/tree/9fc56815121a461cd79dbe7c30f2c59fde7aa938/shifted-prime-parity. Preprint, 9 October 2026.

[7] D. R. Heath-Brown. Burgess's bounds for character sums, 2012. URL https://arxiv.org/abs/1203.5219v2. Version 2, 8 May 2012; introduction, equation (1).

[8] Henryk Iwaniec. On zeros of Dirichlet's L series. *Inventiones Mathematicae*, 23(2):97–104, 1974. doi: 10.1007/BF01405163. URL https://link.springer.com/article/10.1007/BF01405163.

[9] Bryce Kerr. A simpler proof that there are no Landau–Siegel zeros, 2026. URL https://arxiv.org/abs/2610.21099v1. Version 1, 15 October 2026; Theorem 1.

[10] Jared Duker Lichtman. Averages of the Möbius function on shifted primes, 2021. URL https://arxiv.org/abs/2009.08969v2. Version 2, 20 October 2021.

[11] Jared Duker Lichtman and Joni Teräväinen. On the Hardy–Littlewood–Chowla conjecture on average, 2022. URL https://arxiv.org/abs/2111.08912v3. Version 3, 21 June 2022; Corollary 1.4.

[12] Guangshi Lü, Zhiwei Wang, and Jie Wu. On the Hardy–Littlewood–Chowla conjecture, 2026. URL https://arxiv.org/abs/2610.20242v1. Version 1, 14 October 2026.

[13] Alexander P. Mangerel. On a Goldbach-type problem for the Liouville function, 2024. URL https://arxiv.org/abs/2404.12117v2. Version 2, 2 May 2024.

[14] Kaisa Matomäki, Xuancheng Shao, Terence Tao, and Joni Teräväinen. Higher uniformity of arithmetic functions in short intervals I. All intervals. *Forum of Mathematics, Pi*, 11:e29, 2023. doi: 10.1017/fmp.2023.28. URL https://www.cambridge.org/core/journals/forum-of-mathematics-pi/article/higher-uniformity-of-arithmetic-functions-in-short-intervals-i-all-intervals/6DF08E2A3EFABB2DD0DA6CDA38649B18.

[15] Kaisa Matomäki, Mayank Pandey, Javier Pliego, Joni Teräväinen, and Mengdi Wang. Higher order uniformity of the primes and cancellation of the Möbius function in shorter intervals, 2026. URL https://arxiv.org/abs/2610.09567v1. Version 1, 7 October 2026.

[16] Hugh L. Montgomery and Robert C. Vaughan. *Multiplicative Number Theory I: Classical Theory*, volume 97 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 2006. doi: 10.1017/CBO9780511618314. URL https://www.cambridge.org/core/books/multiplicative-number-theory-i/4E45519B26115AEEA4839C6C38206ACD.

[17] OpenAI. The Poisson–Dirichlet law for prime predecessors, September 2026. URL https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/The-Poisson-Dirichlet-Law-for-Prime-Predecessors-September-24-2026. 24 September 2026, 74 pages. Repository commit adc7f1241b42e322a6451854ab7e4b4c146bf78a.

[18] OpenAI. The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane $\Re s > 7/8$, September 2026. URL https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c

146bf78a/preprints/The-Quasi-Riemann-Hypothesis-September-30-2026/paper.pdf. Family 003 accompanying manuscript, 30 September 2026; Section 1, Theorem 1.1, p. 4. Repository commit adc7f1241b42e322a6451854ab7e4b4c146bf78a.

[19] Terence Tao and Joni Teräväinen. The Hardy–Littlewood–Chowla conjecture in the presence of a Siegel zero, 2022. URL https://arxiv.org/abs/2109.06291v2. Version 2, 6 June 2022; Theorem 1.6.

Use of AI

The proofs and the text of this paper were produced with AI systems under the author's direction: version 1 with GPT-6.1 Sol, Claude Fable 5.1 and Claude Opus 5.5, and the additions of version 2 with GPT-6 Astra, GPT-6.1 Sol and Claude Opus 5.5. The proofs were checked by independent reviews with AI models (for version 2, GPT-6.1 Sol, GPT-6 Astra and Gemini 4 Argon), and finite identities and models by the programs in the ancillary files. No human expert has reviewed the proofs. The author takes responsibility for the content.